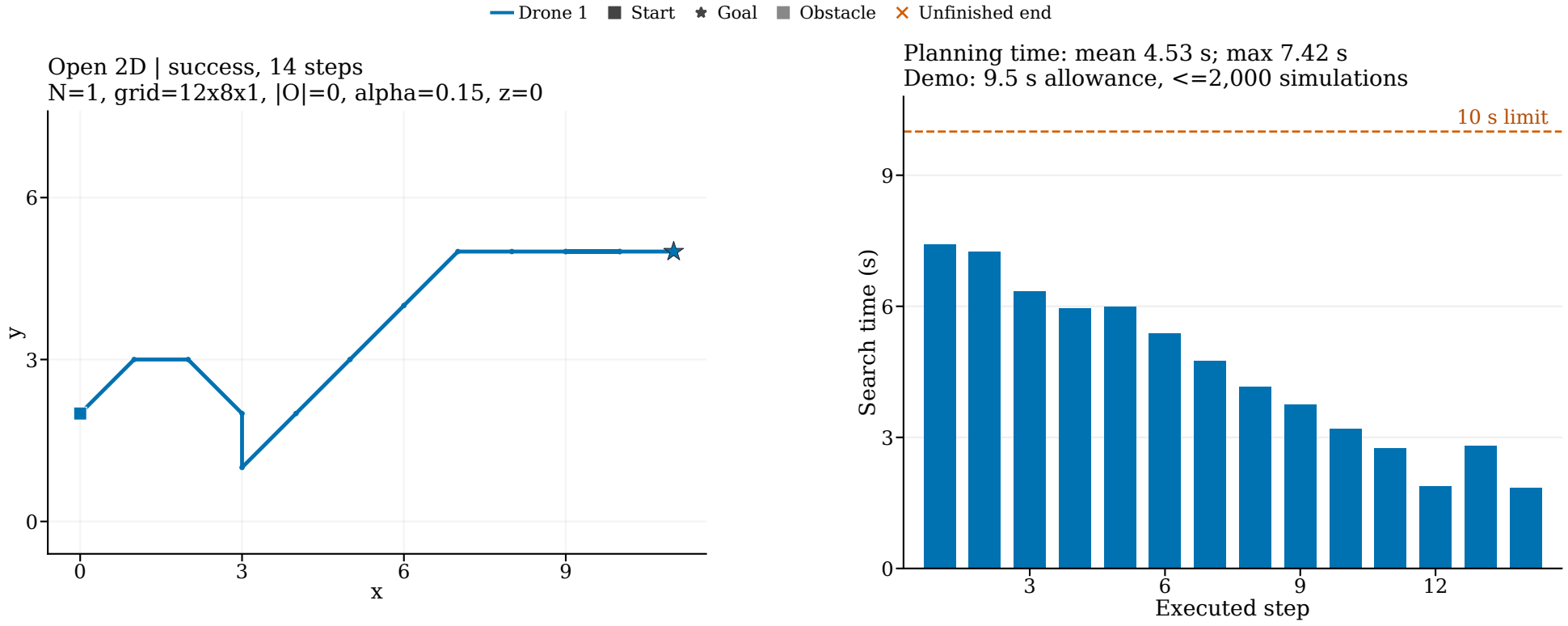
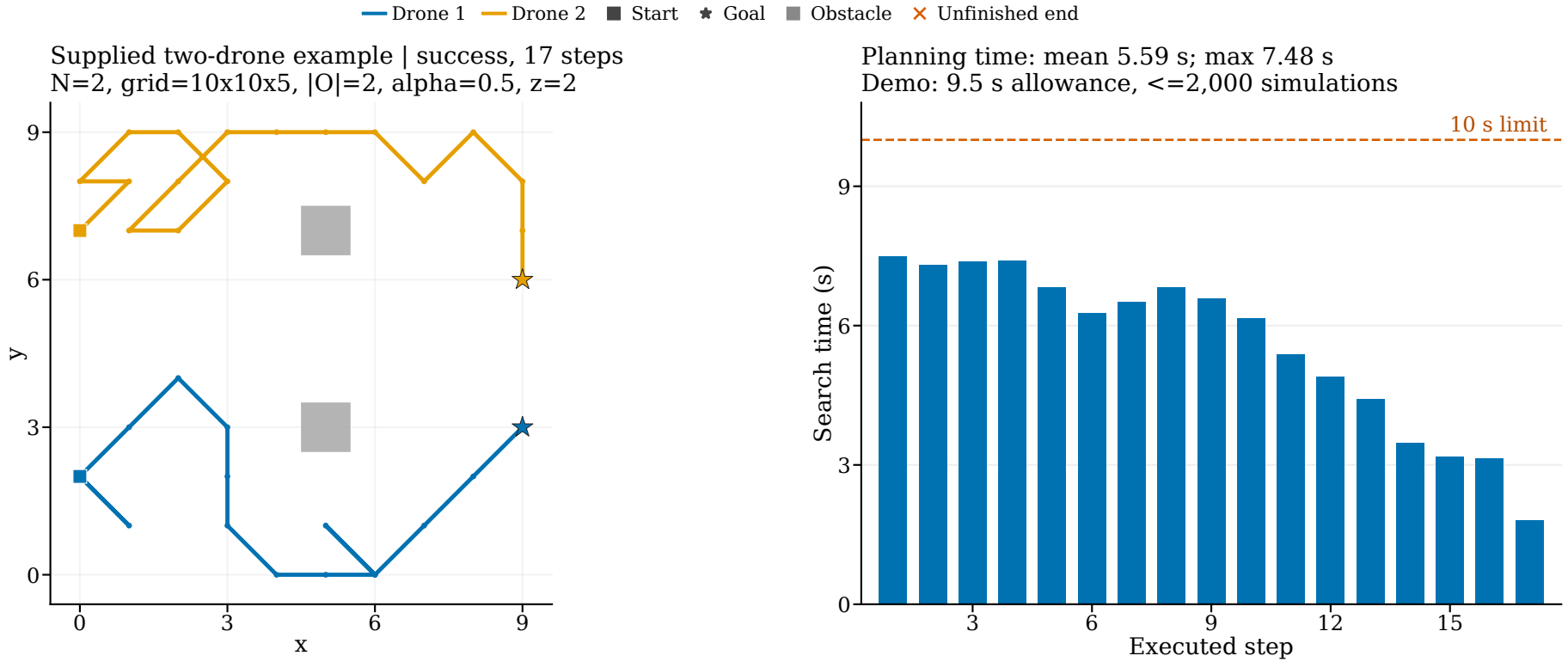
demo open 2d



Saved full-solver trace, seed 0: results/demo_open_2d_default.json
Crosses mark returned positions of unfinished drones; collision targets are not logged.

example config

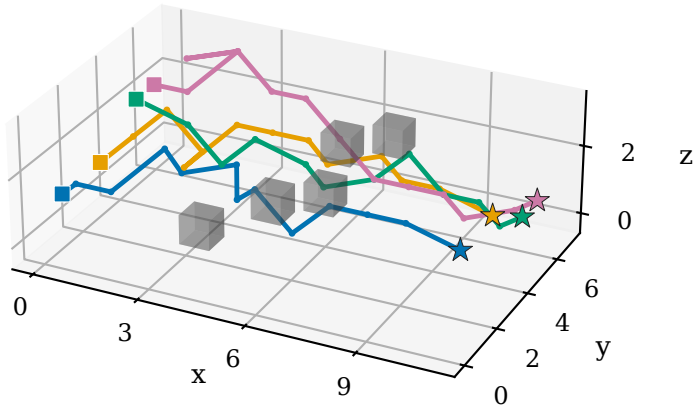


Saved full-solver trace, seed 0: results/example_config_default.json
Crosses mark returned positions of unfinished drones; collision targets are not logged.

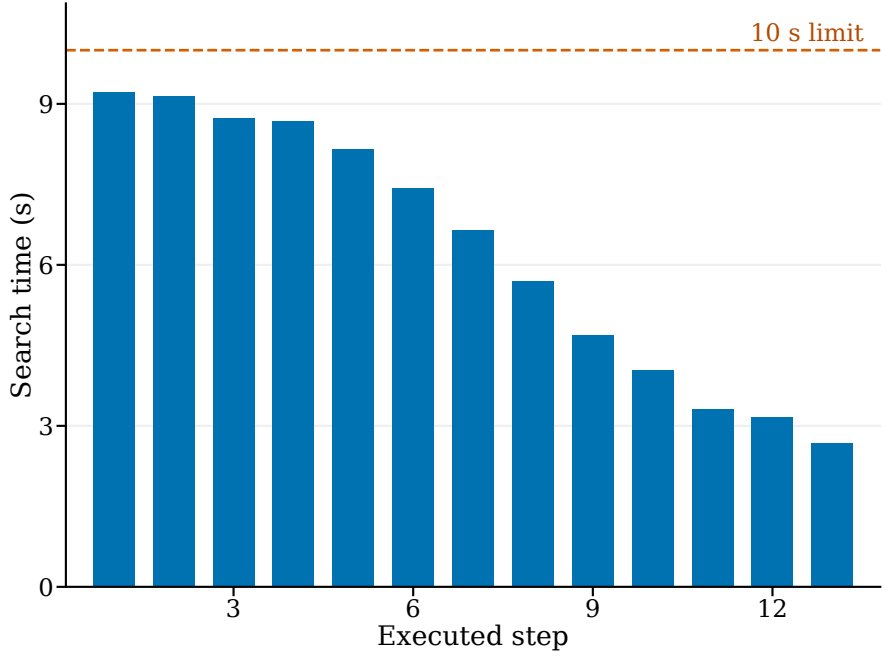
demo four 3d

Drone 1 Drone 2 Drone 3 Drone 4 Start Goal Obstacle Unfinished end

Four drones in 3D | success, 13 steps
N=4, grid=12x8x4, |O|=5, alpha=0.2

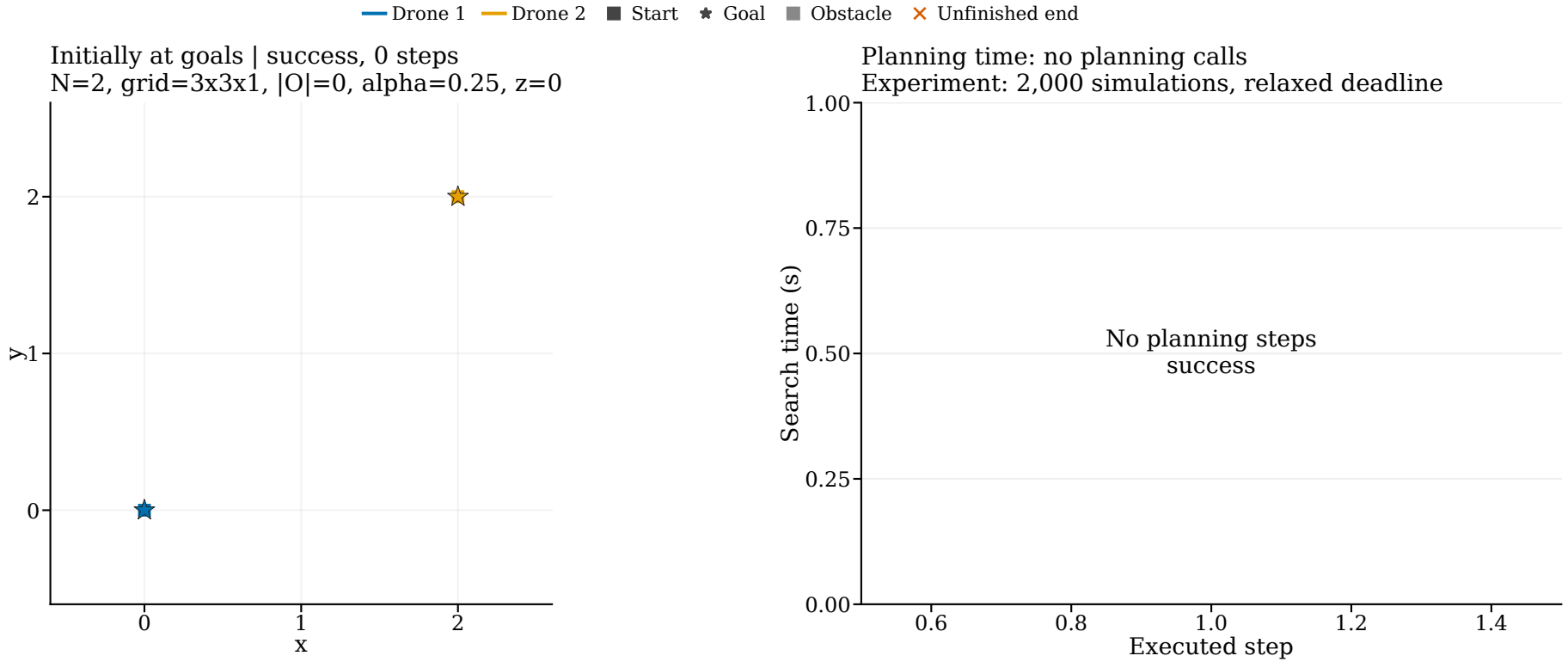


Planning time: mean 6.27 s; max 9.21 s
Demo: 9.5 s allowance, <=2,000 simulations

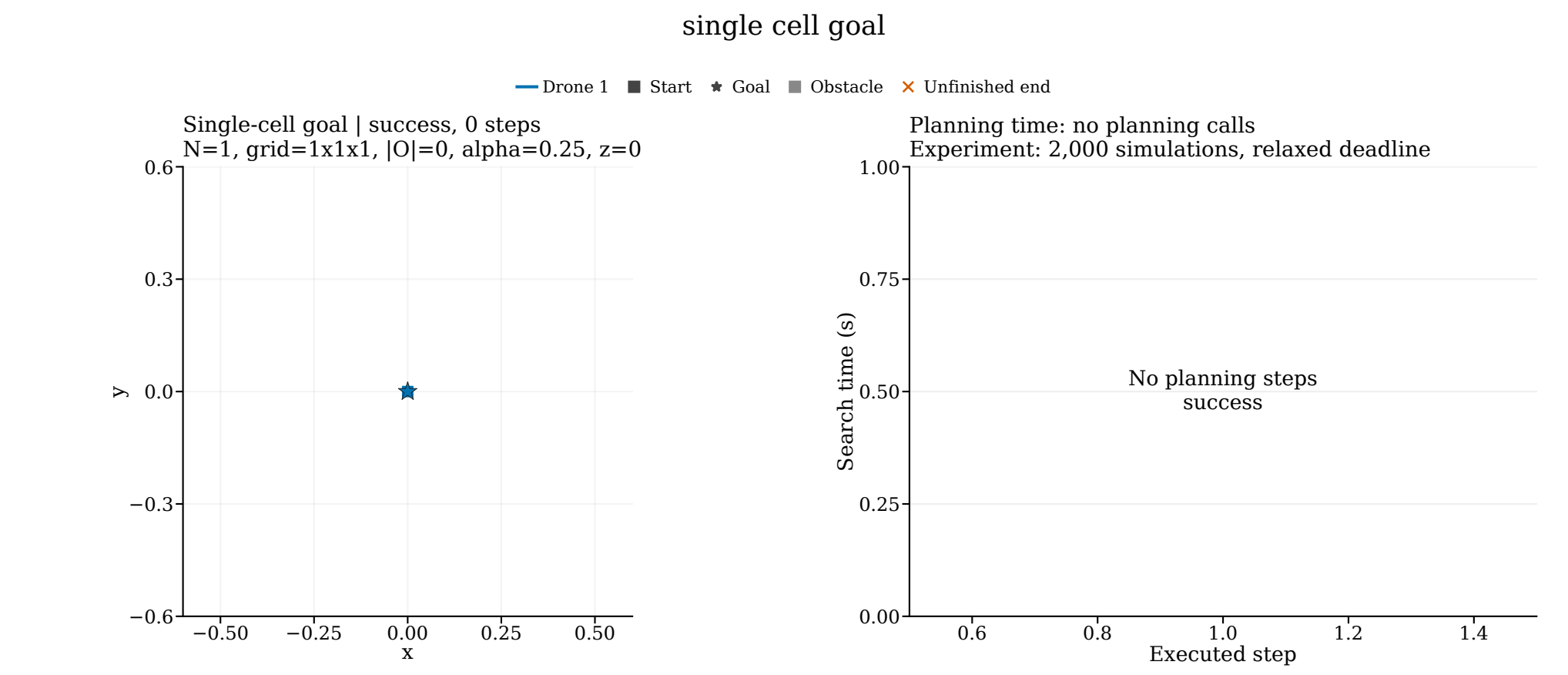


Saved full-solver trace, seed 0: results/demo_four_3d_default.json
Crosses mark returned positions of unfinished drones; collision targets are not logged.

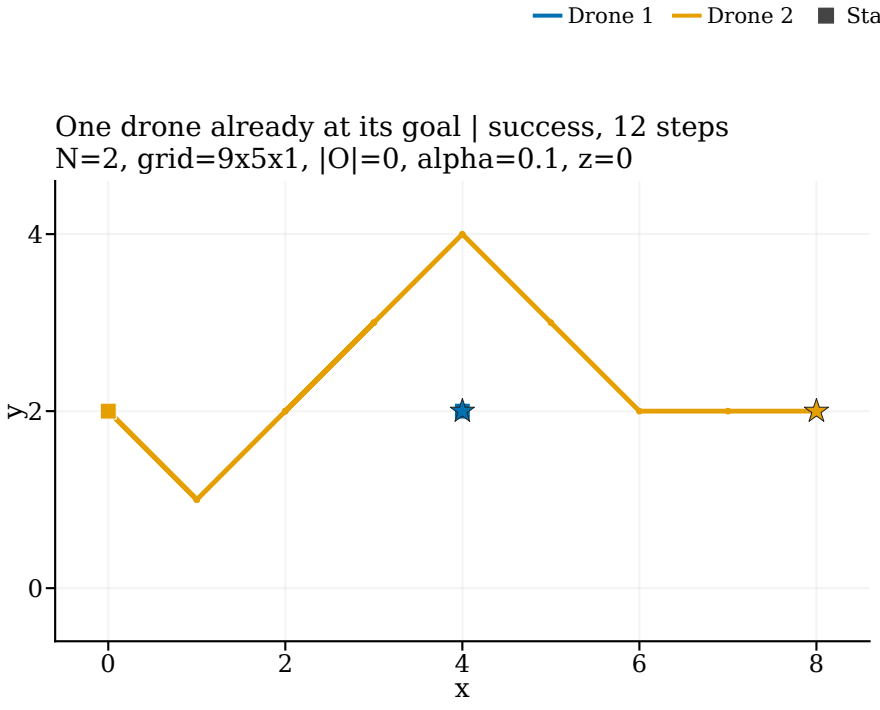
already at goals

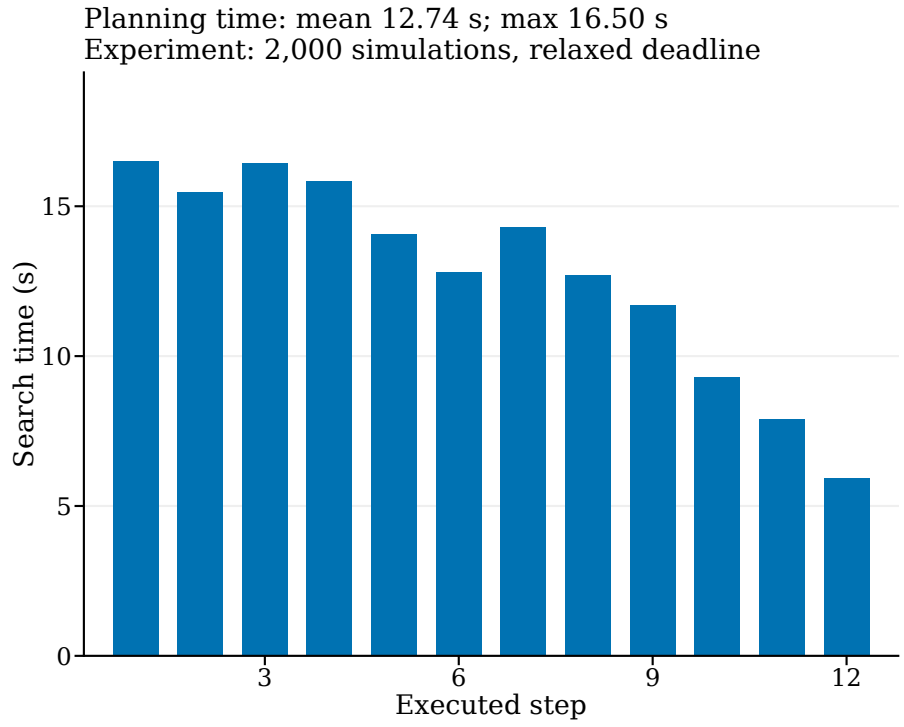


Saved full-solver trace, seed 0: results/trajectories/regression_already_at_goals_full_2000.json
Crosses mark returned positions of unfinished drones; collision targets are not logged.


single cell goal
Drone 1
Start
Goal
Obstacle
Unfinished end
Single-cell goal | success, 0 steps
N=1, grid=1x1x1, |O|=0, alpha=0.25, z=0
y
x
Planning time: no planning calls
Experiment: 2,000 simulations, relaxed deadline
Search time (s)
No planning steps
success
Executed step
Saved full-solver trace, seed 0: results/trajectories/regression_single_cell_goal_full_2000.json
Crosses mark returned positions of unfinished drones; collision targets are not logged.

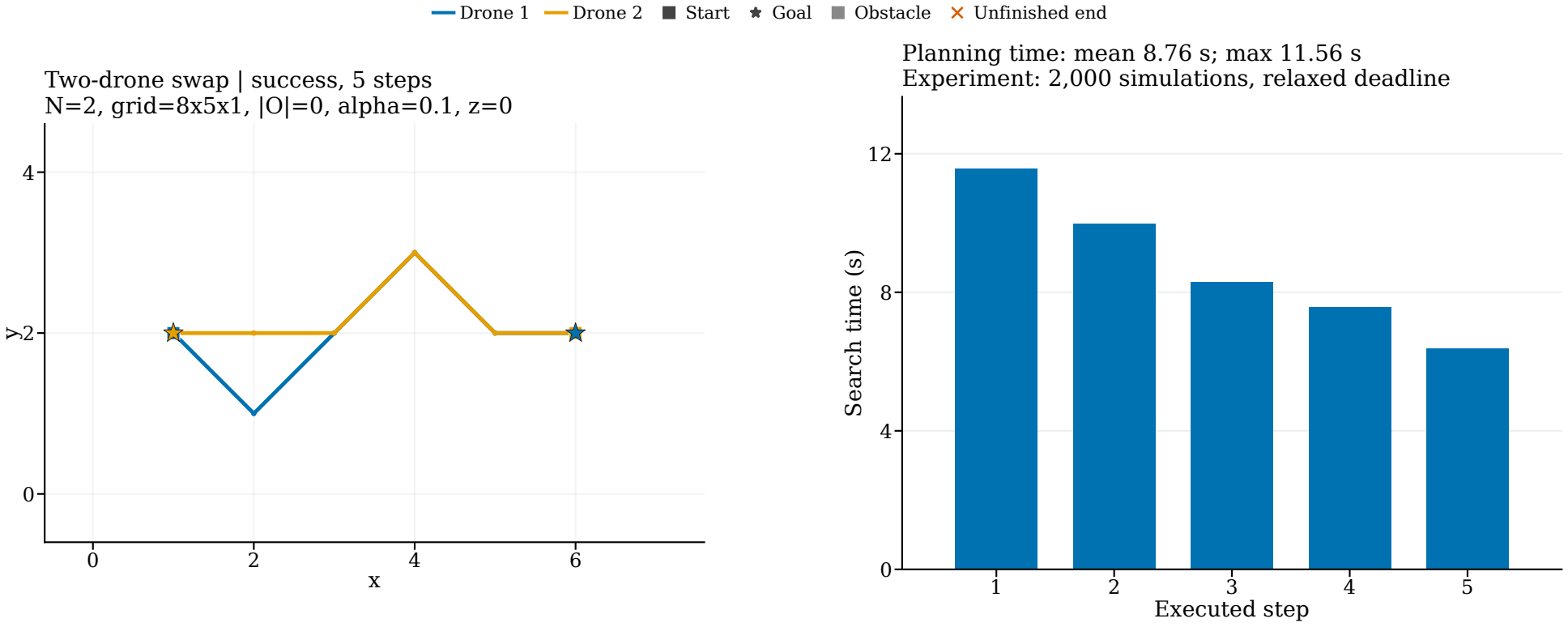
one reached





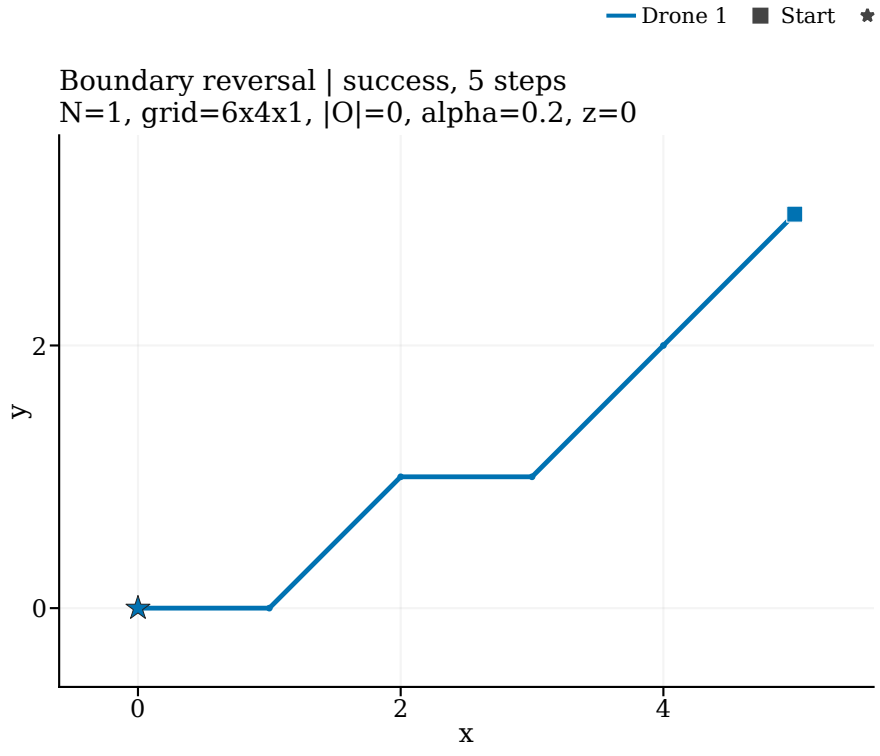
Saved full-solver trace, seed 0: results/trajectories/regression_one_reached_full_2000.json
Crosses mark returned positions of unfinished drones; collision targets are not logged.

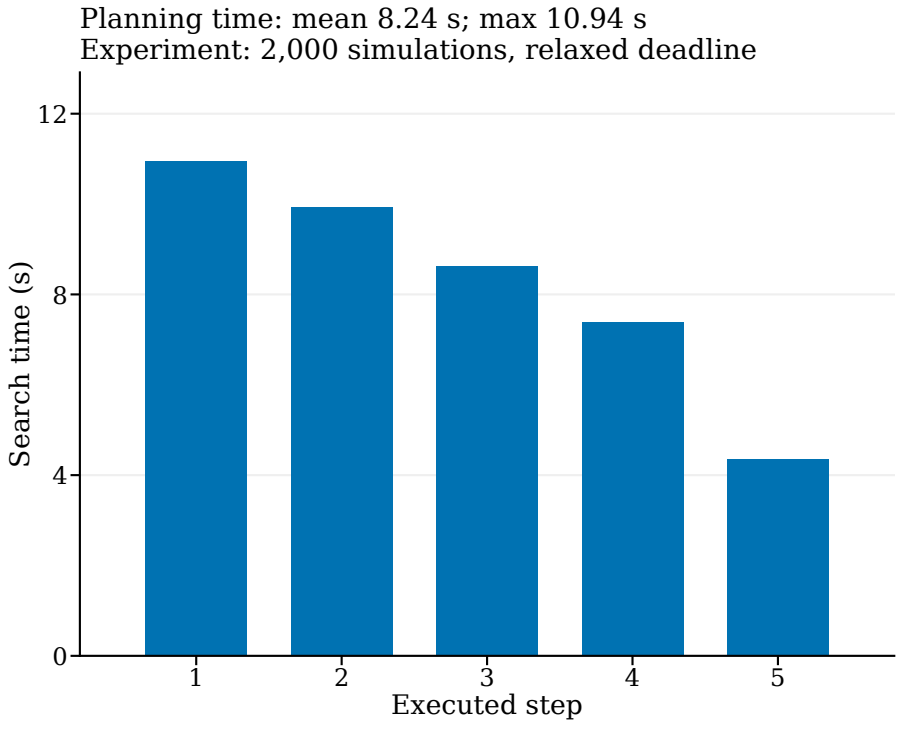
swap 2d



Saved full-solver trace, seed 0: results/trajectories/regression_swap_2d_full_2000.json
Crosses mark returned positions of unfinished drones; collision targets are not logged.

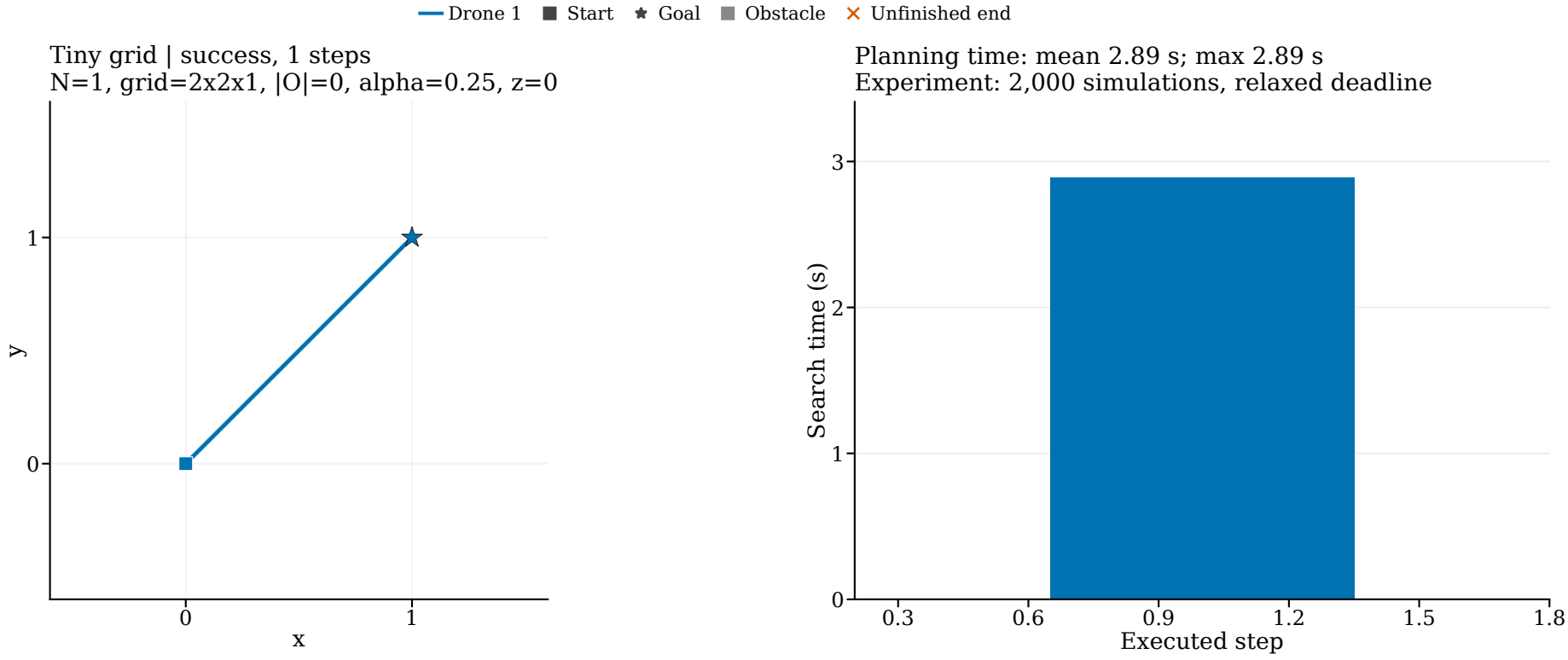
boundary reverse





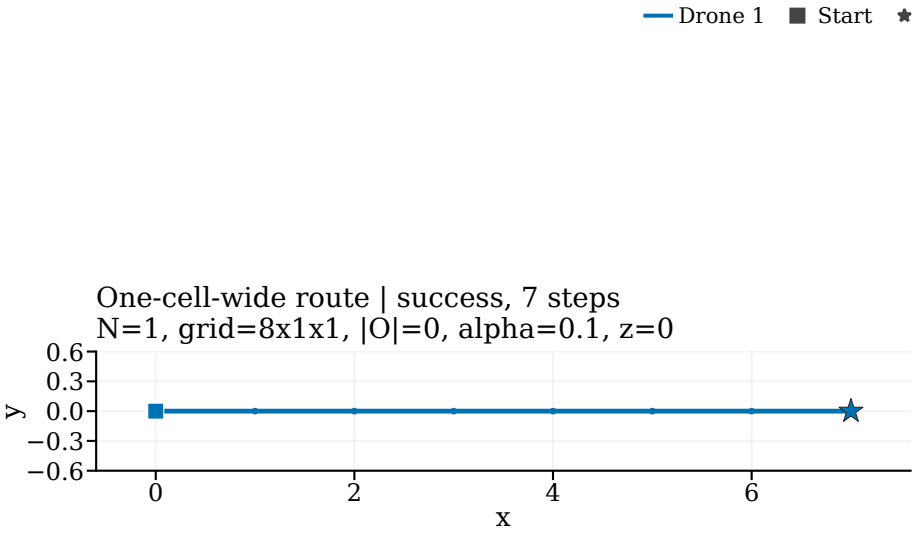
Saved full-solver trace, seed 0: results/trajectories/regression_boundary_reverse_full_2000.json
Crosses mark returned positions of unfinished drones; collision targets are not logged.

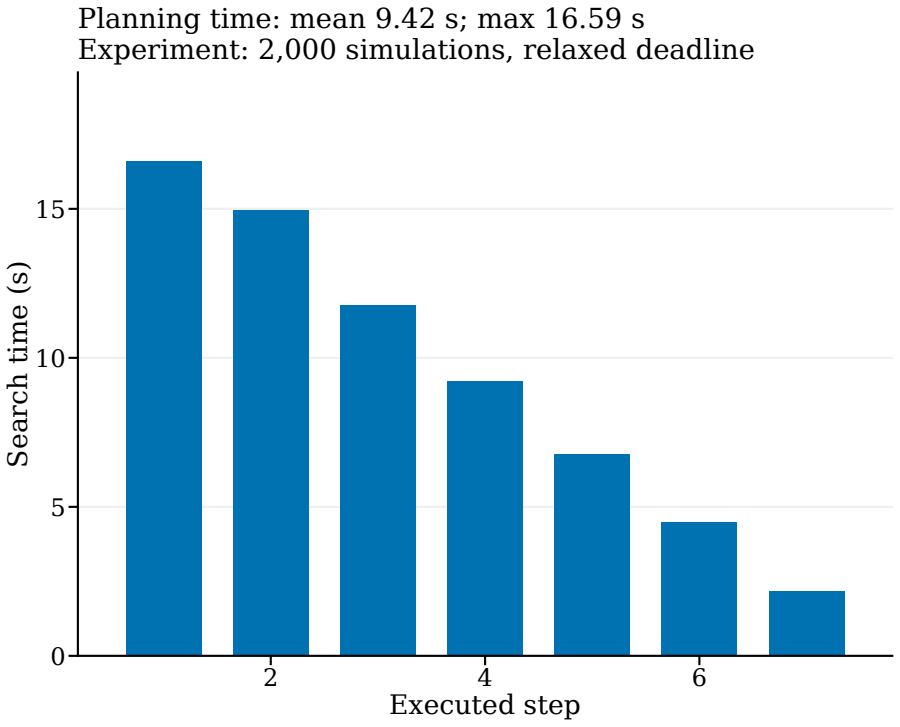
tiny 2d



Saved full-solver trace, seed 0: results/trajectories/regression_tiny_2d_full_2000.json
Crosses mark returned positions of unfinished drones; collision targets are not logged.

thin 1d



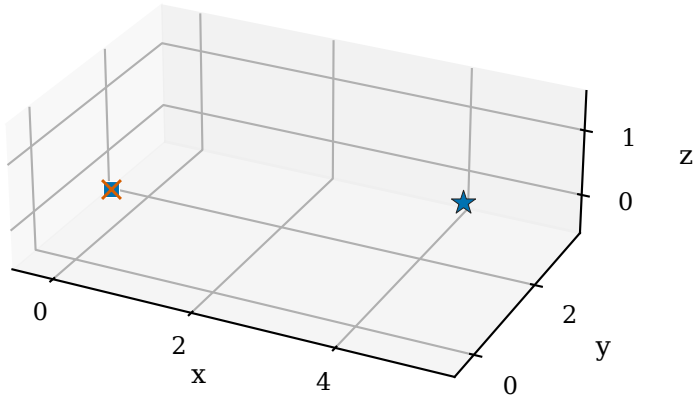


Saved full-solver trace, seed 0: results/trajectories/regression_thin_1d_full_2000.json
Crosses mark returned positions of unfinished drones; collision targets are not logged.

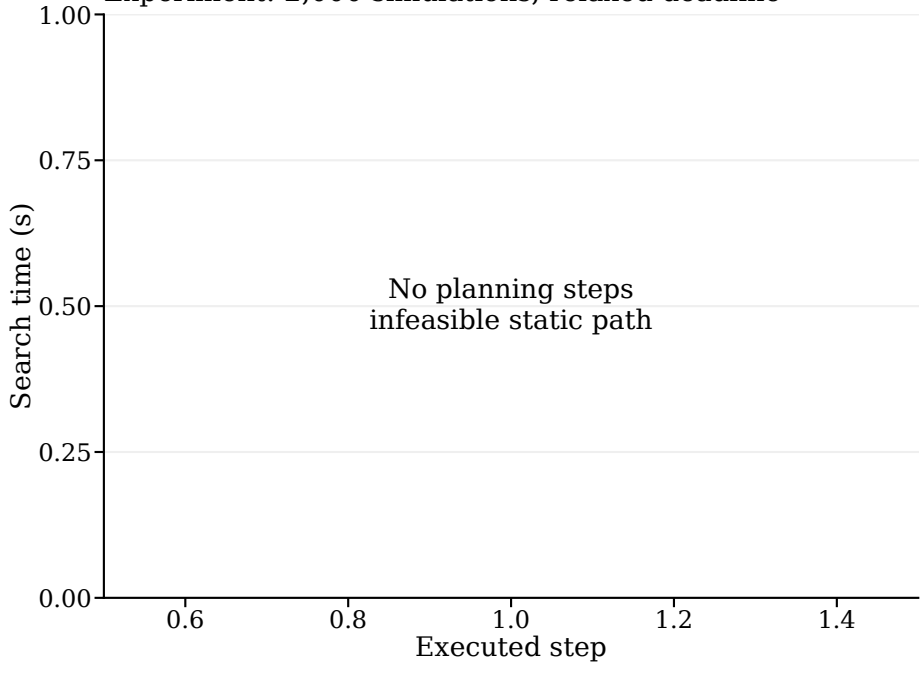
infeasible altitude

Drone 1 Start Goal Obstacle Unfinished end

Unreachable altitude | reject: static path, 0 steps
N=1, grid=6x4x2, |O|=0, alpha=0.25

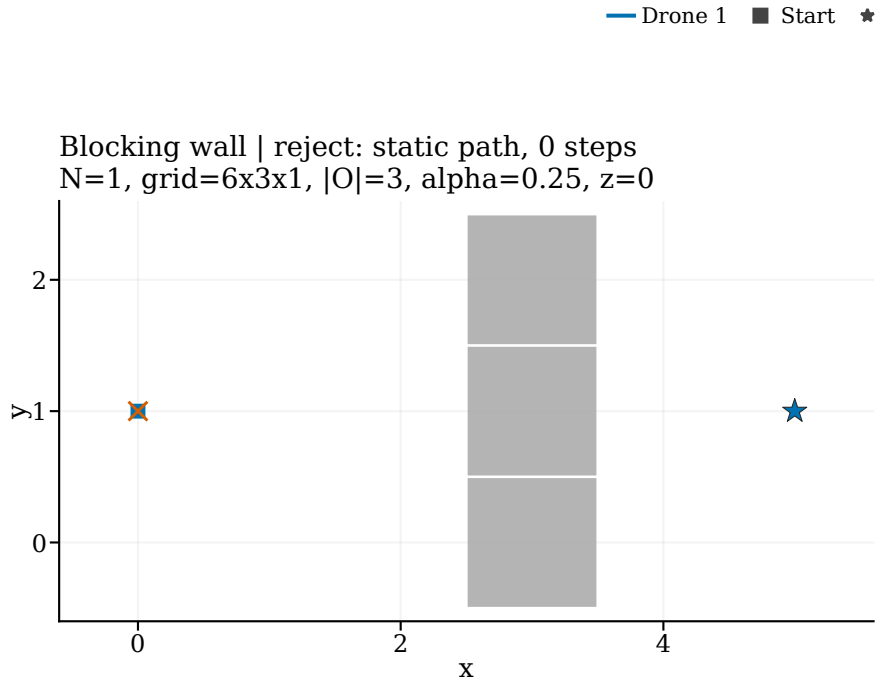


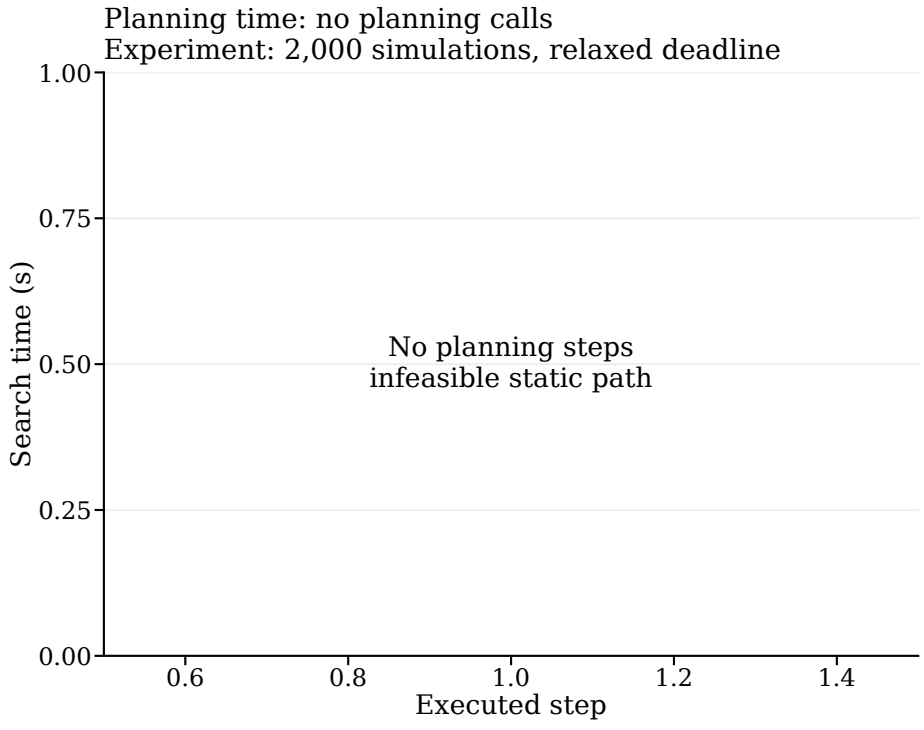
Planning time: no planning calls
Experiment: 2,000 simulations, relaxed deadline



Saved full-solver trace, seed 0: results/trajectories/regression_infeasible_altitude_full_2000.json
Crosses mark returned positions of unfinished drones; collision targets are not logged.

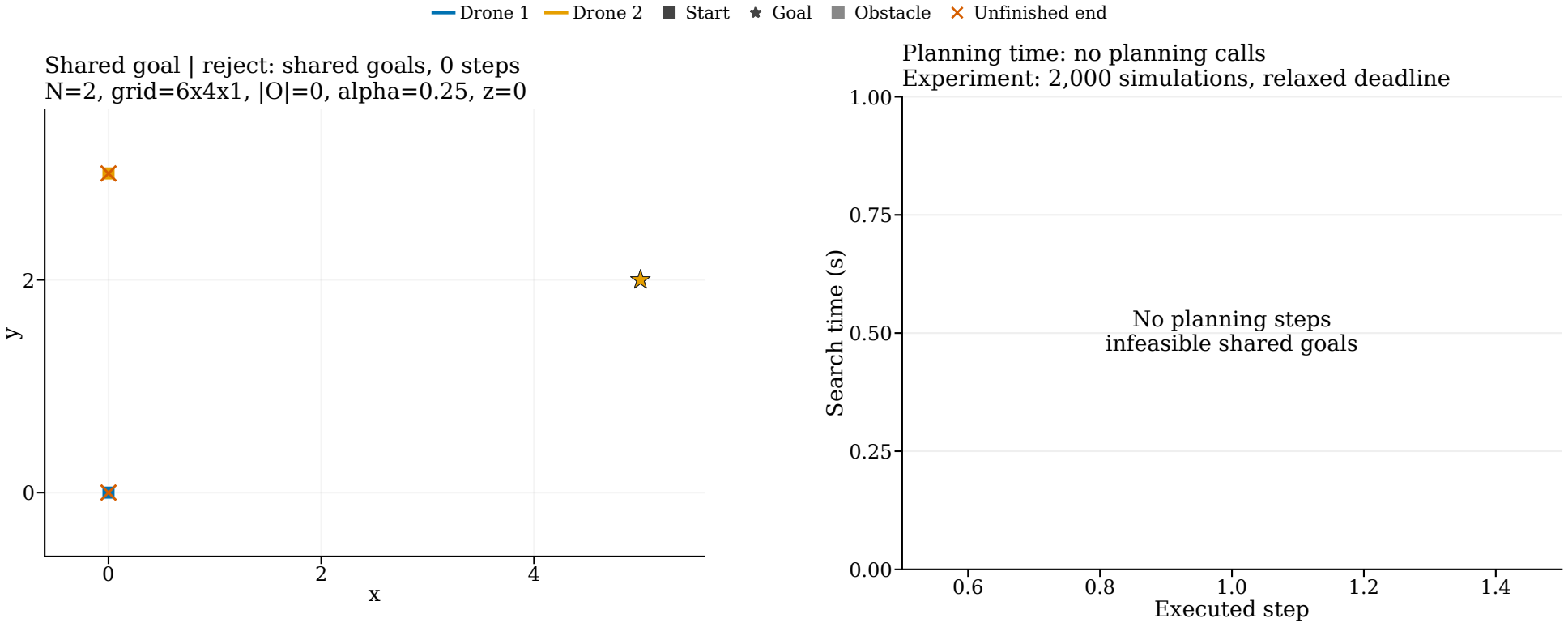
infeasible wall





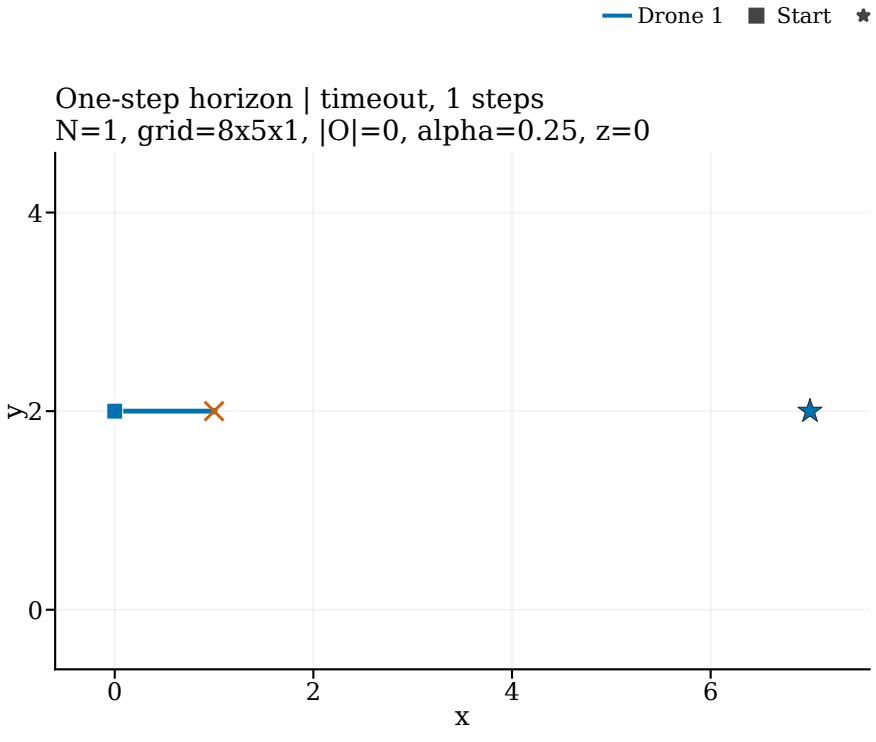
Saved full-solver trace, seed 0: results/trajectories/regression_infeasible_wall_full_2000.json
Crosses mark returned positions of unfinished drones; collision targets are not logged.

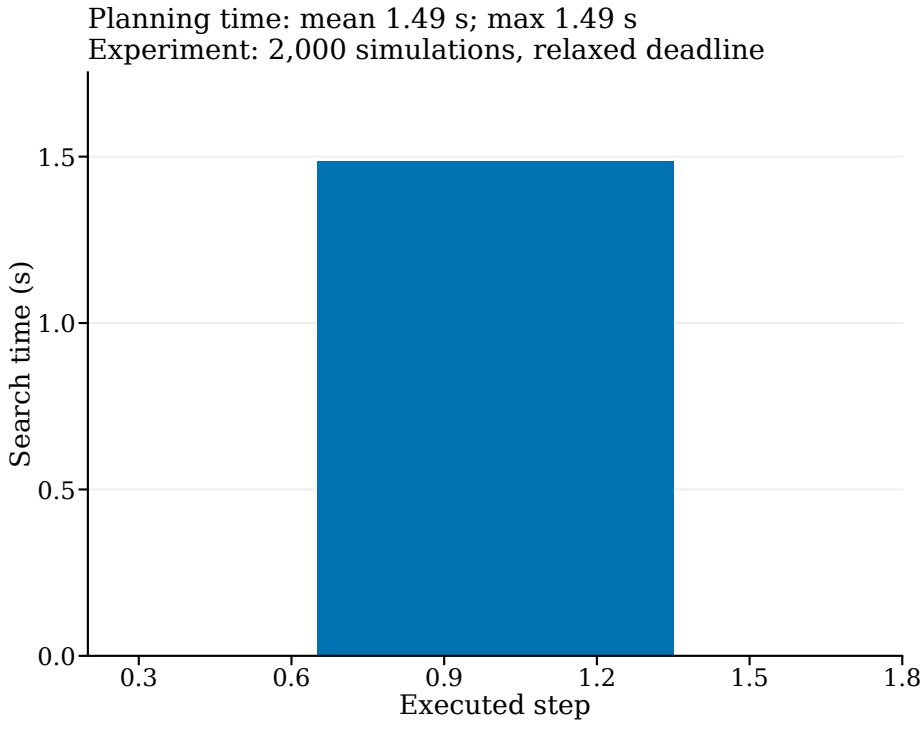
infeasible shared goal



Saved full-solver trace, seed 0: results/trajectories/regression_infeasible_shared_goal_full_2000.json
Crosses mark returned positions of unfinished drones; collision targets are not logged.

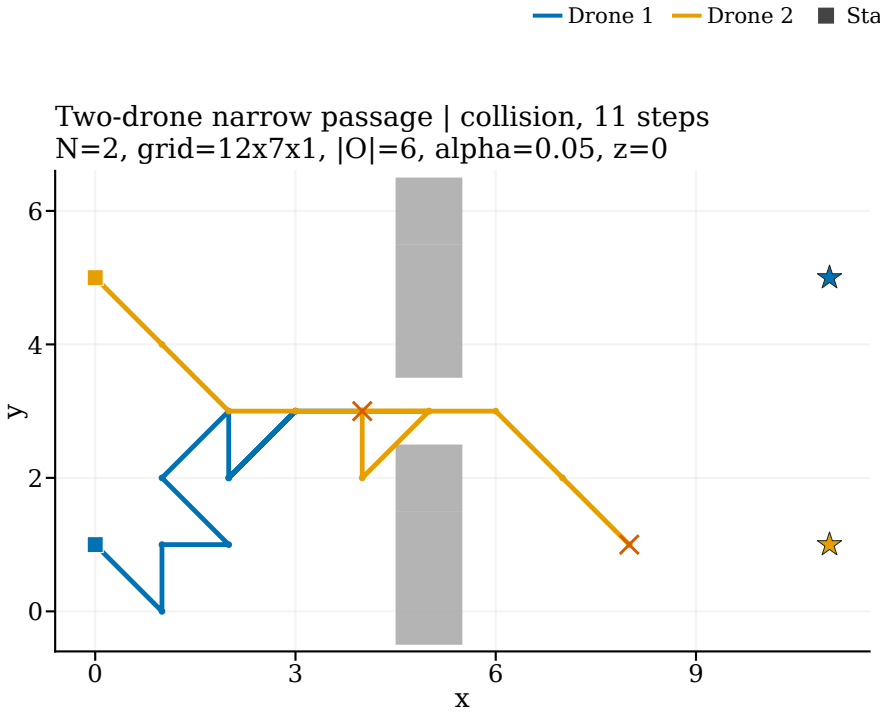
horizon one

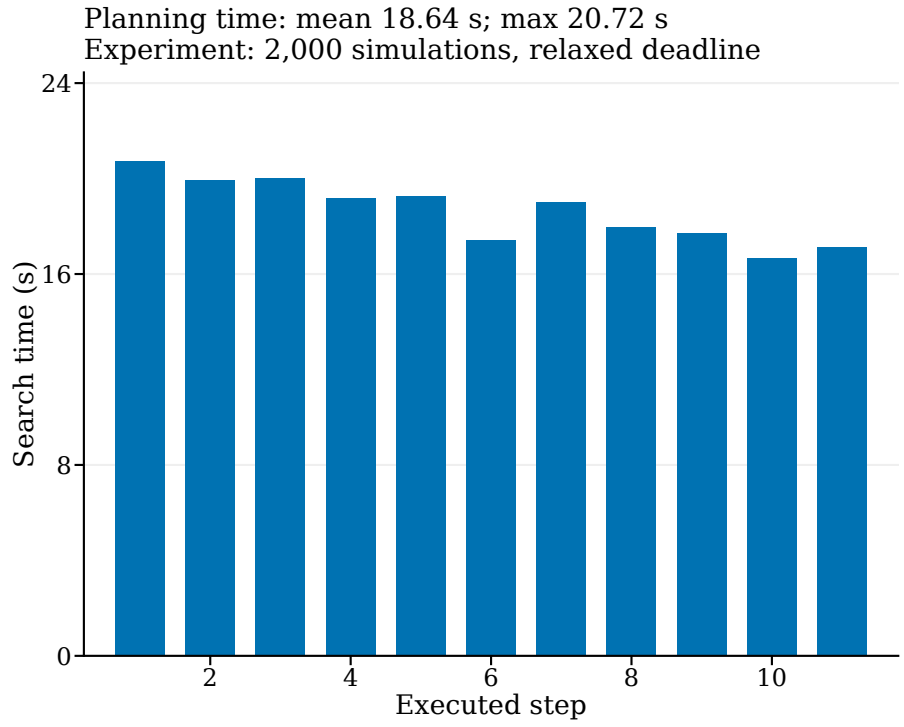




Saved full-solver trace, seed 0: results/trajectories/regression_horizon_one_full_2000.json
Crosses mark returned positions of unfinished drones; collision targets are not logged.

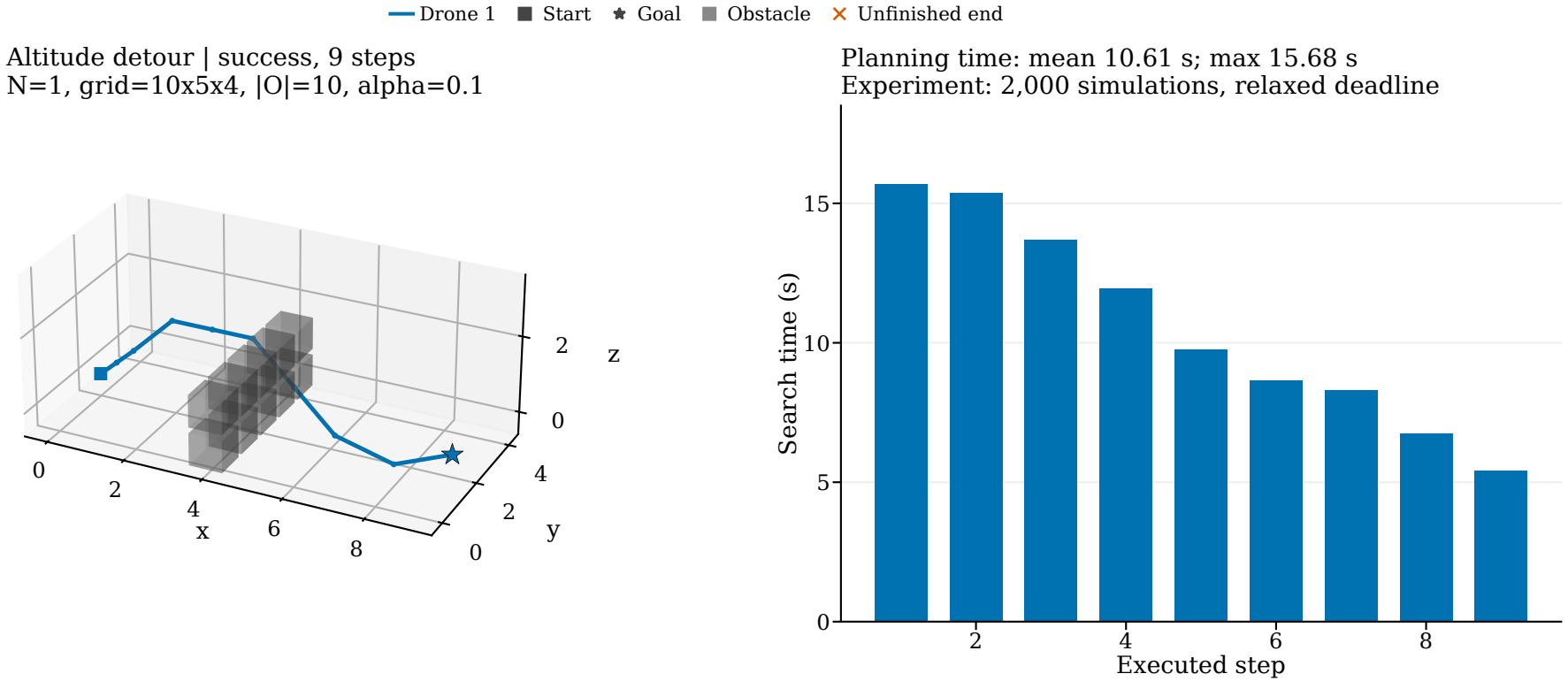
narrow passage





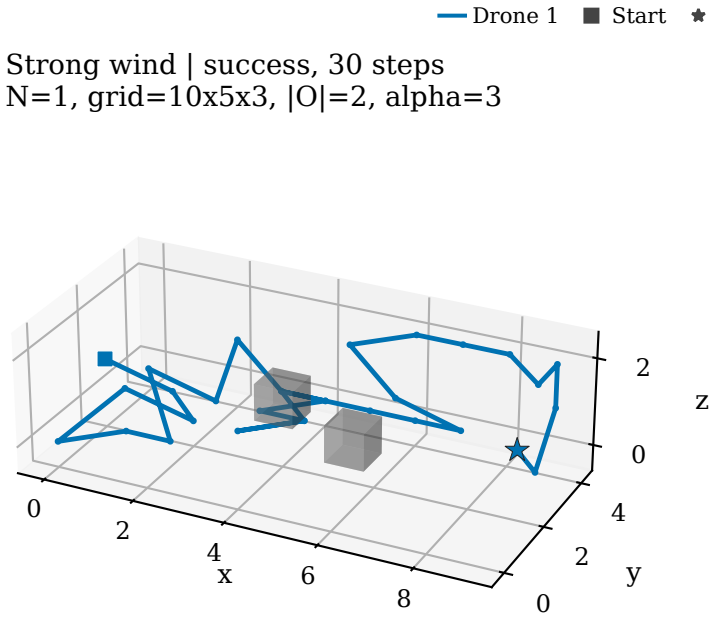
Saved full-solver trace, seed 0: results/trajectories/regression_narrow_passage_full_2000.json
Crosses mark returned positions of unfinished drones; collision targets are not logged.

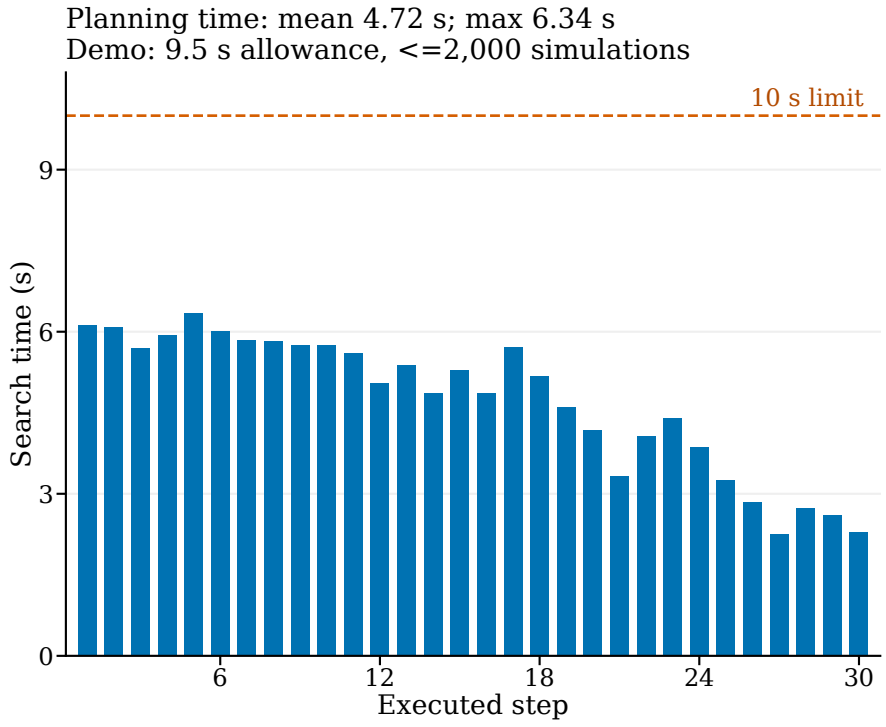
altitude detour



Saved full-solver trace, seed 0: results/trajectories/regression_altitude_detour_full_2000.json
Crosses mark returned positions of unfinished drones; collision targets are not logged.

strong wind



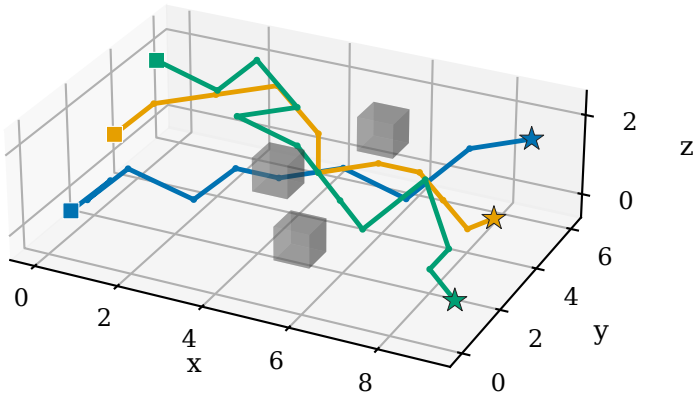


Saved full-solver trace, seed 0: results/strong_wind_default.json
Crosses mark returned positions of unfinished drones; collision targets are not logged.

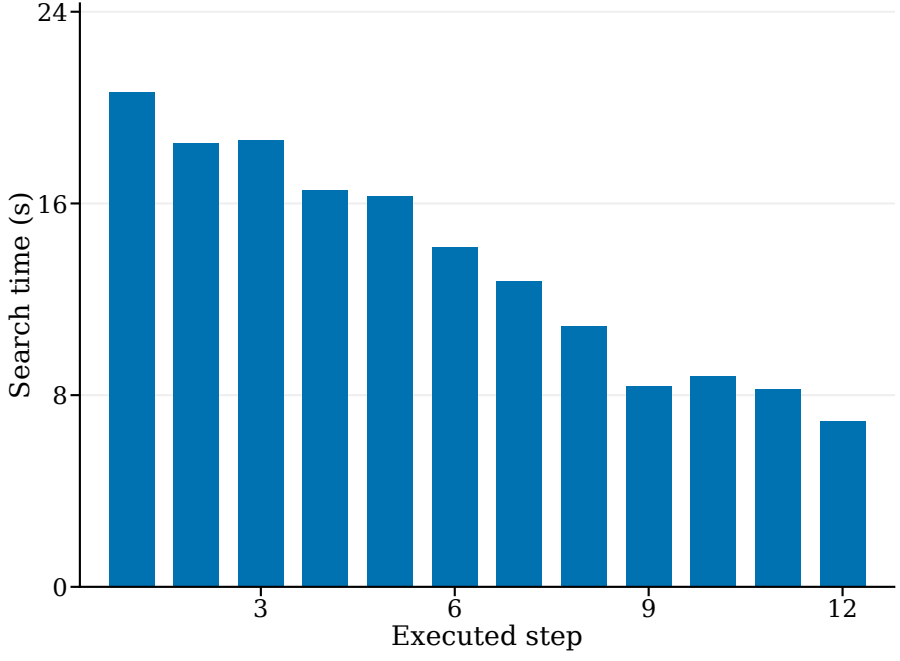
three 3d

Drone 1 Drone 2 Drone 3 Start Goal Obstacle Unfinished end

Three drones in 3D | success, 12 steps
N=3, grid=10x7x3, |O|=3, alpha=0.15

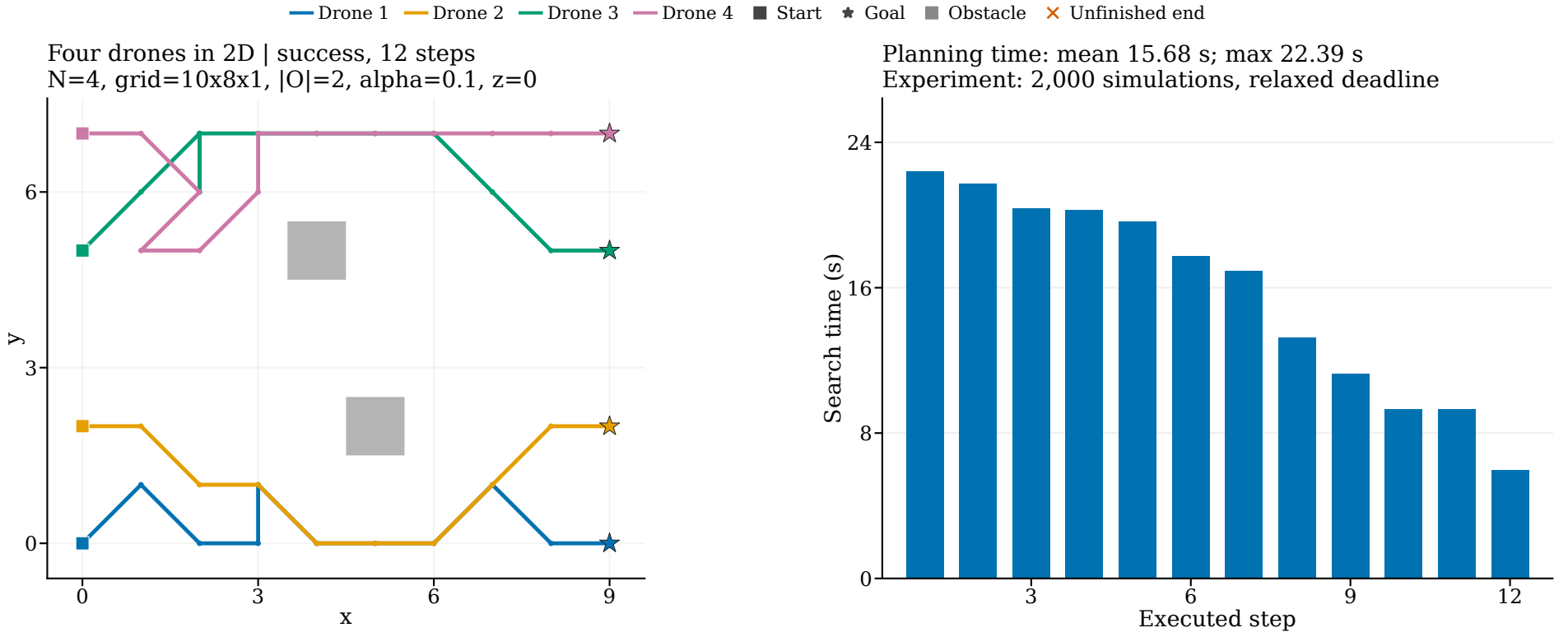


Planning time: mean 13.39 s; max 20.63 s
Experiment: 2,000 simulations, relaxed deadline



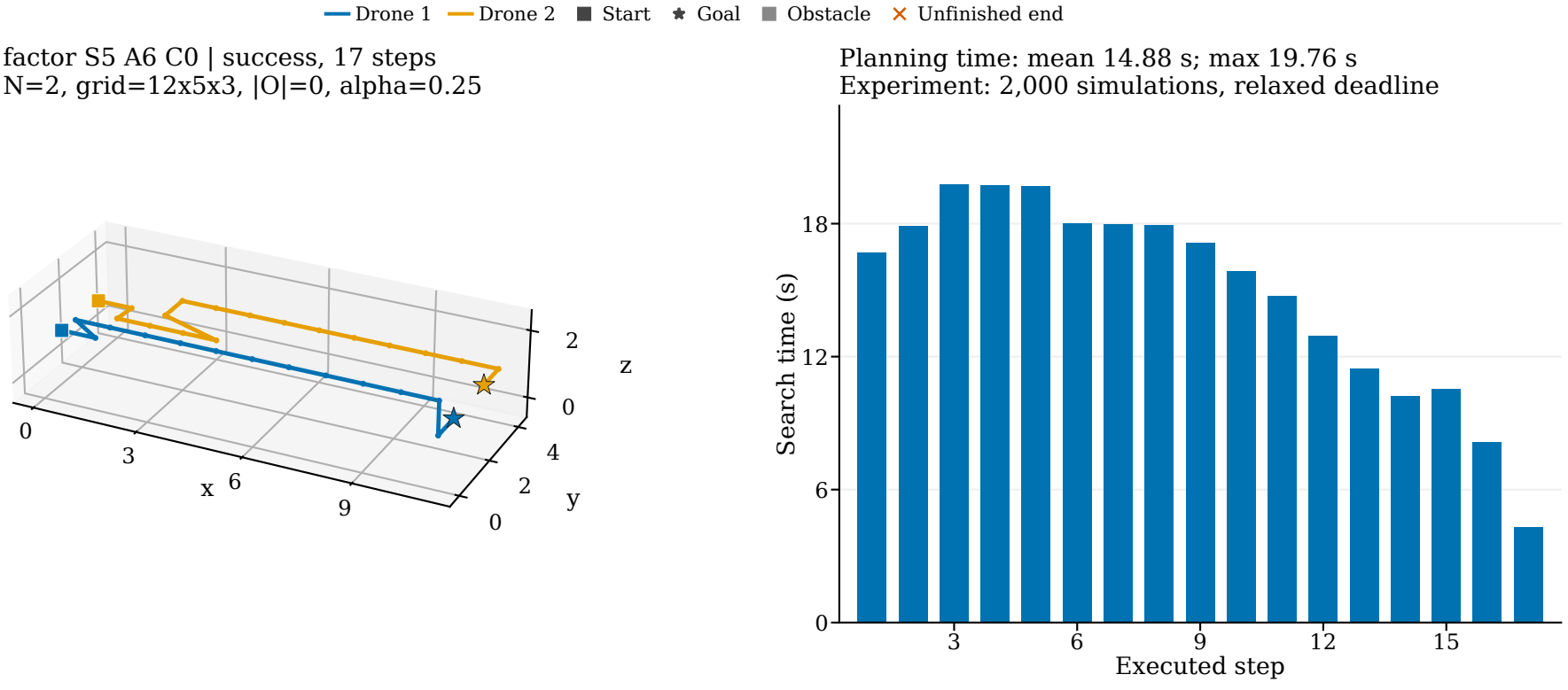
Saved full-solver trace, seed 0: results/trajectories/regression_three_3d_full_2000.json
Crosses mark returned positions of unfinished drones; collision targets are not logged.

four 2d



Saved full-solver trace, seed 0: results/trajectories/regression_four_2d_full_2000.json
Crosses mark returned positions of unfinished drones; collision targets are not logged.

factor S5 A6 C0

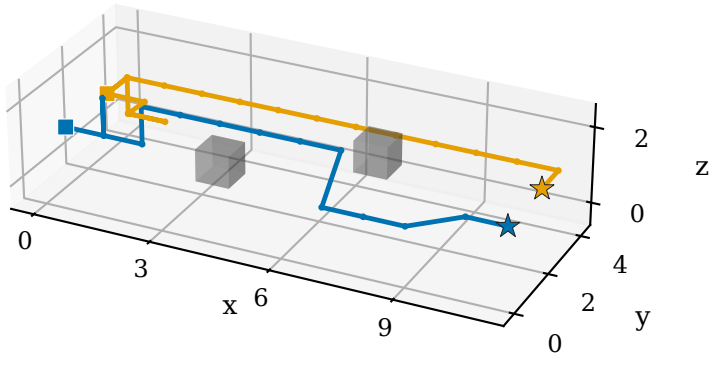


Saved full-solver trace, seed 0: results/trajectories/factorial_factor_S5_A6_C0_full_2000.json
Crosses mark returned positions of unfinished drones; collision targets are not logged.

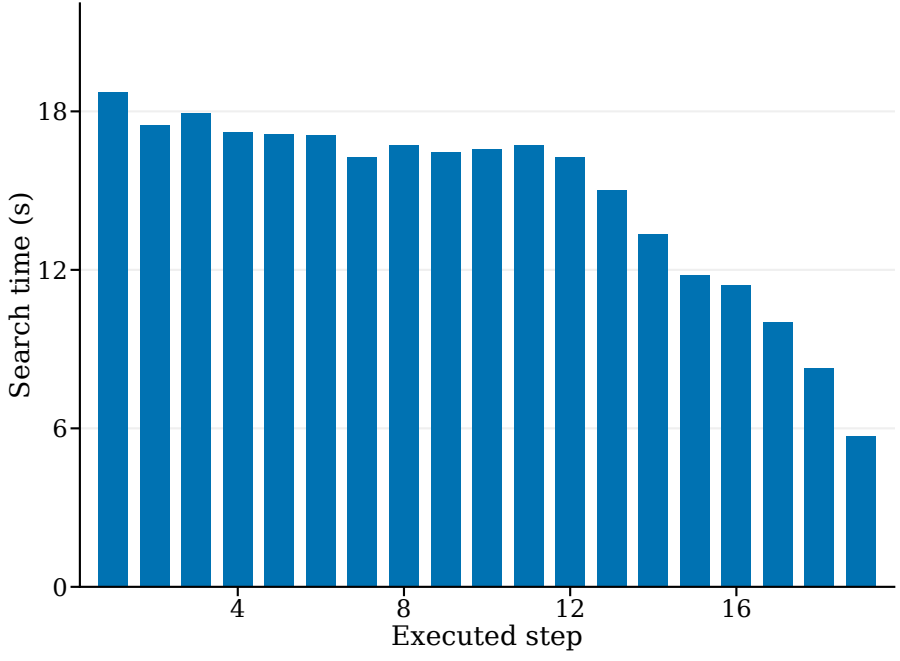
factor S5 A6 C2

Drone 1 Drone 2 Start Goal Obstacle Unfinished end

factor S5 A6 C2 | success, 19 steps
N=2, grid=12x5x3, |O|=2, alpha=0.25

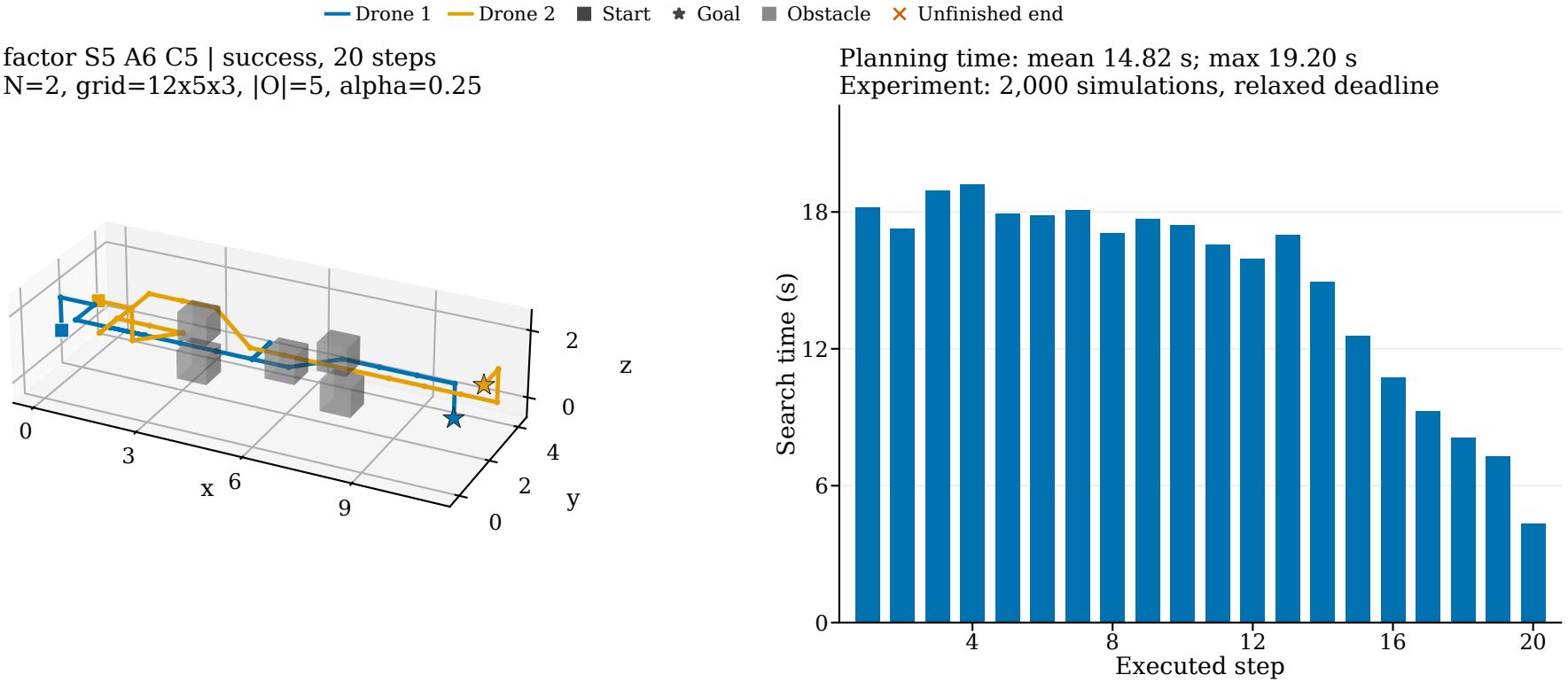


Planning time: mean 14.73 s; max 18.72 s
Experiment: 2,000 simulations, relaxed deadline



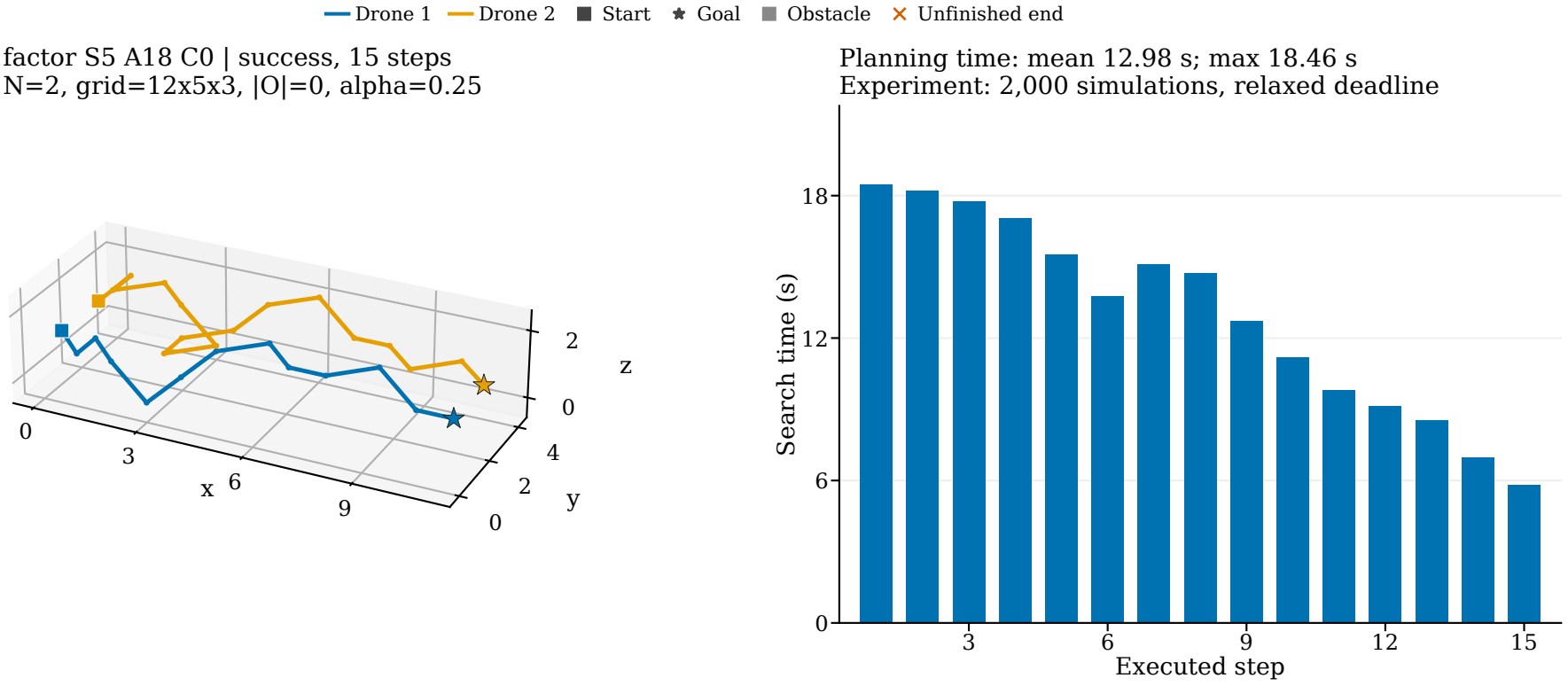
Saved full-solver trace, seed 0: results/trajectories/factorial_factor_S5_A6_C2_full_2000.json
Crosses mark returned positions of unfinished drones; collision targets are not logged.

factor S5 A6 C5



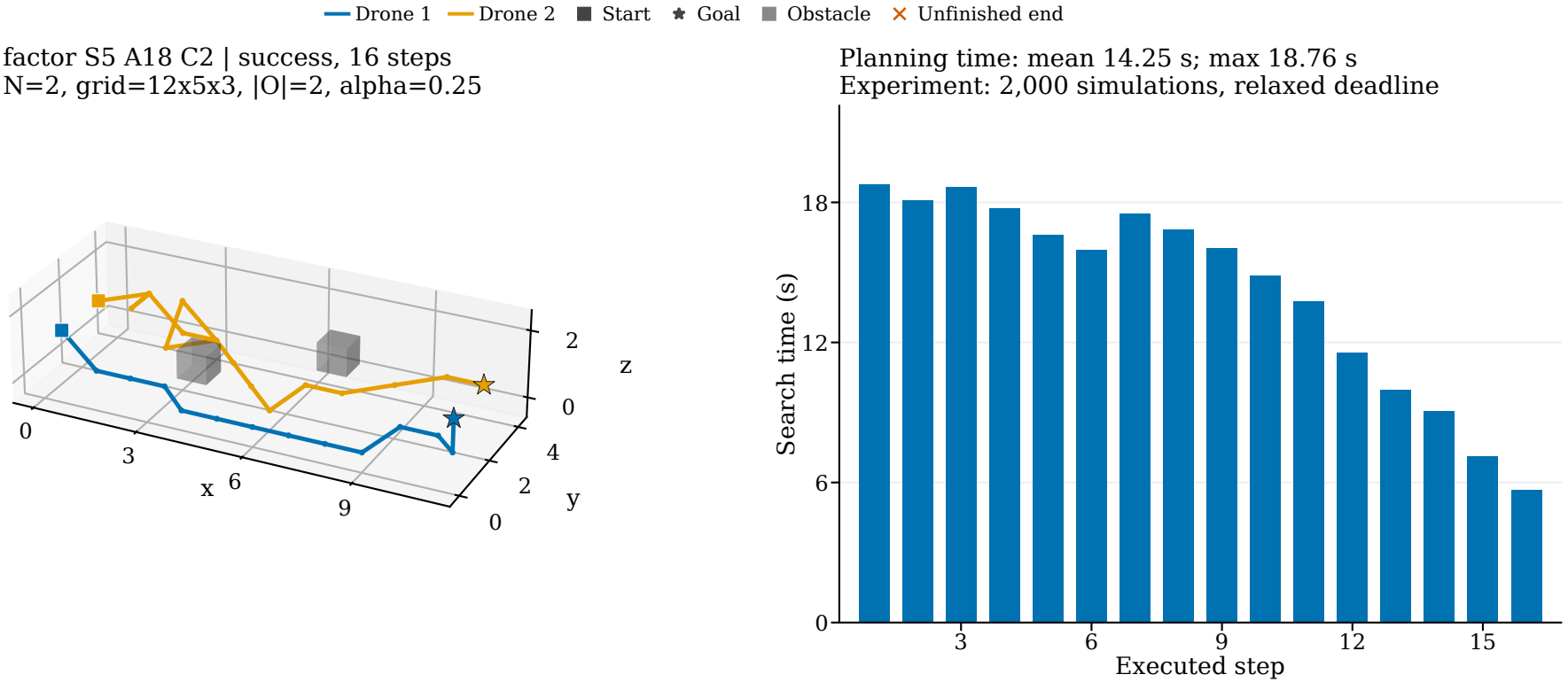
Saved full-solver trace, seed 0: results/trajectories/factorial_factor_S5_A6_C5_full_2000.json
Crosses mark returned positions of unfinished drones; collision targets are not logged.

factor S5 A18 C0



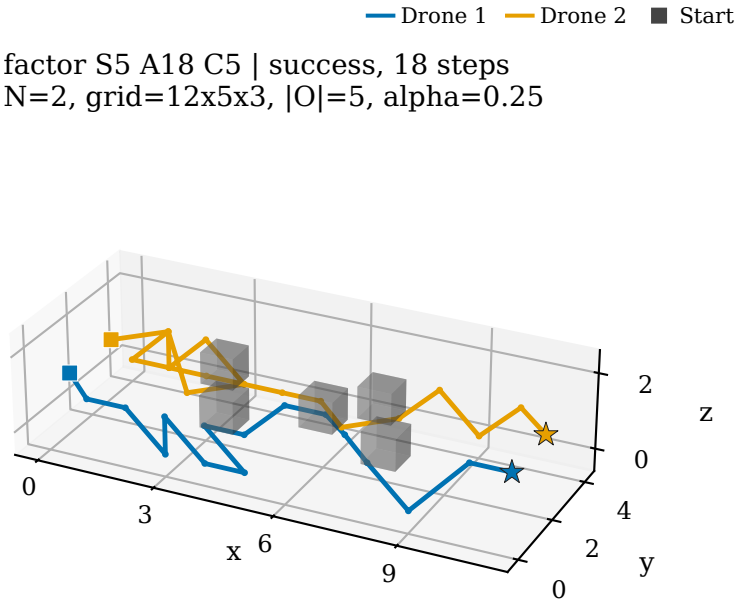
Saved full-solver trace, seed 0: results/trajectories/factorial_factor_S5_A18_C0_full_2000.json
Crosses mark returned positions of unfinished drones; collision targets are not logged.

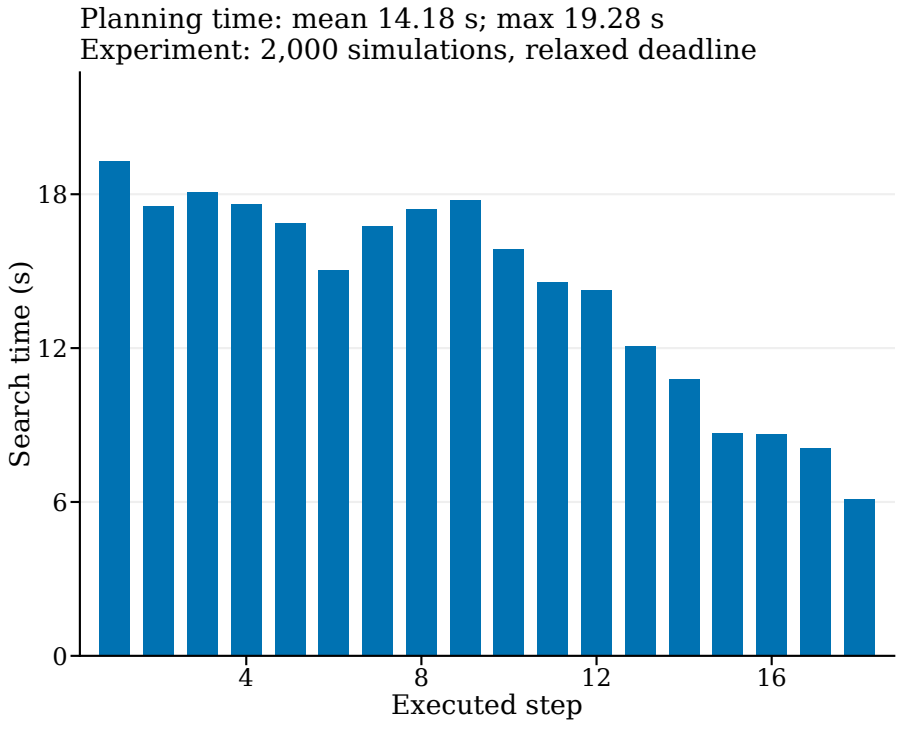
factor S5 A18 C2



Saved full-solver trace, seed 0: results/trajectories/factorial_factor_S5_A18_C2_full_2000.json
Crosses mark returned positions of unfinished drones; collision targets are not logged.

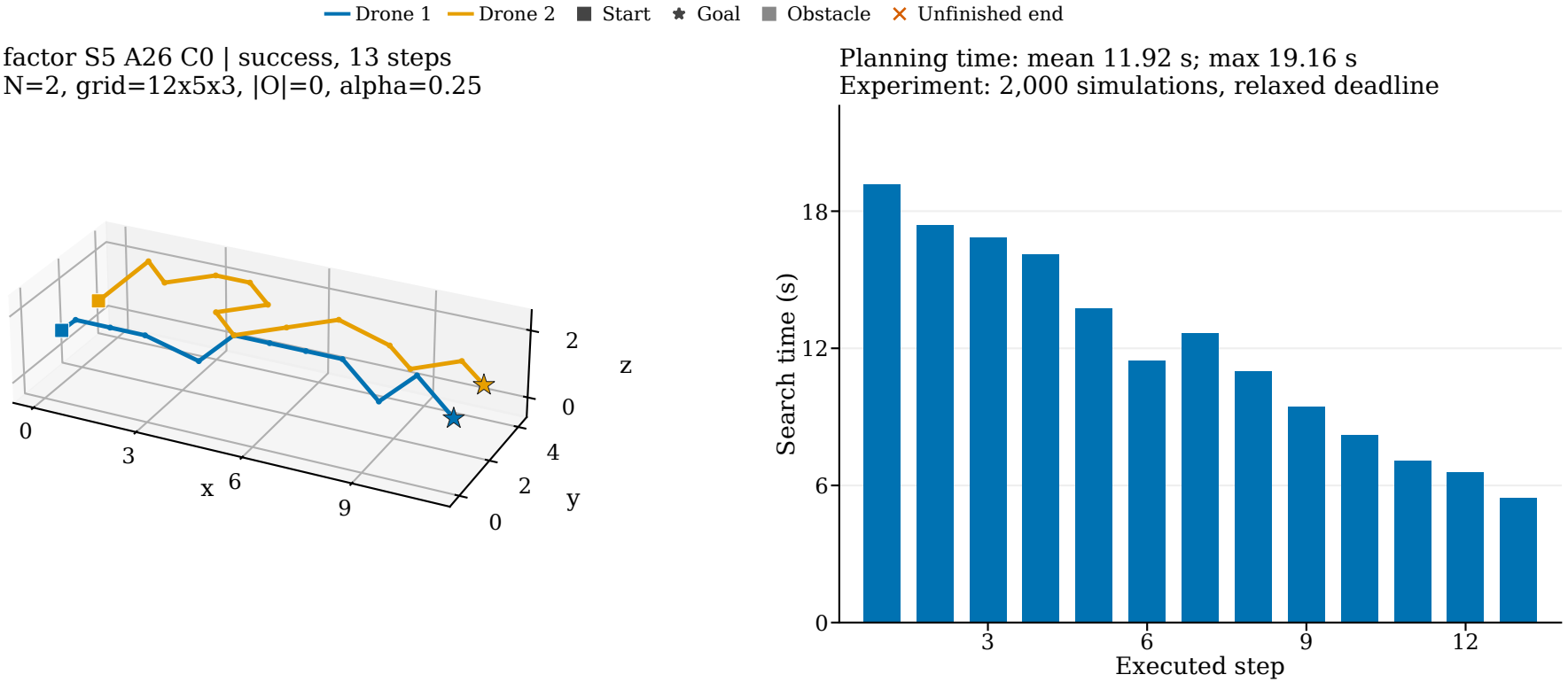
factor S5 A18 C5





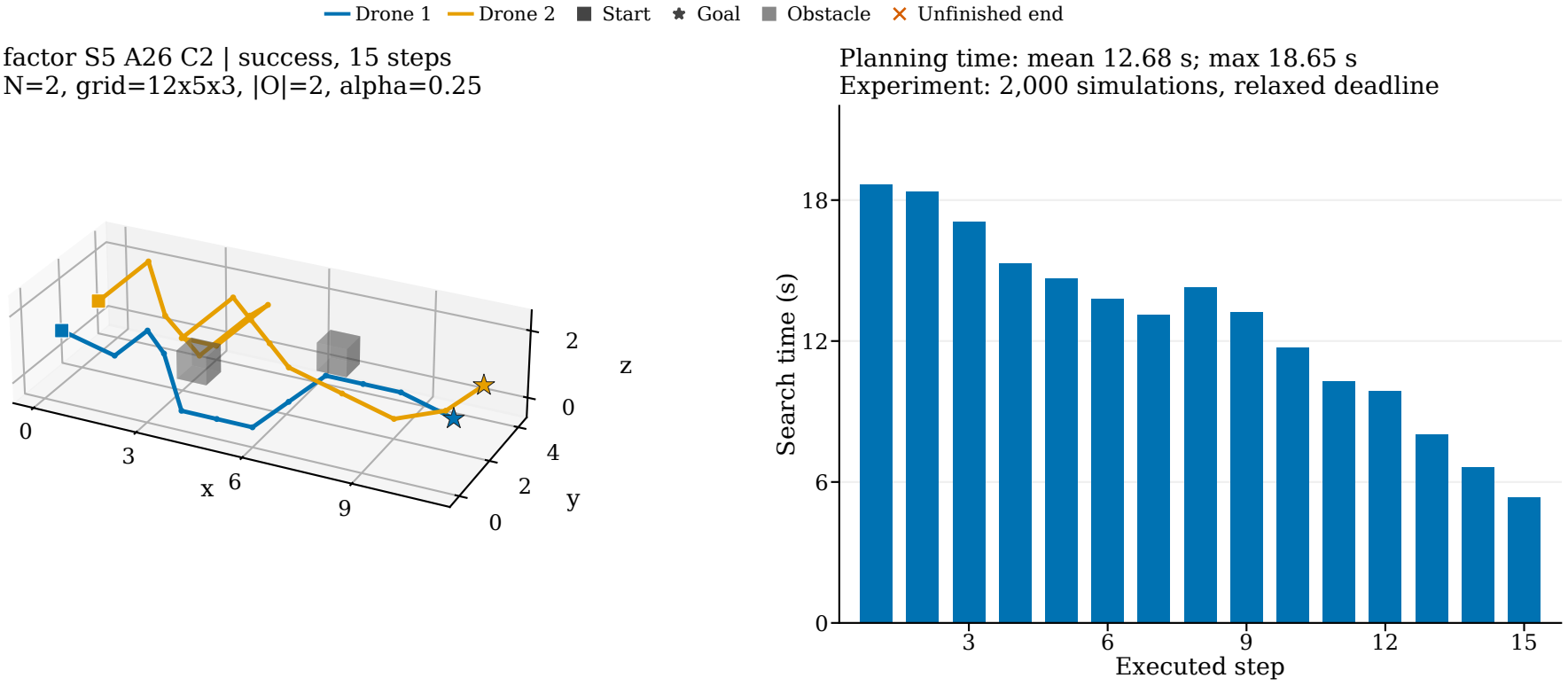
Saved full-solver trace, seed 0: results/trajectories/factorial_factor_S5_A18_C5_full_2000.json
Crosses mark returned positions of unfinished drones; collision targets are not logged.

factor S5 A26 C0



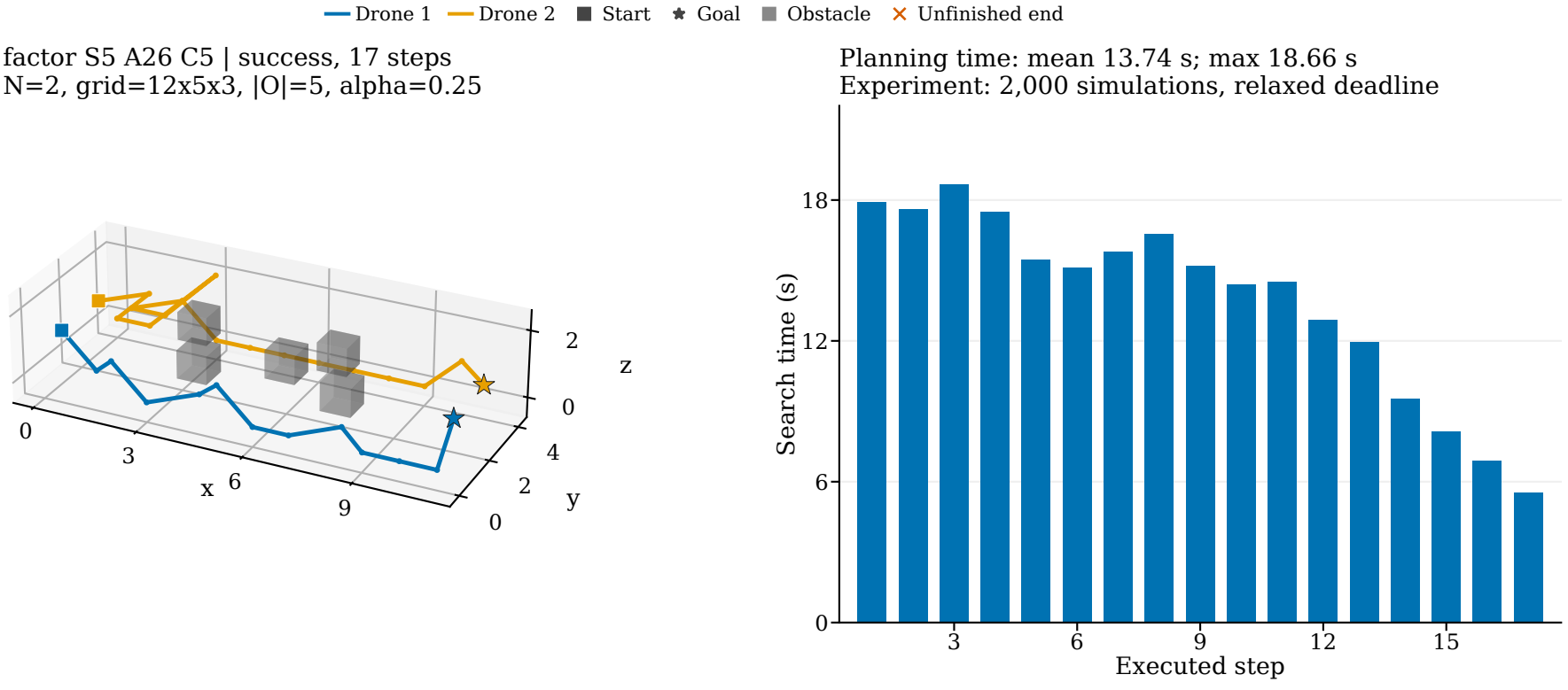
Saved full-solver trace, seed 0: results/trajectories/factorial_factor_S5_A26_C0_full_2000.json
Crosses mark returned positions of unfinished drones; collision targets are not logged.

factor S5 A26 C2



Saved full-solver trace, seed 0: results/trajectories/factorial_factor_S5_A26_C2_full_2000.json
Crosses mark returned positions of unfinished drones; collision targets are not logged.

factor S5 A26 C5

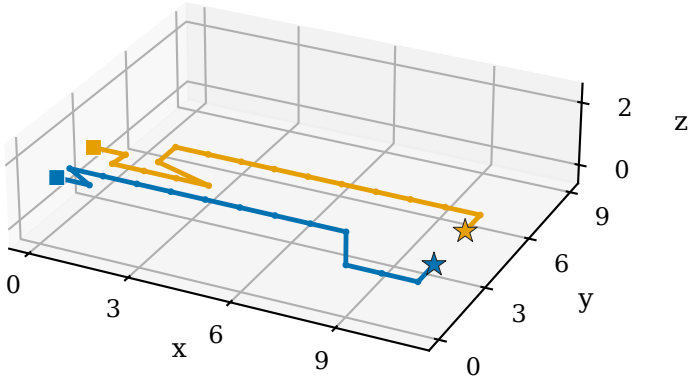


Saved full-solver trace, seed 0: results/trajectories/factorial_factor_S5_A26_C5_full_2000.json
Crosses mark returned positions of unfinished drones; collision targets are not logged.

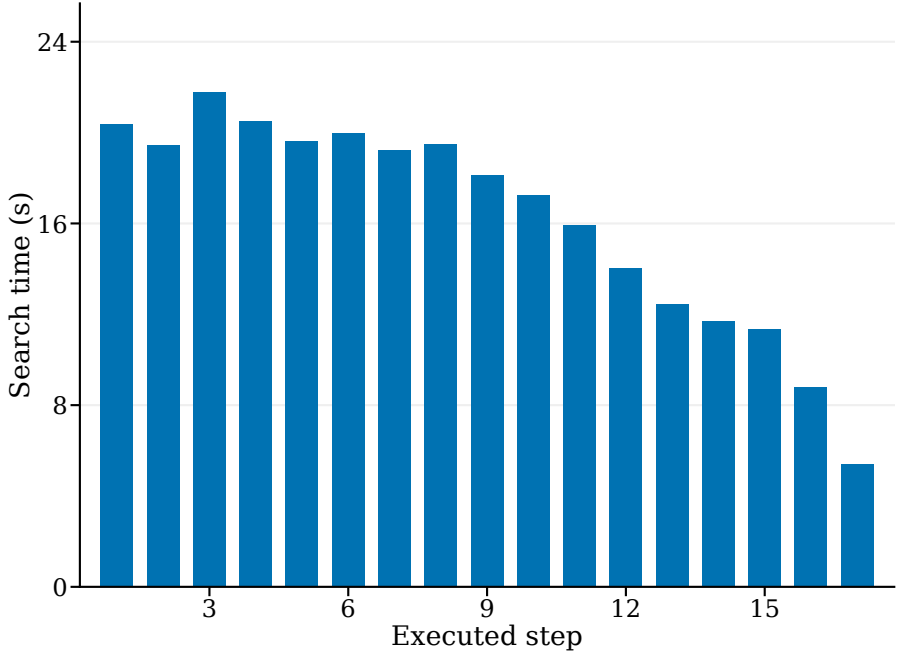
factor S10 A6 C0



factor S10 A6 C0 | success, 17 steps
N=2, grid=12x10x3, |O|=0, alpha=0.25



Planning time: mean 16.19 s; max 21.77 s
Experiment: 2,000 simulations, relaxed deadline

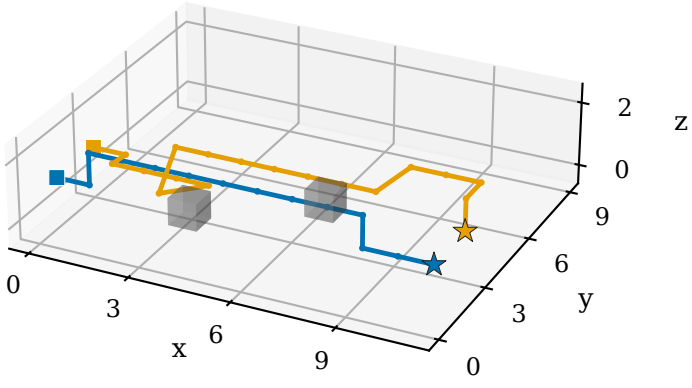


Saved full-solver trace, seed 0: results/trajectories/factorial_factor_S10_A6_C0_full_2000.json
Crosses mark returned positions of unfinished drones; collision targets are not logged.

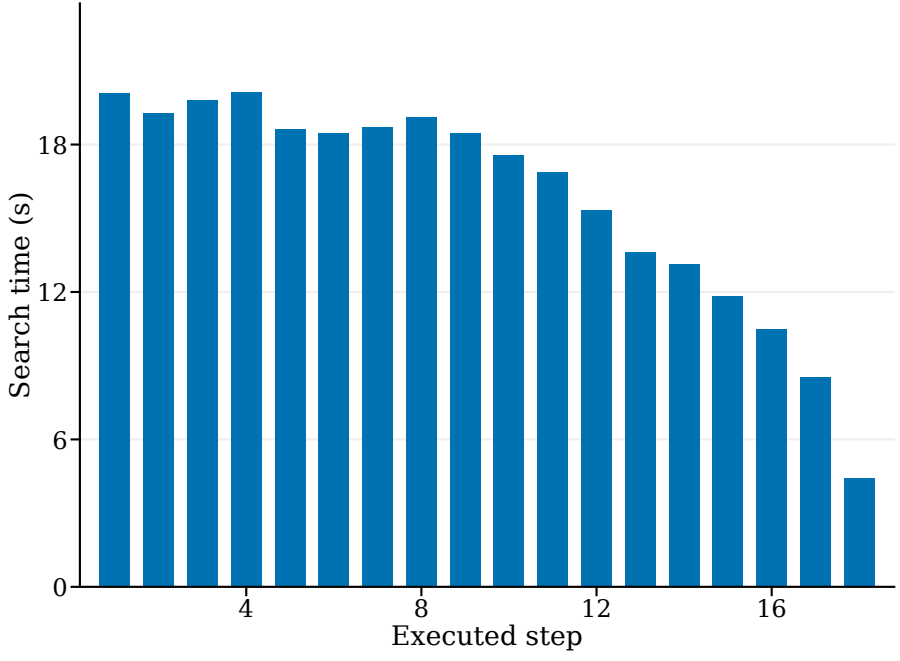
factor S10 A6 C2



factor S10 A6 C2 | success, 18 steps
N=2, grid=12x10x3, |O|=2, alpha=0.25

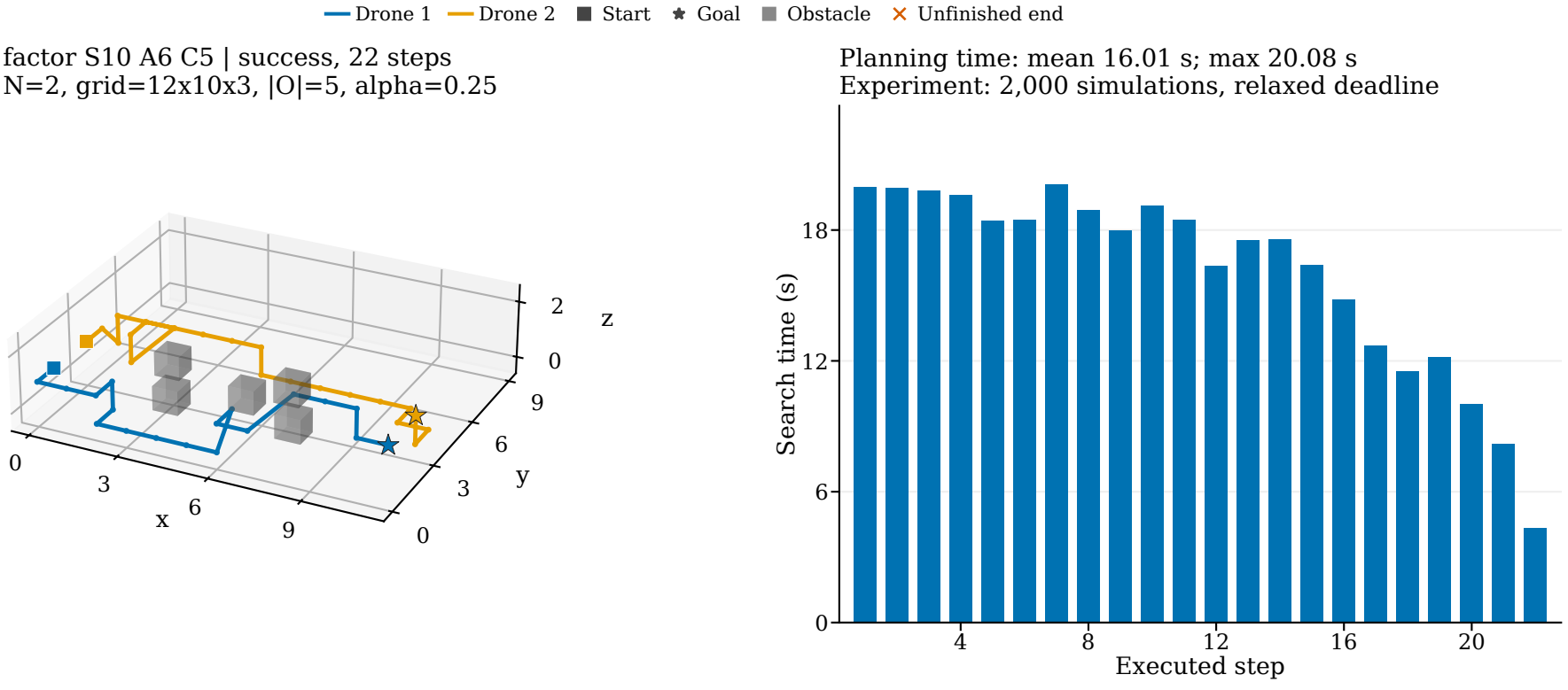


Planning time: mean 15.80 s; max 20.12 s
Experiment: 2,000 simulations, relaxed deadline



Saved full-solver trace, seed 0: results/trajectories/factorial_factor_S10_A6_C2_full_2000.json
Crosses mark returned positions of unfinished drones; collision targets are not logged.

factor S10 A6 C5

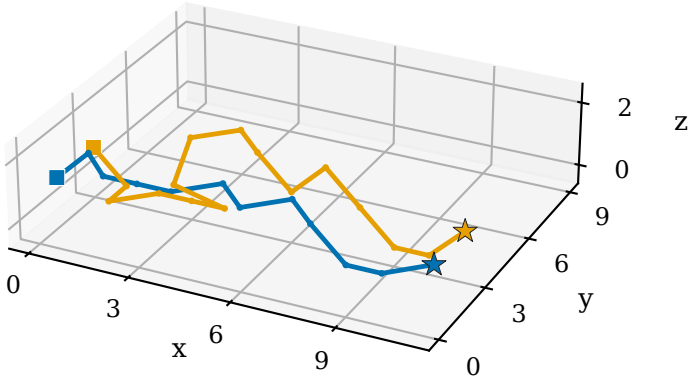


Saved full-solver trace, seed 0: results/trajectories/factorial_factor_S10_A6_C5_full_2000.json
Crosses mark returned positions of unfinished drones; collision targets are not logged.

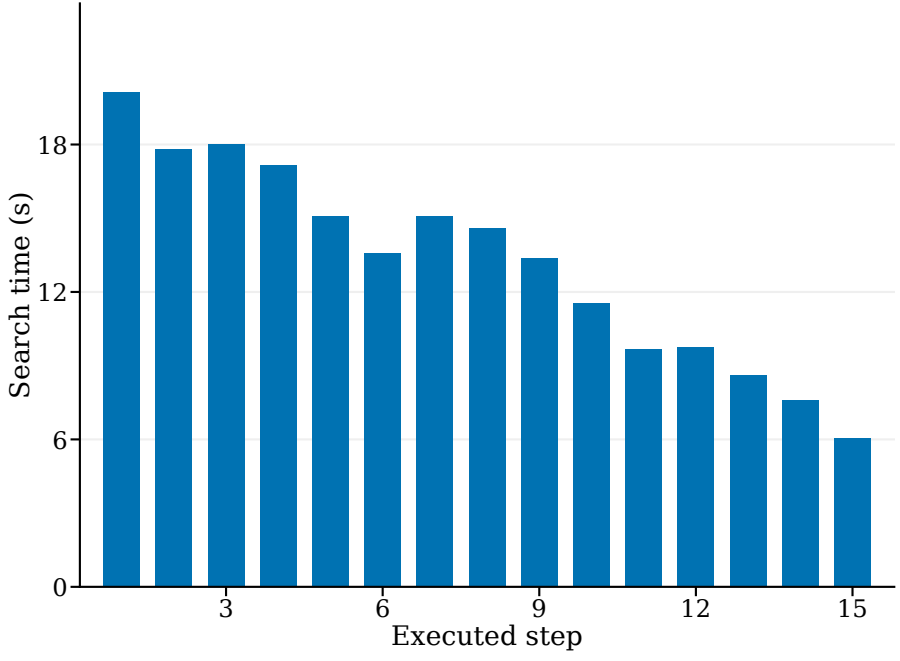
factor S10 A18 C0



factor S10 A18 C0 | success, 15 steps
N=2, grid=12x10x3, |O|=0, alpha=0.25

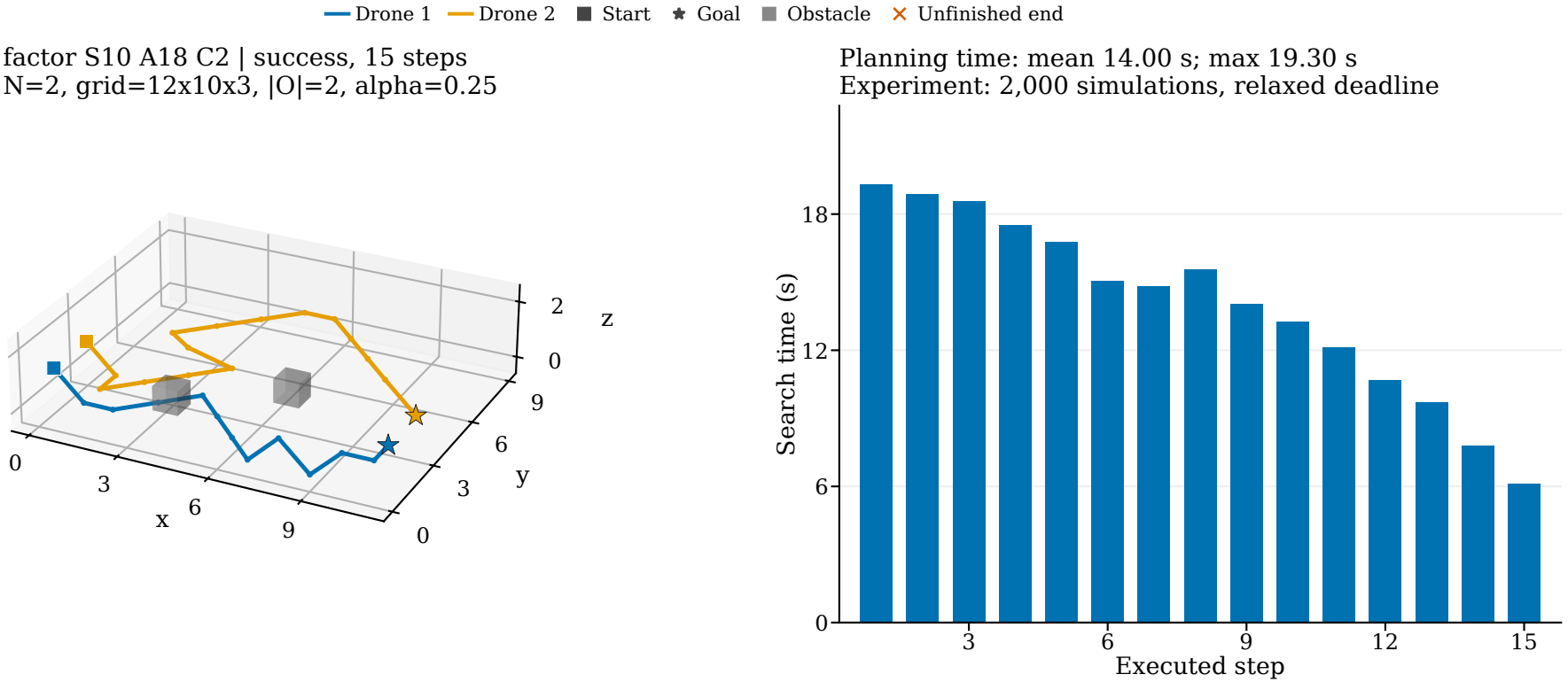


Planning time: mean 13.19 s; max 20.12 s
Experiment: 2,000 simulations, relaxed deadline



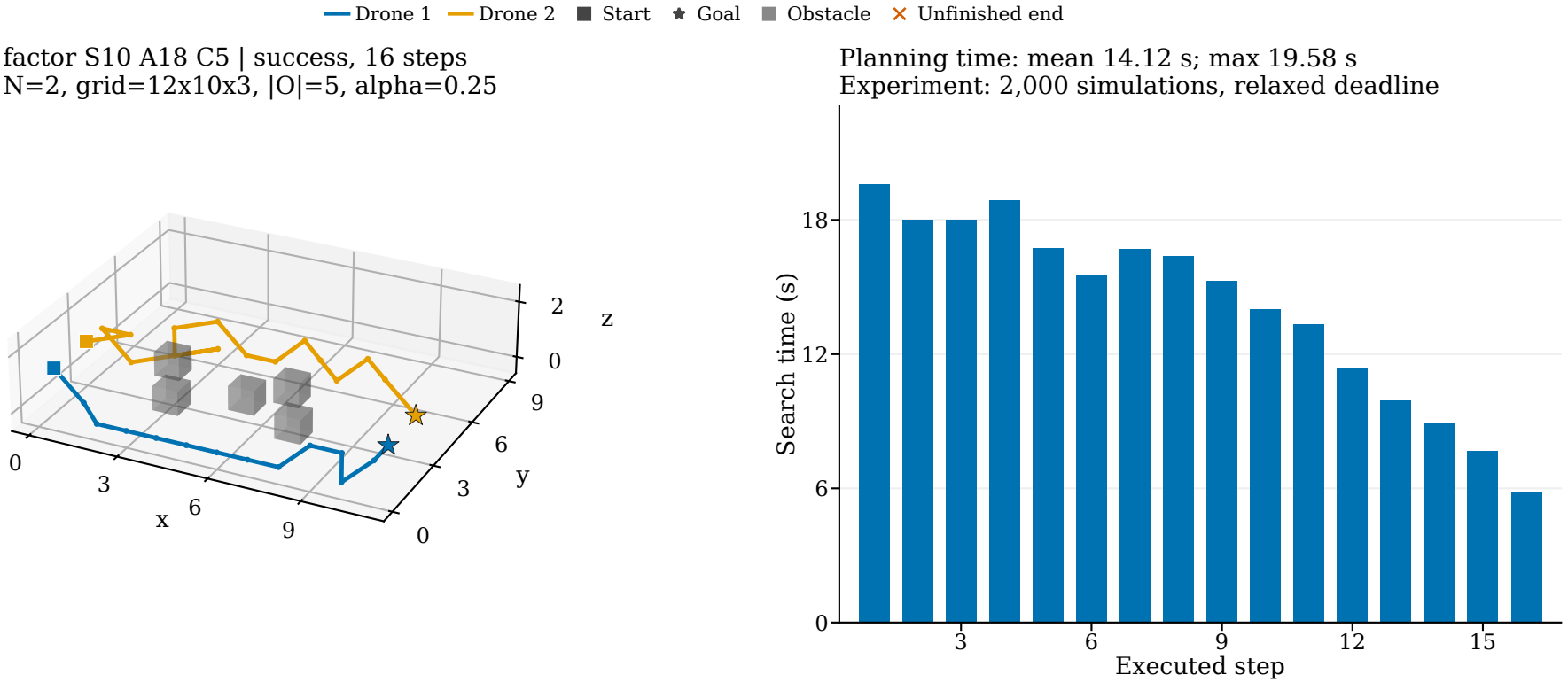
Saved full-solver trace, seed 0: results/trajectories/factorial_factor_S10_A18_C0_full_2000.json
Crosses mark returned positions of unfinished drones; collision targets are not logged.

factor S10 A18 C2



Saved full-solver trace, seed 0: results/trajectories/factorial_factor_S10_A18_C2_full_2000.json
Crosses mark returned positions of unfinished drones; collision targets are not logged.

factor S10 A18 C5

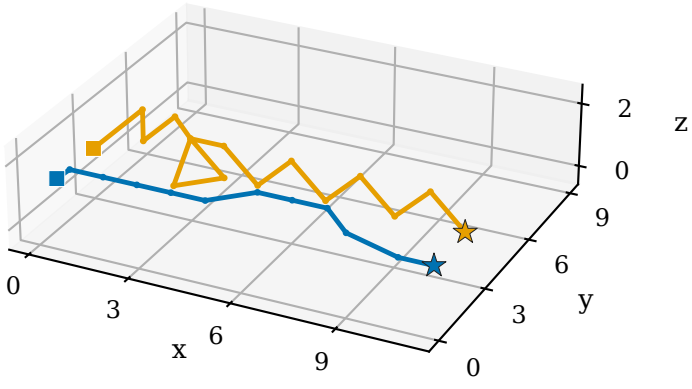


Saved full-solver trace, seed 0: results/trajectories/factorial_factor_S10_A18_C5_full_2000.json
Crosses mark returned positions of unfinished drones; collision targets are not logged.

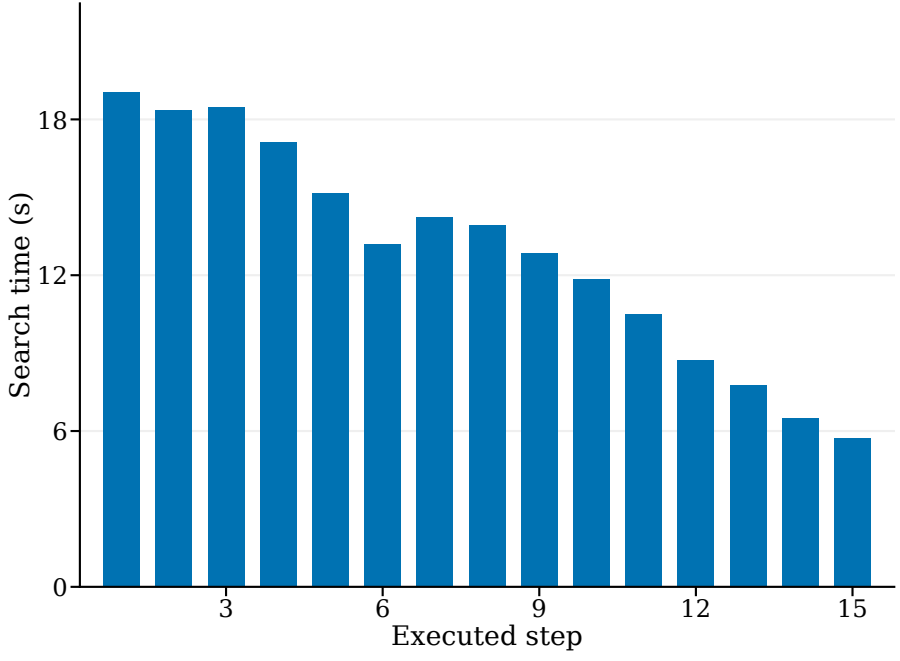
factor S10 A26 C0

Drone 1 Drone 2 Start Goal Obstacle Unfinished end

factor S10 A26 C0 | success, 15 steps
N=2, grid=12x10x3, |O|=0, alpha=0.25

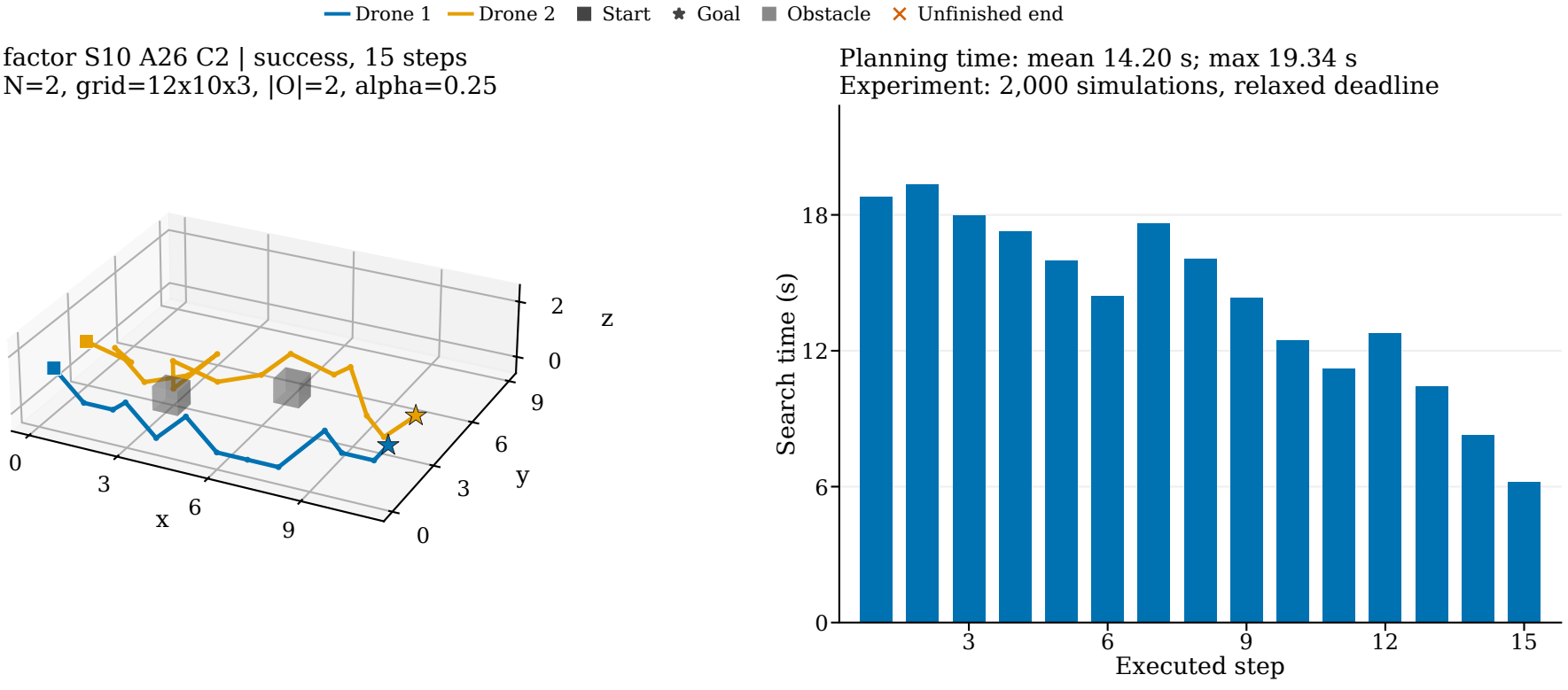


Planning time: mean 12.89 s; max 19.04 s
Experiment: 2,000 simulations, relaxed deadline



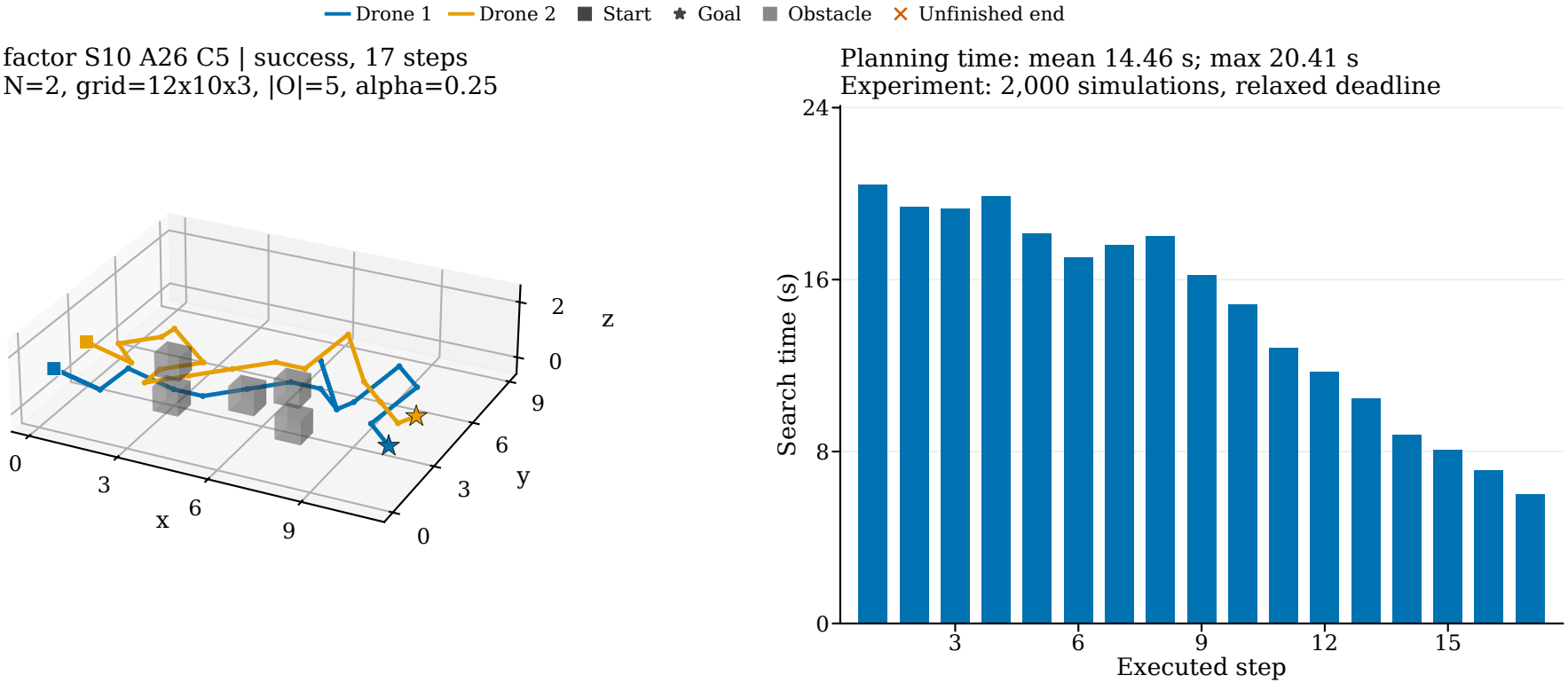
Saved full-solver trace, seed 0: results/trajectories/factorial_factor_S10_A26_C0_full_2000.json
Crosses mark returned positions of unfinished drones; collision targets are not logged.

factor S10 A26 C2



Saved full-solver trace, seed 0: results/trajectories/factorial_factor_S10_A26_C2_full_2000.json
Crosses mark returned positions of unfinished drones; collision targets are not logged.

factor S10 A26 C5

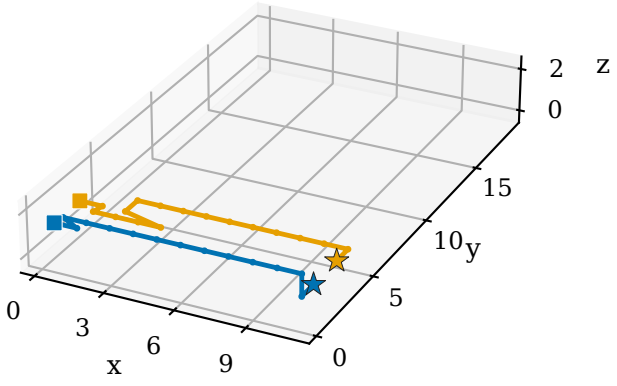


Saved full-solver trace, seed 0: results/trajectories/factorial_factor_S10_A26_C5_full_2000.json
Crosses mark returned positions of unfinished drones; collision targets are not logged.

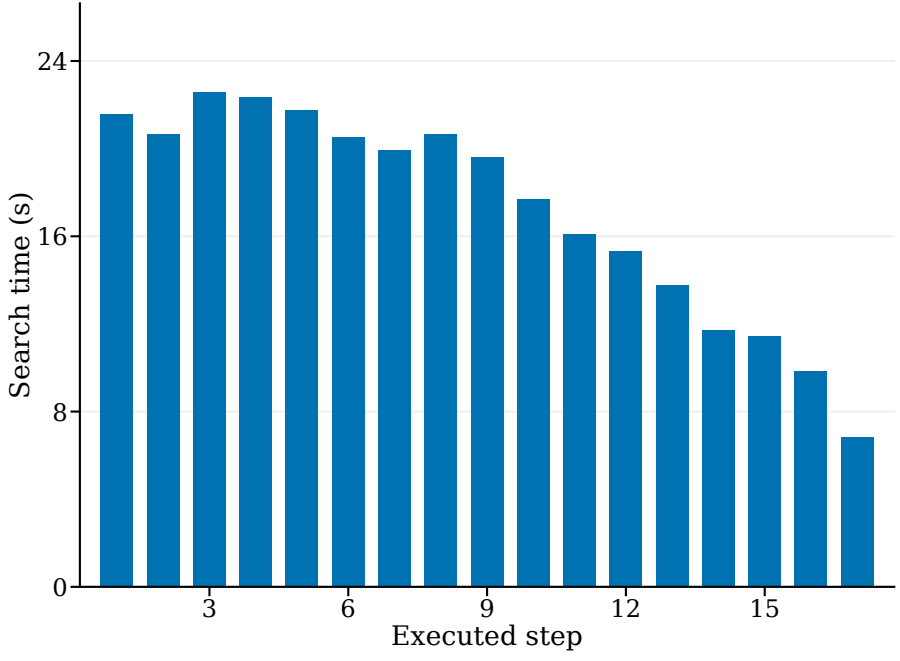
factor S20 A6 C0

Drone 1 Drone 2 Start Goal Obstacle Unfinished end

factor S20 A6 C0 | success, 17 steps
N=2, grid=12x20x3, |O|=0, alpha=0.25

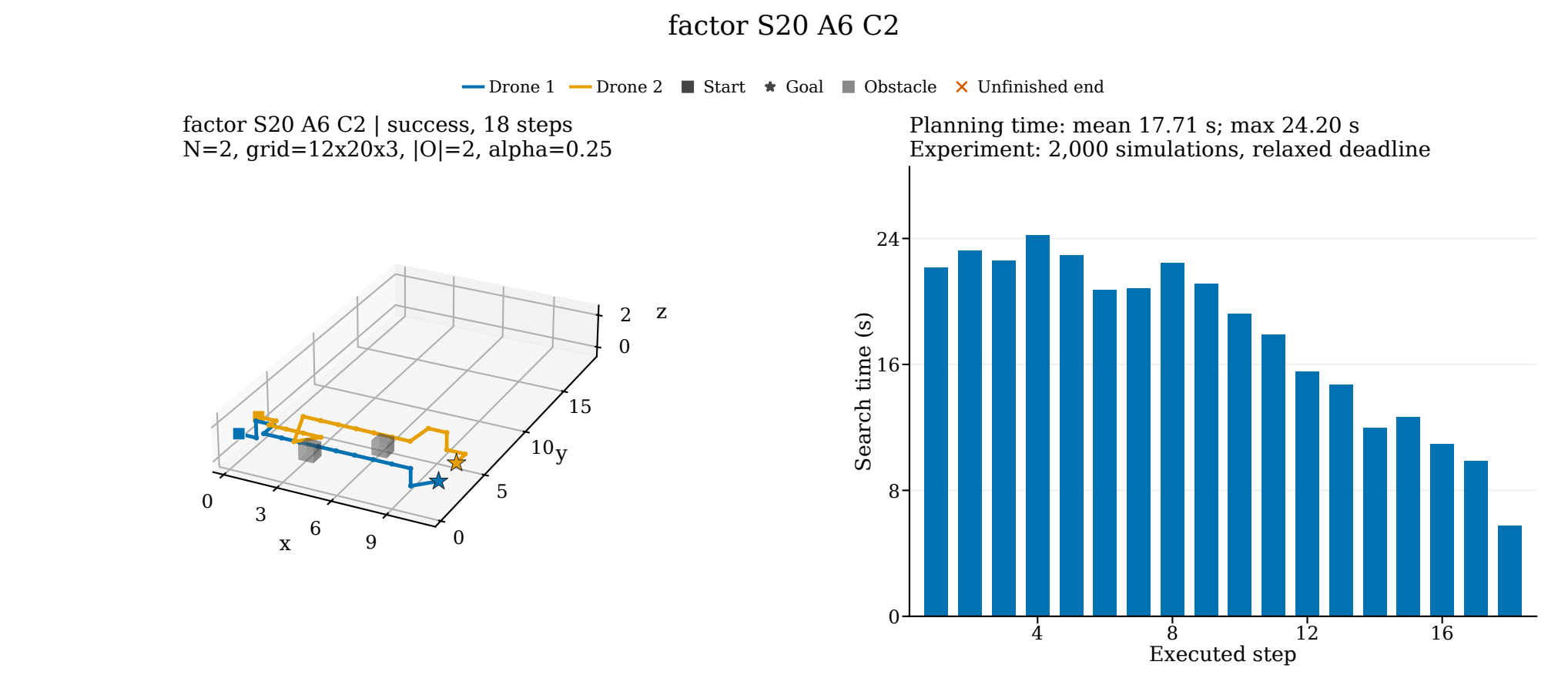


Planning time: mean 17.19 s; max 22.57 s
Experiment: 2,000 simulations, relaxed deadline



Saved full-solver trace, seed 0: results/trajectories/factorial_factor_S20_A6_C0_full_2000.json
Crosses mark returned positions of unfinished drones; collision targets are not logged.

factor S20 A6 C2

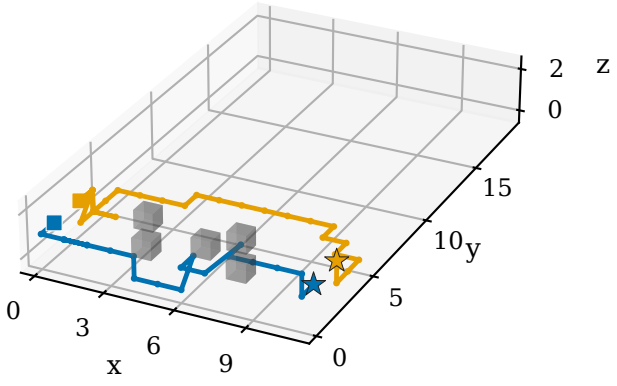


Saved full-solver trace, seed 0: results/trajectories/factorial_factor_S20_A6_C2_full_2000.json
Crosses mark returned positions of unfinished drones; collision targets are not logged.

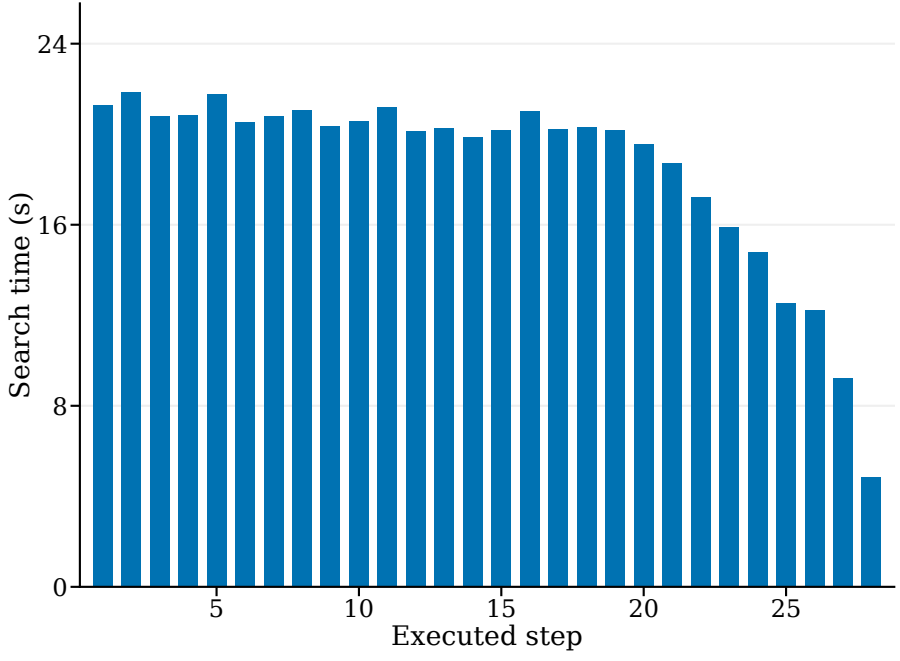
factor S20 A6 C5

Drone 1 | Drone 2 | Start | Goal | Obstacle | Unfinished end

factor S20 A6 C5 | success, 28 steps
N=2, grid=12x20x3, |O|=5, alpha=0.25



Planning time: mean 18.50 s; max 21.85 s
Experiment: 2,000 simulations, relaxed deadline

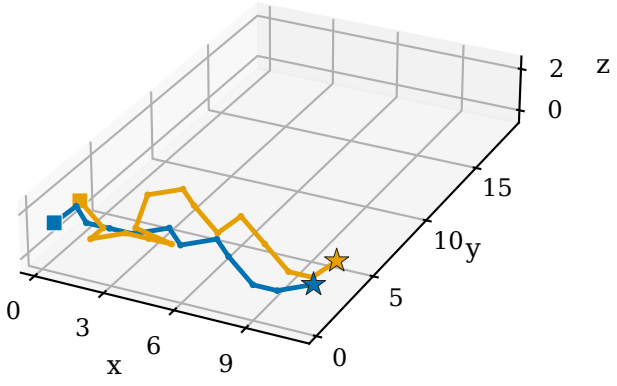


Saved full-solver trace, seed 0: results/trajectories/factorial_factor_S20_A6_C5_full_2000.json
Crosses mark returned positions of unfinished drones; collision targets are not logged.

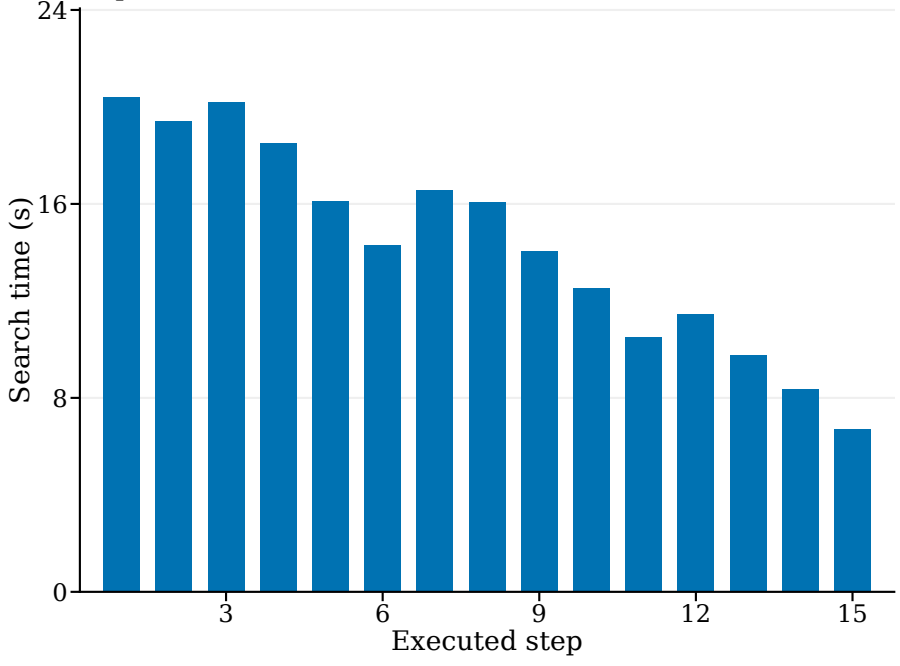
factor S20 A18 C0

Drone 1 Drone 2 Start Goal Obstacle Unfinished end

factor S20 A18 C0 | success, 15 steps
N=2, grid=12x20x3, |O|=0, alpha=0.25

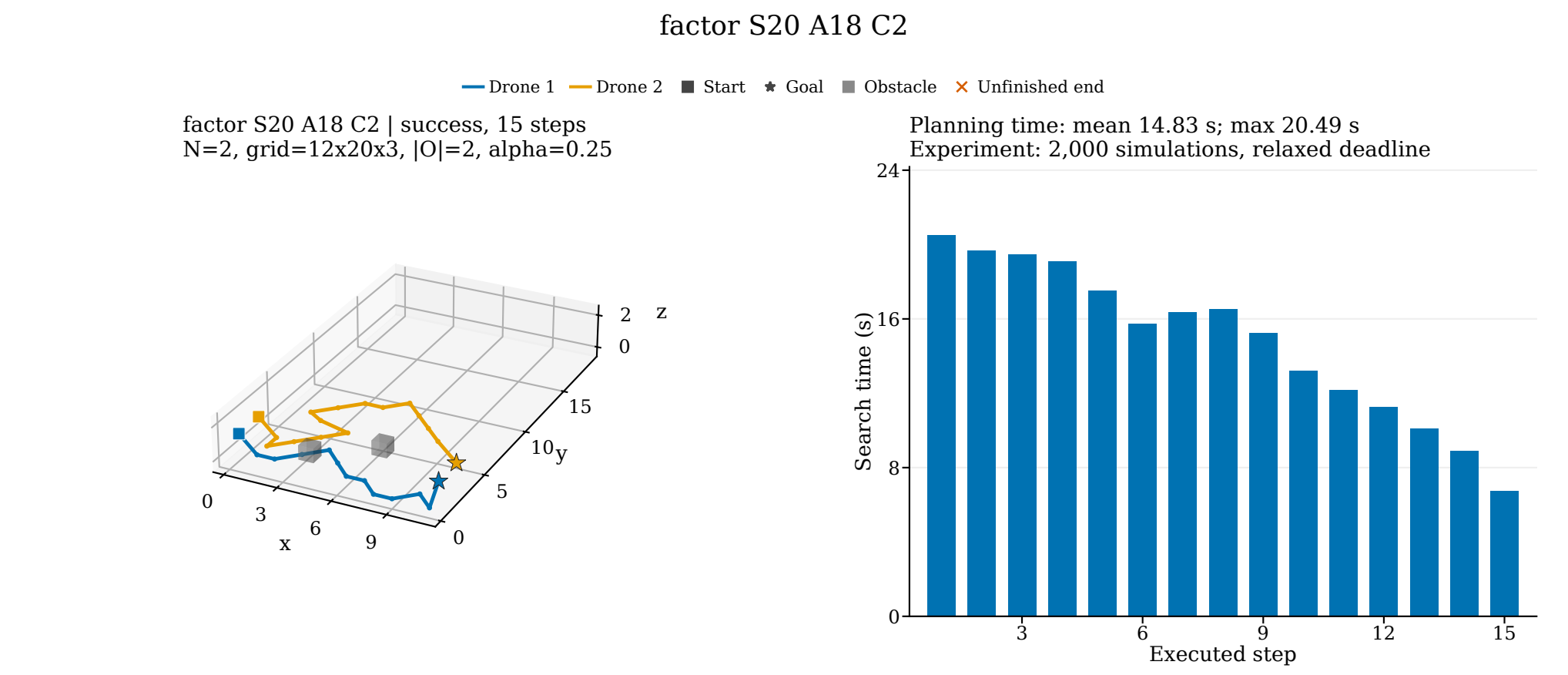


Planning time: mean 14.32 s; max 20.39 s
Experiment: 2,000 simulations, relaxed deadline



Saved full-solver trace, seed 0: results/trajectories/factorial_factor_S20_A18_C0_full_2000.json
Crosses mark returned positions of unfinished drones; collision targets are not logged.

factor S20 A18 C2

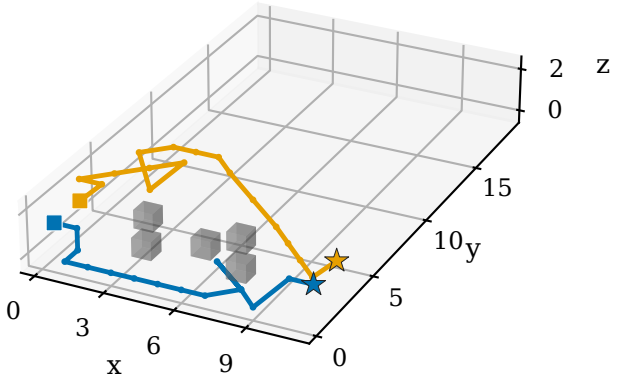


Saved full-solver trace, seed 0: results/trajectories/factorial_factor_S20_A18_C2_full_2000.json
Crosses mark returned positions of unfinished drones; collision targets are not logged.

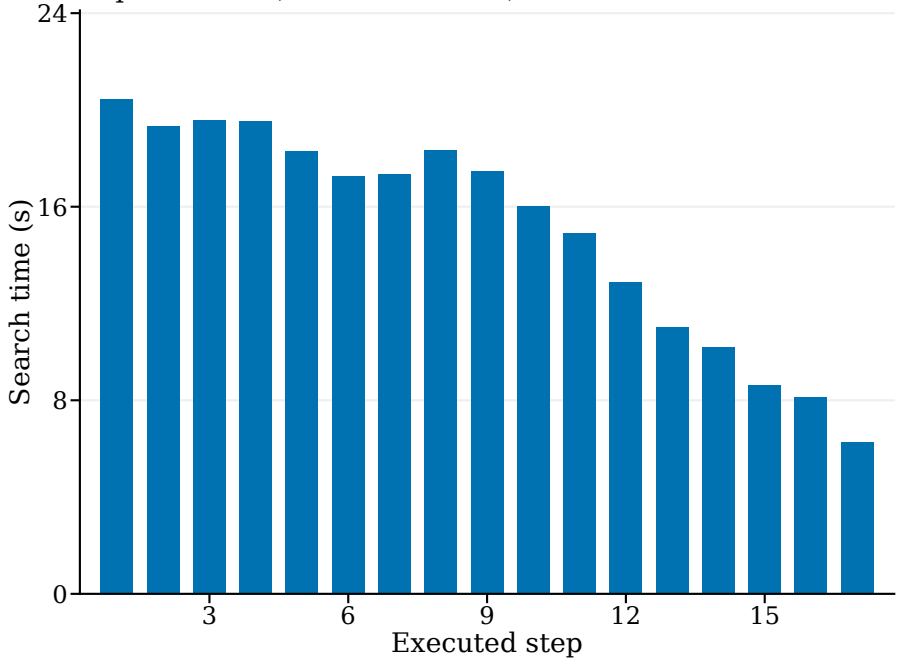
factor S20 A18 C5

Drone 1 Drone 2 Start Goal Obstacle Unfinished end

factor S20 A18 C5 | success, 17 steps
N=2, grid=12x20x3, |O|=5, alpha=0.25



Planning time: mean 15.03 s; max 20.43 s
Experiment: 2,000 simulations, relaxed deadline

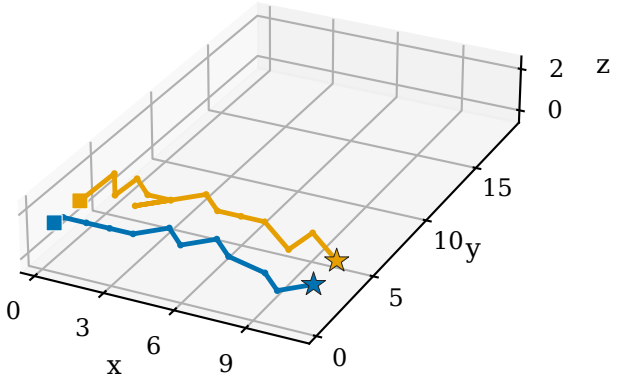


Saved full-solver trace, seed 0: results/trajectories/factorial_factor_S20_A18_C5_full_2000.json
Crosses mark returned positions of unfinished drones; collision targets are not logged.

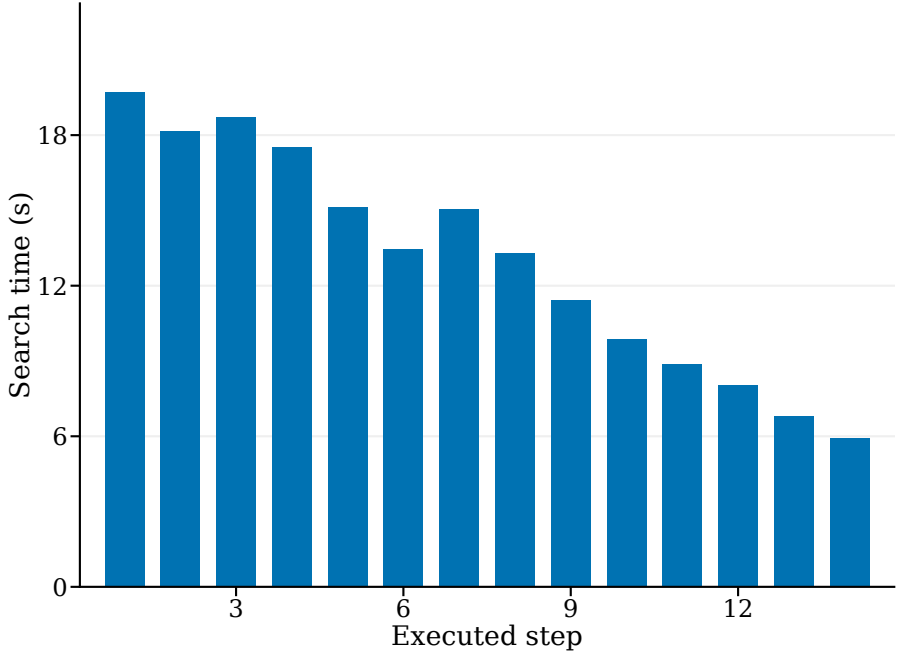
factor S20 A26 C0

Drone 1 Drone 2 Start Goal Obstacle Unfinished end

factor S20 A26 C0 | success, 14 steps
N=2, grid=12x20x3, |O|=0, alpha=0.25



Planning time: mean 12.99 s; max 19.70 s
Experiment: 2,000 simulations, relaxed deadline

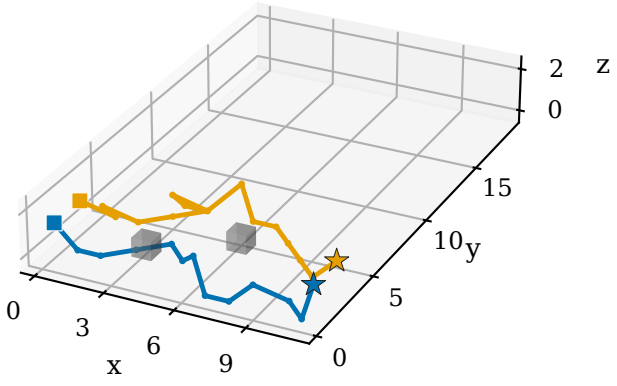


Saved full-solver trace, seed 0: results/trajectories/factorial_factor_S20_A26_C0_full_2000.json
Crosses mark returned positions of unfinished drones; collision targets are not logged.

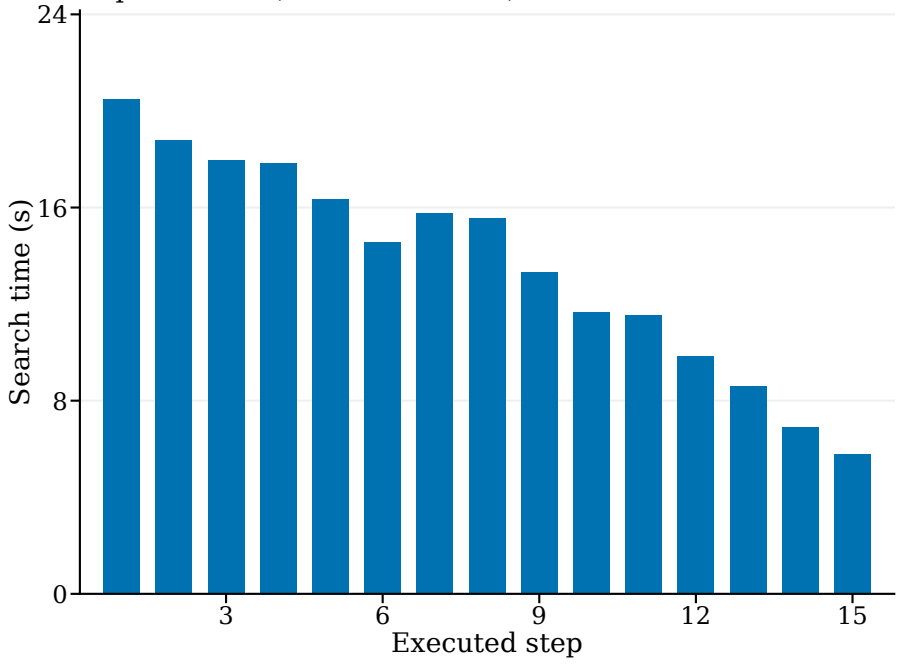
factor S20 A26 C2

Drone 1 Drone 2 Start Goal Obstacle Unfinished end

factor S20 A26 C2 | success, 15 steps
N=2, grid=12x20x3, |O|=2, alpha=0.25

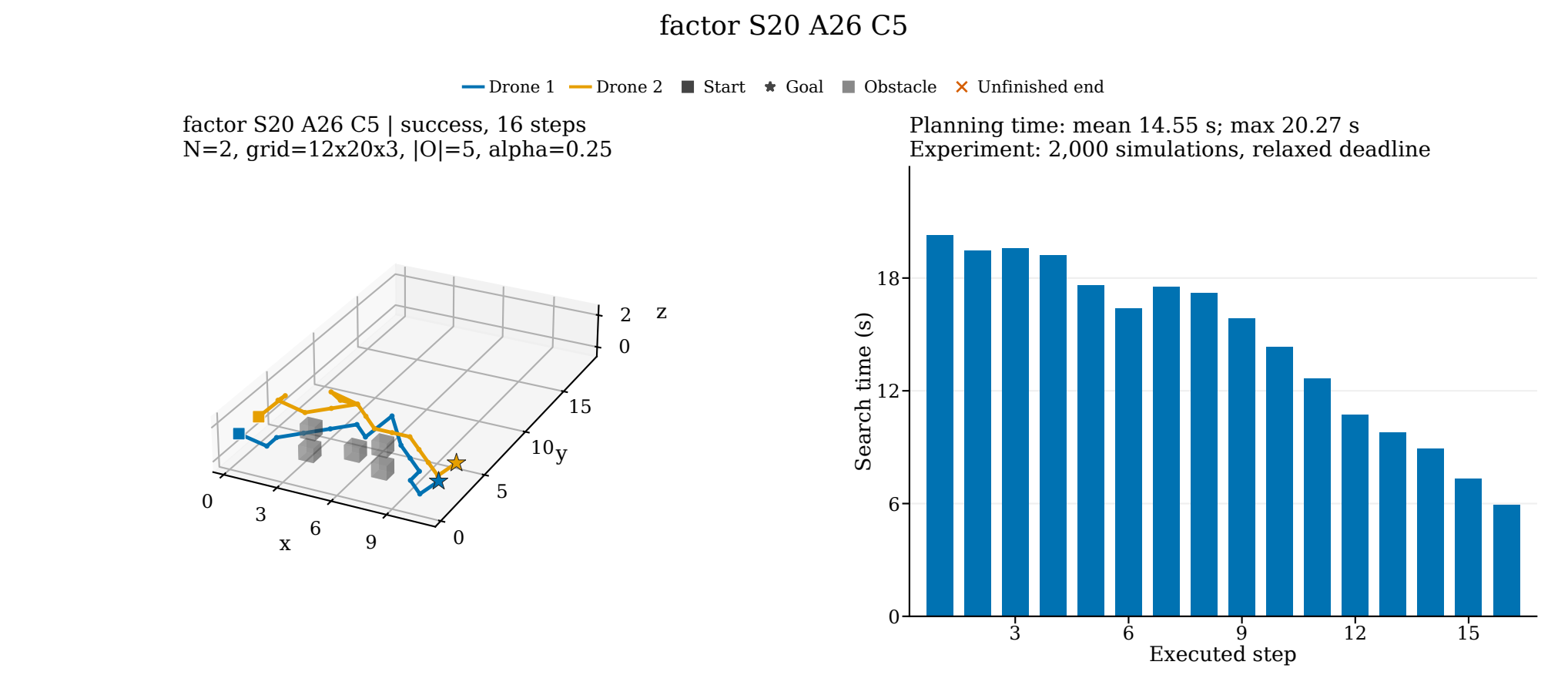


Planning time: mean 13.66 s; max 20.48 s
Experiment: 2,000 simulations, relaxed deadline



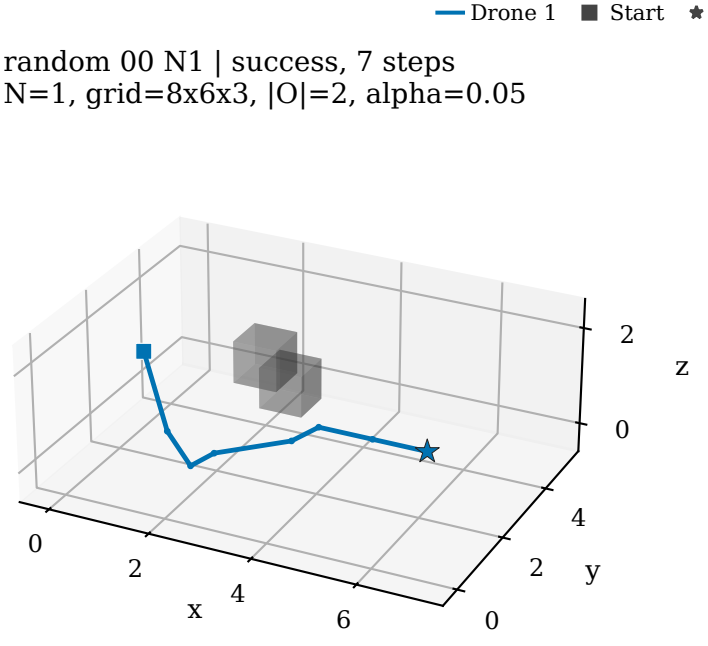
Saved full-solver trace, seed 0: results/trajectories/factorial_factor_S20_A26_C2_full_2000.json
Crosses mark returned positions of unfinished drones; collision targets are not logged.

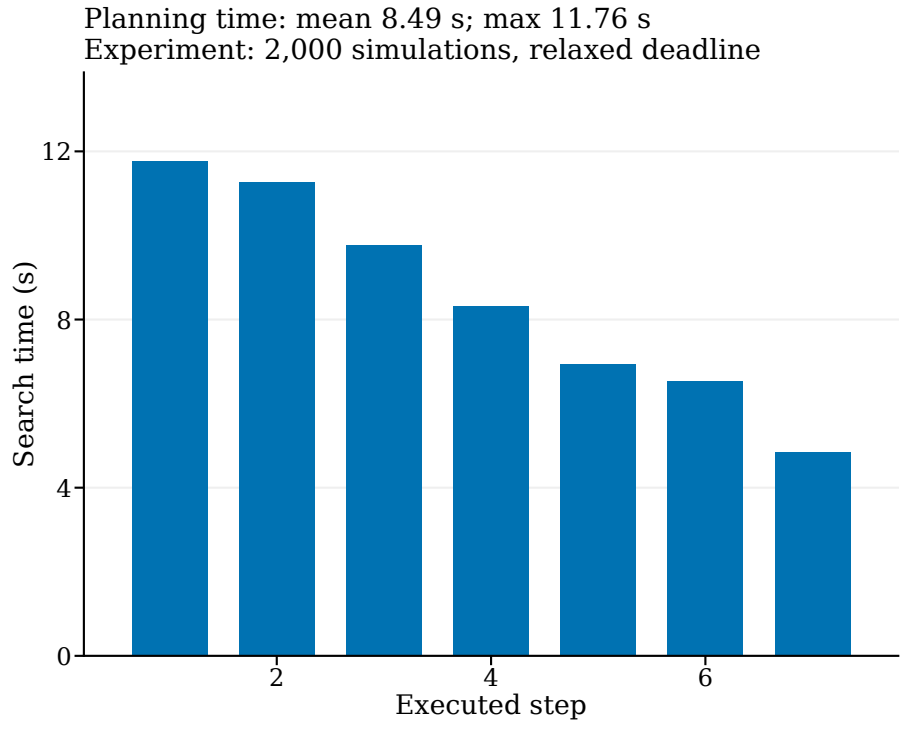
factor S20 A26 C5



Saved full-solver trace, seed 0: results/trajectories/factorial_factor_S20_A26_C5_full_2000.json
Crosses mark returned positions of unfinished drones; collision targets are not logged.

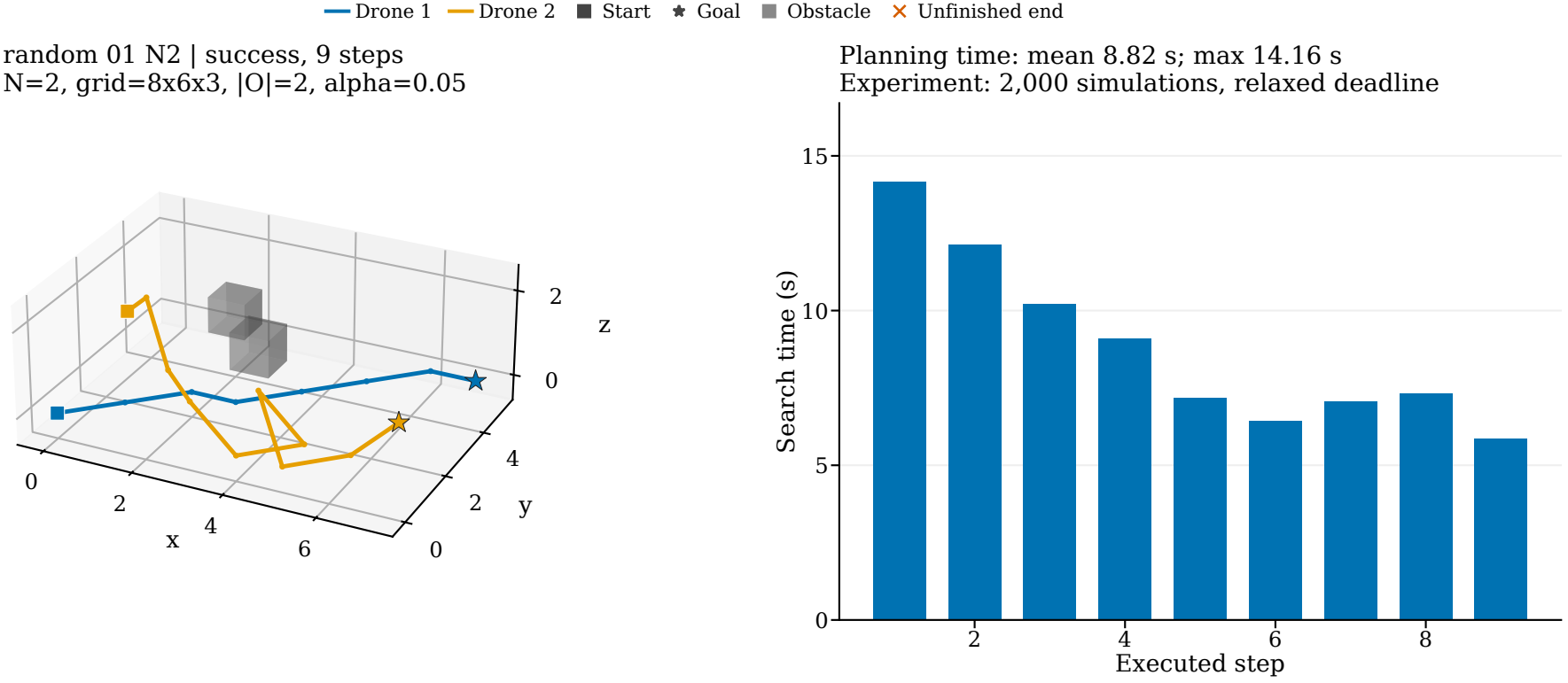
random 00 N1





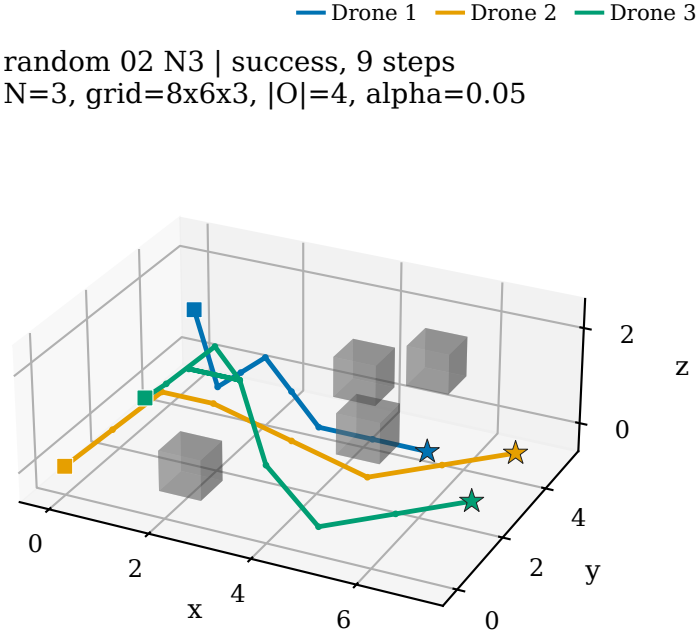
Saved full-solver trace, seed 0: results/trajectories/random_random_00_N1_full_2000.json
Crosses mark returned positions of unfinished drones; collision targets are not logged.

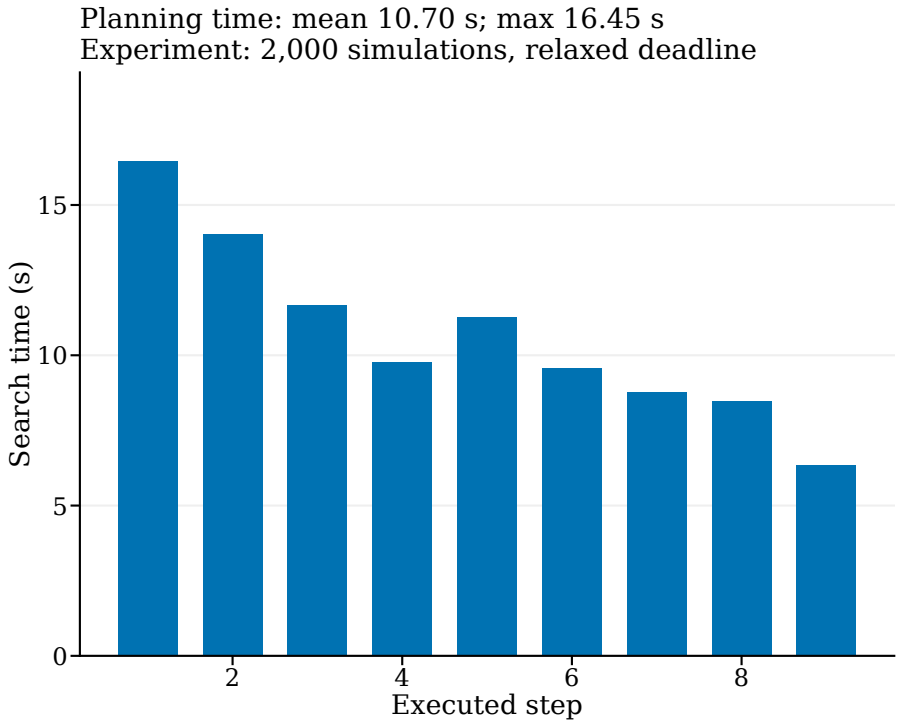
random 01 N2



Saved full-solver trace, seed 0: results/trajectories/random_random_01_N2_full_2000.json
Crosses mark returned positions of unfinished drones; collision targets are not logged.

random 02 N3



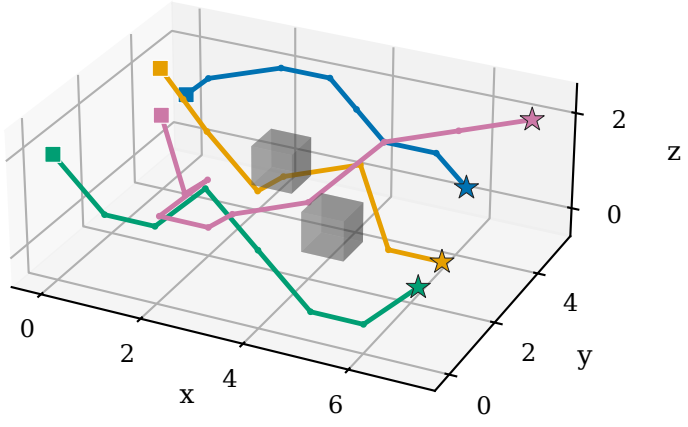


Saved full-solver trace, seed 0: results/trajectories/random_random_02_N3_full_2000.json
Crosses mark returned positions of unfinished drones; collision targets are not logged.

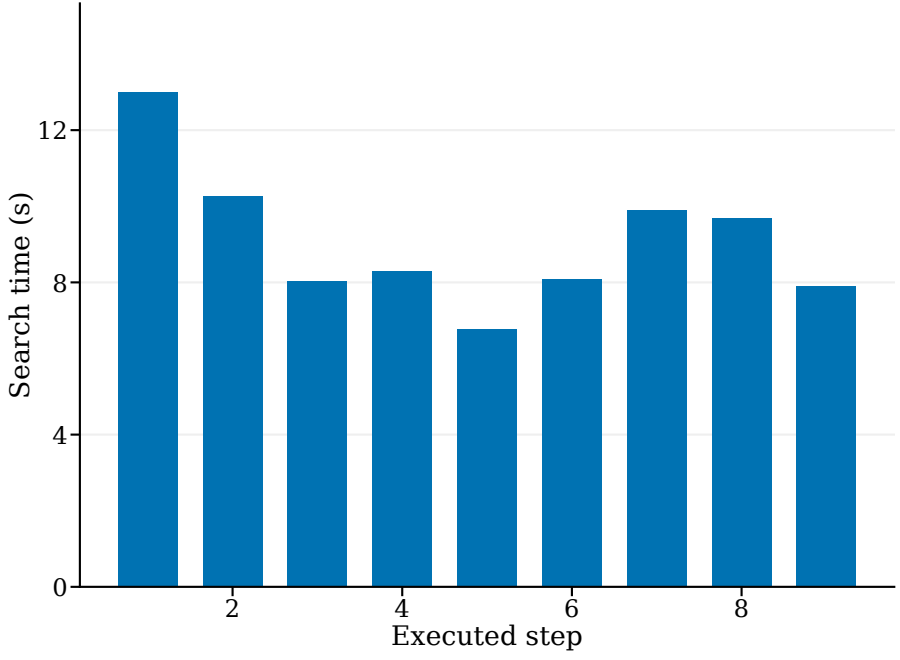
random 03 N4



random 03 N4 | success, 9 steps
N=4, grid=8x6x3, |O|=2, alpha=0.05

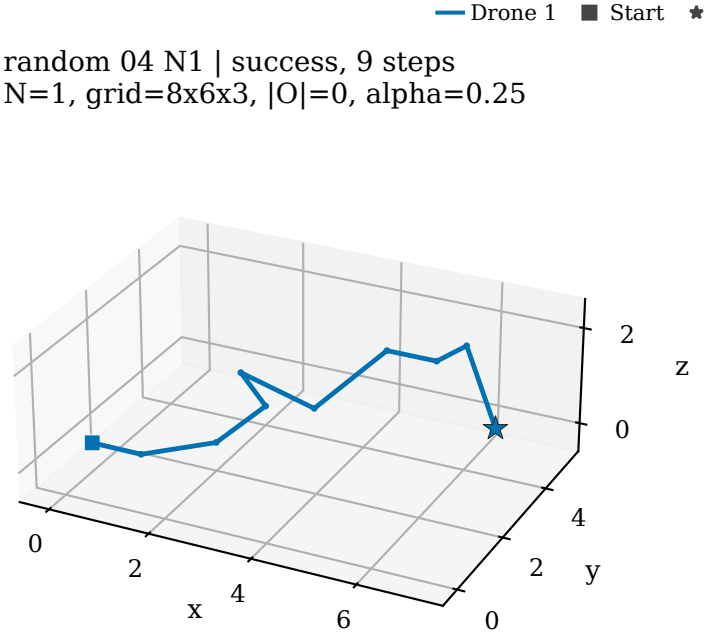


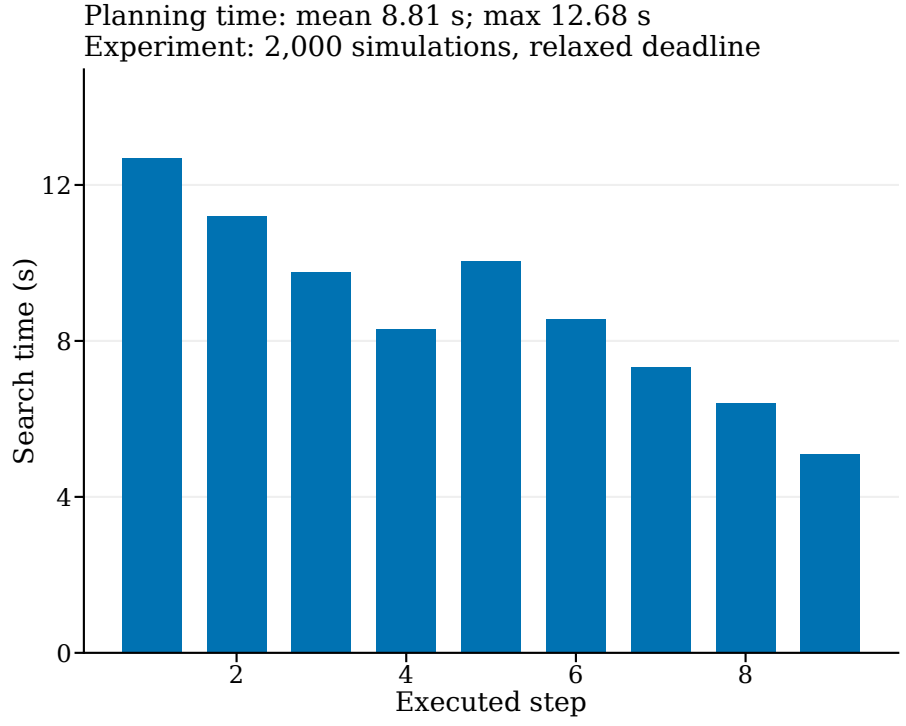
Planning time: mean 9.09 s; max 12.99 s
Experiment: 2,000 simulations, relaxed deadline



Saved full-solver trace, seed 0: results/trajectories/random_random_03_N4_full_2000.json
Crosses mark returned positions of unfinished drones; collision targets are not logged.

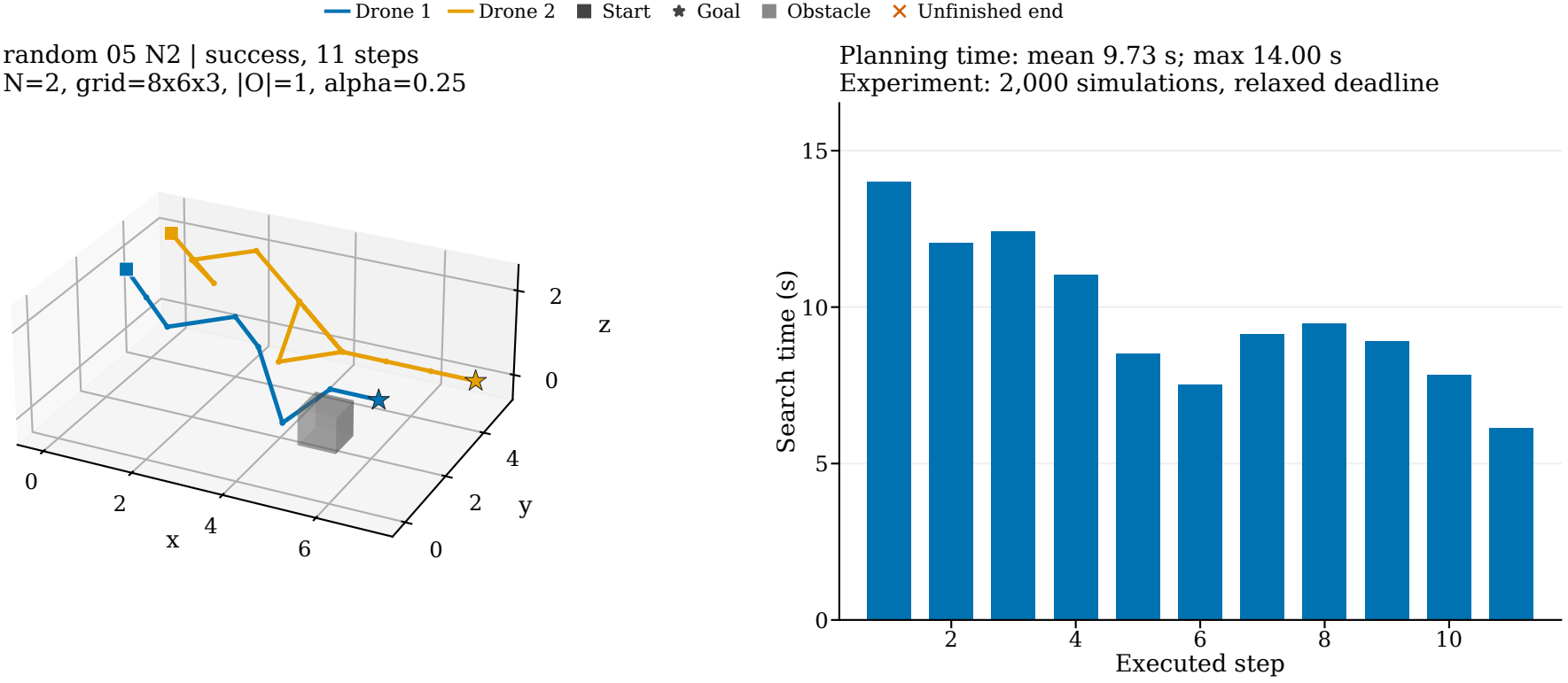
random 04 N1





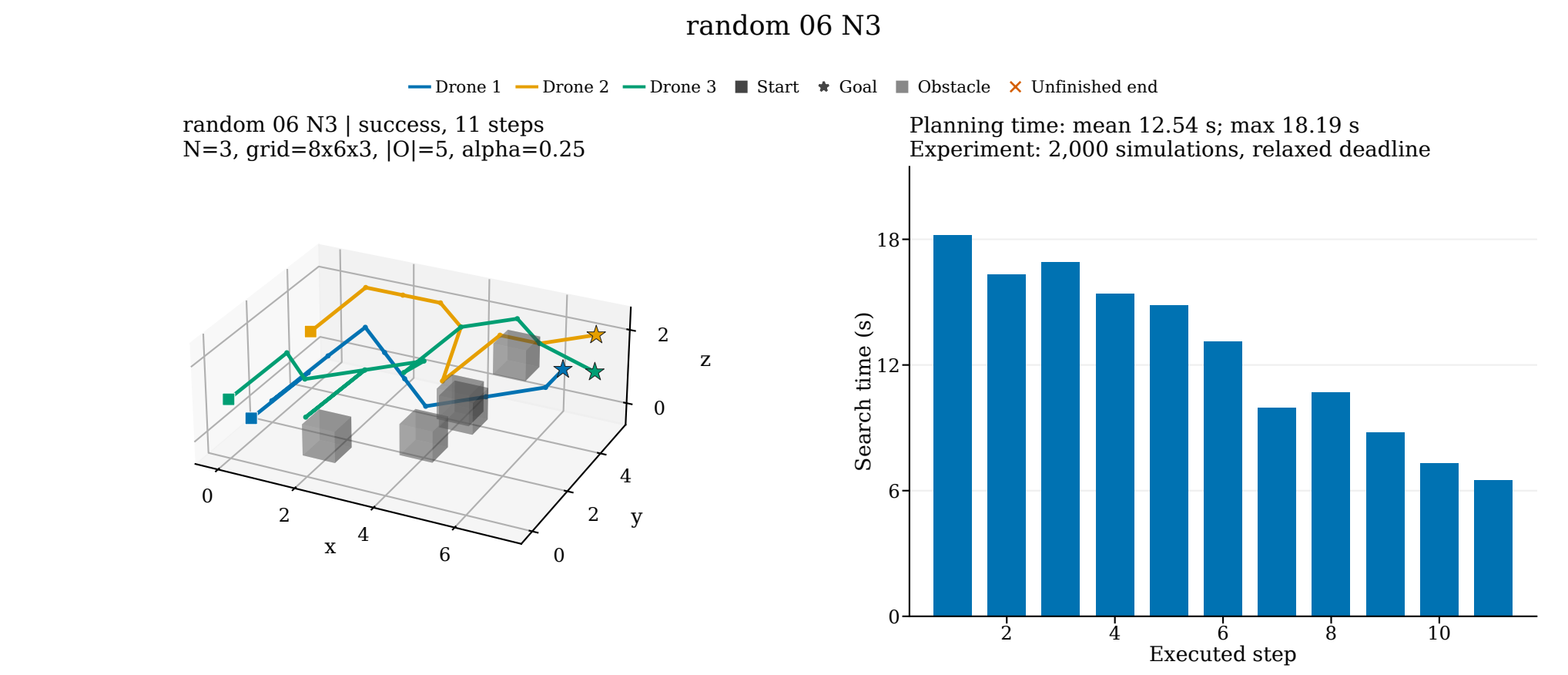
Saved full-solver trace, seed 0: results/trajectories/random_random_04_N1_full_2000.json
Crosses mark returned positions of unfinished drones; collision targets are not logged.

random 05 N2



Saved full-solver trace, seed 0: results/trajectories/random_random_05_N2_full_2000.json
Crosses mark returned positions of unfinished drones; collision targets are not logged.

random 06 N3

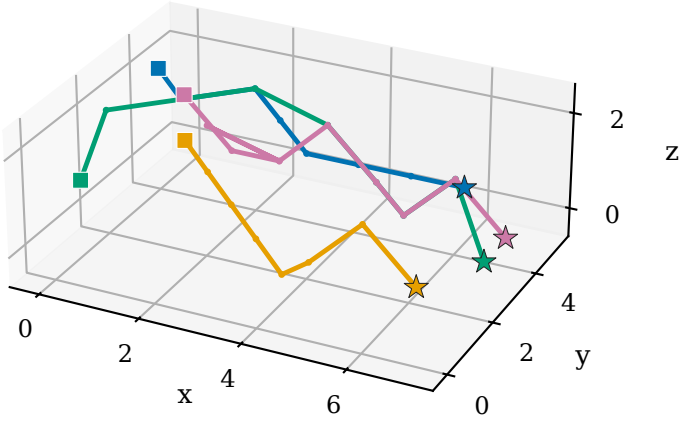


Saved full-solver trace, seed 0: results/trajectories/random_random_06_N3_full_2000.json
Crosses mark returned positions of unfinished drones; collision targets are not logged.

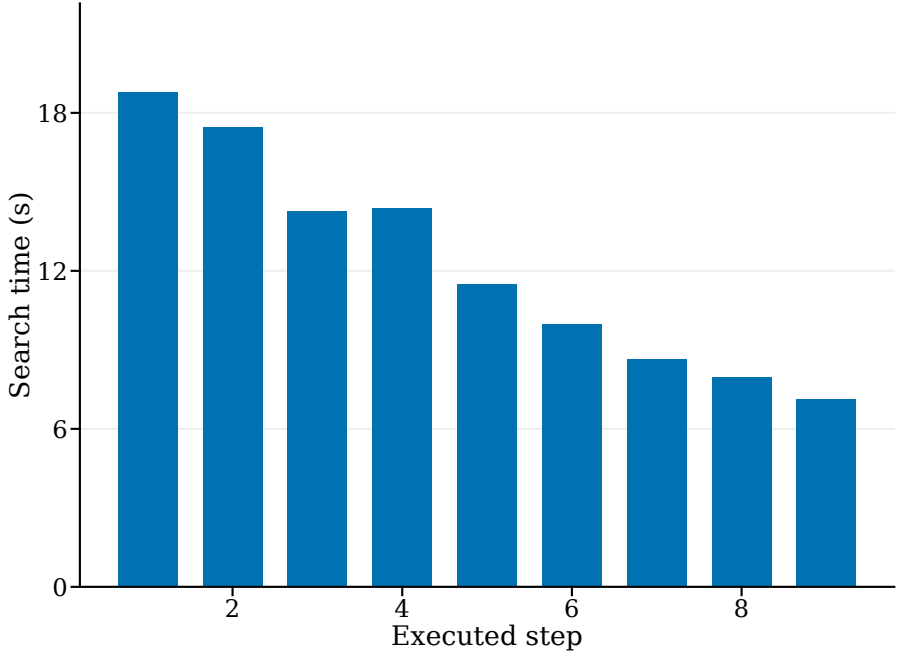
random 07 N4

Drone 1 Drone 2 Drone 3 Drone 4 Start Goal Obstacle Unfinished end

random 07 N4 | success, 9 steps
N=4, grid=8x6x3, |O|=0, alpha=0.25

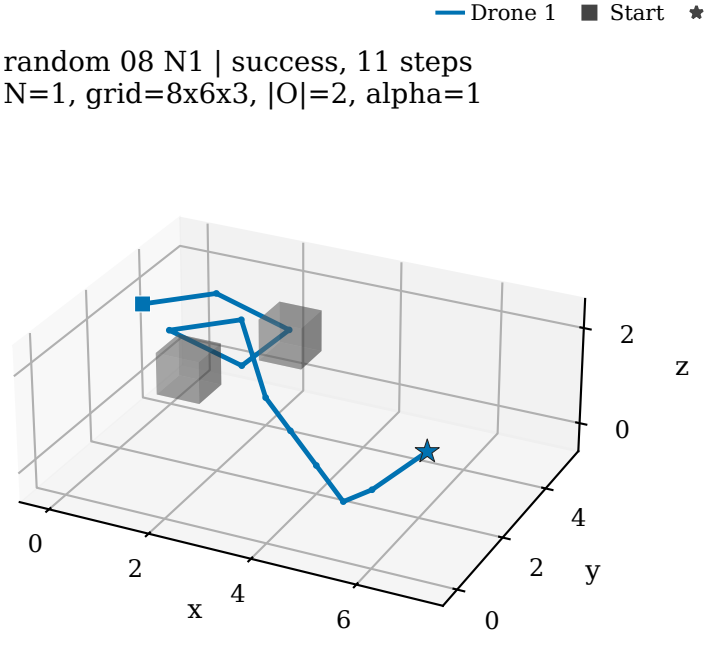


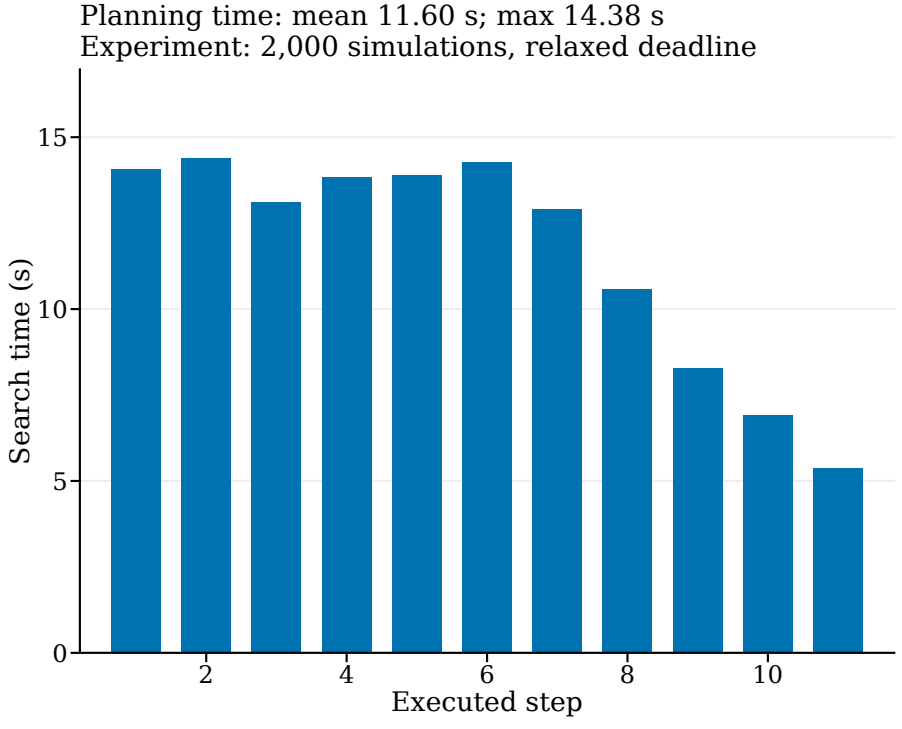
Planning time: mean 12.23 s; max 18.78 s
Experiment: 2,000 simulations, relaxed deadline



Saved full-solver trace, seed 0: results/trajectories/random_random_07_N4_full_2000.json
Crosses mark returned positions of unfinished drones; collision targets are not logged.

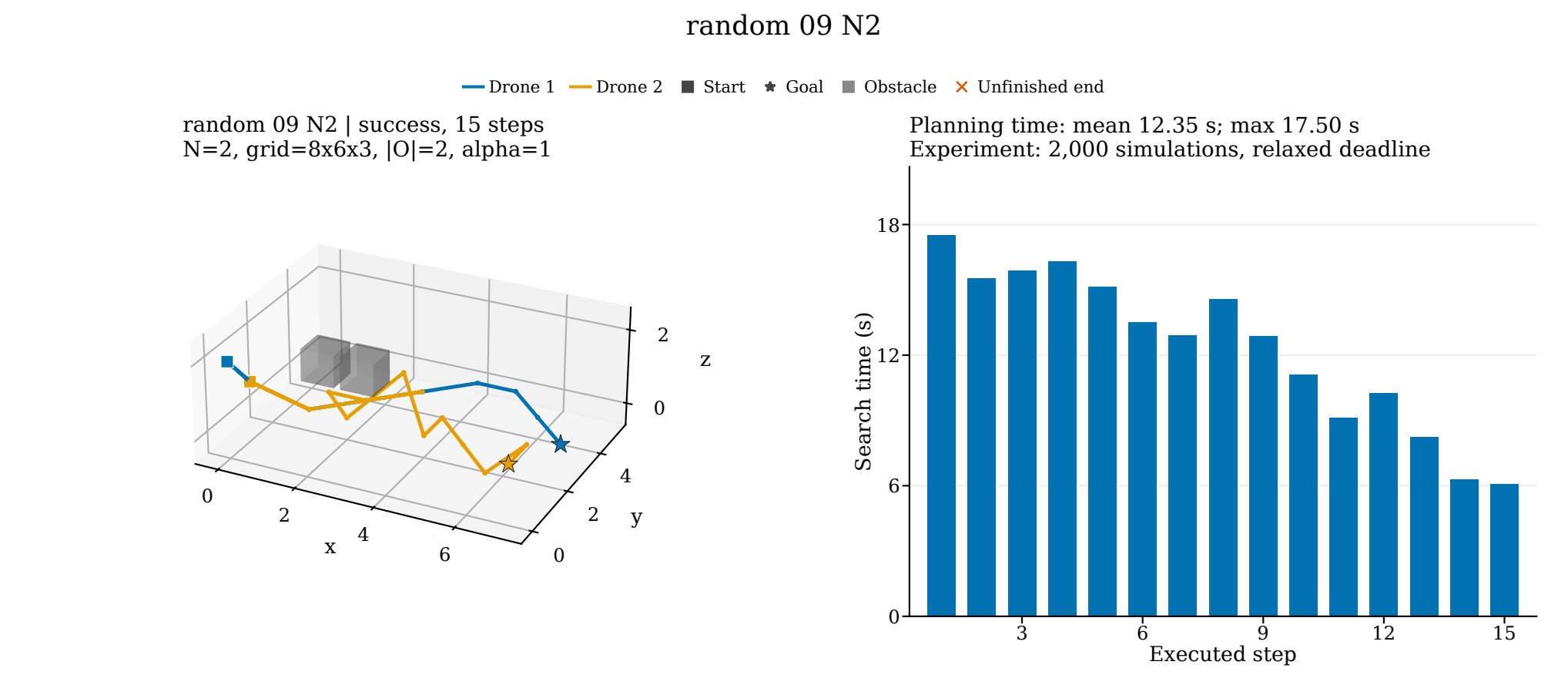
random 08 N1





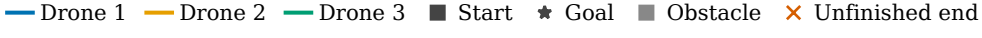
Saved full-solver trace, seed 0: results/trajectories/random_random_08_N1_full_2000.json
Crosses mark returned positions of unfinished drones; collision targets are not logged.

random 09 N2

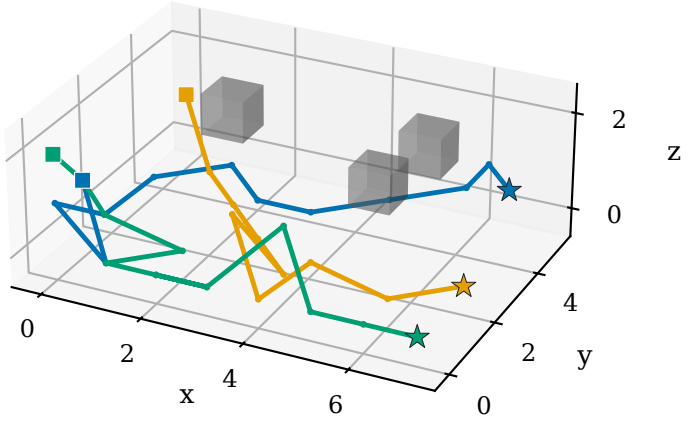


Saved full-solver trace, seed 0: results/trajectories/random_random_09_N2_full_2000.json
Crosses mark returned positions of unfinished drones; collision targets are not logged.

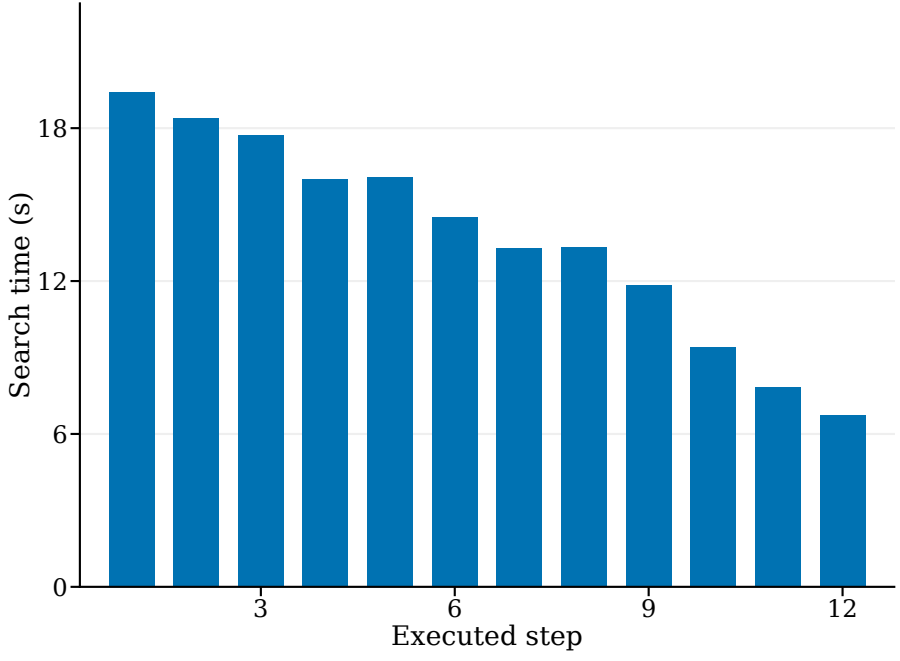
random 10 N3



random 10 N3 | success, 12 steps
N=3, grid=8x6x3, |O|=3, alpha=1



Planning time: mean 13.71 s; max 19.40 s
Experiment: 2,000 simulations, relaxed deadline

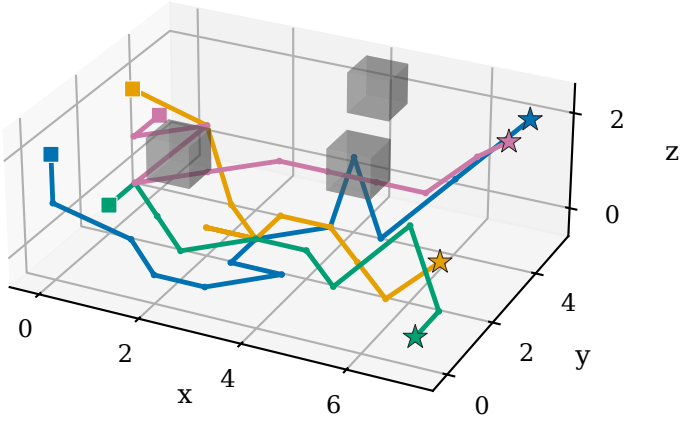


Saved full-solver trace, seed 0: results/trajectories/random_random_10_N3_full_2000.json
Crosses mark returned positions of unfinished drones; collision targets are not logged.

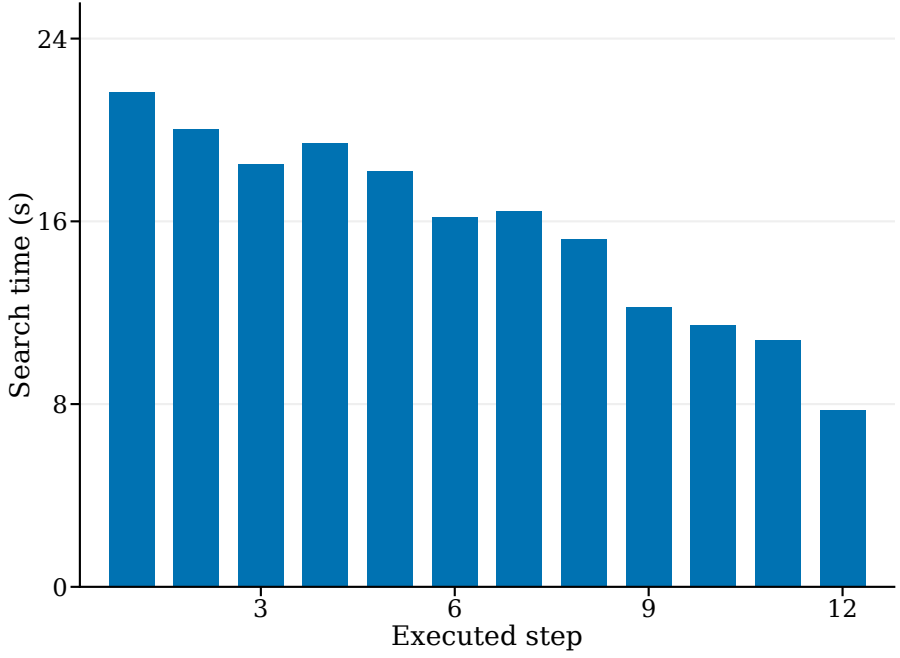
random 11 N4

Drone 1 Drone 2 Drone 3 Drone 4 Start Goal Obstacle Unfinished end

random 11 N4 | success, 12 steps
N=4, grid=8x6x3, |O|=3, alpha=1



Planning time: mean 15.65 s; max 21.64 s
Experiment: 2,000 simulations, relaxed deadline

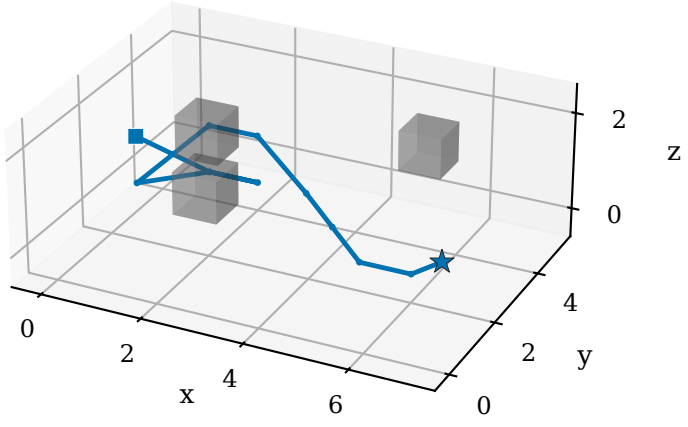


Saved full-solver trace, seed 0: results/trajectories/random_random_11_N4_full_2000.json
Crosses mark returned positions of unfinished drones; collision targets are not logged.

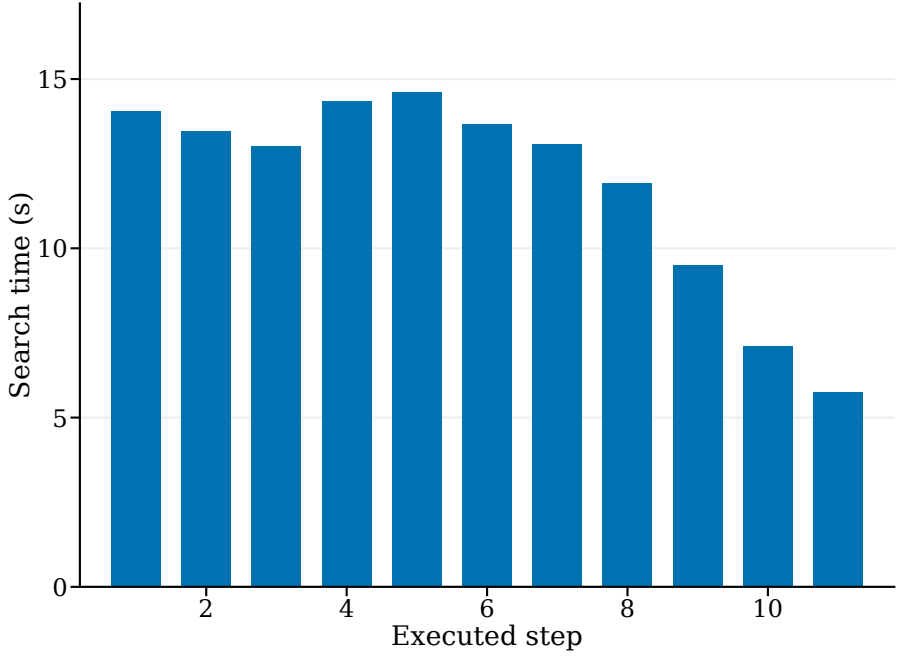
random 12 N1

Drone 1 Start Goal Obstacle Unfinished end

random 12 N1 | success, 11 steps
N=1, grid=8x6x3, |O|=3, alpha=2



Planning time: mean 11.86 s; max 14.60 s
Experiment: 2,000 simulations, relaxed deadline

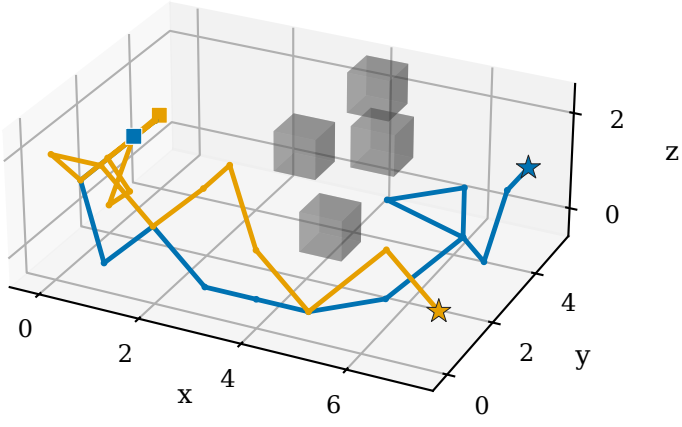


Saved full-solver trace, seed 0: results/trajectories/random_random_12_N1_full_2000.json
Crosses mark returned positions of unfinished drones; collision targets are not logged.

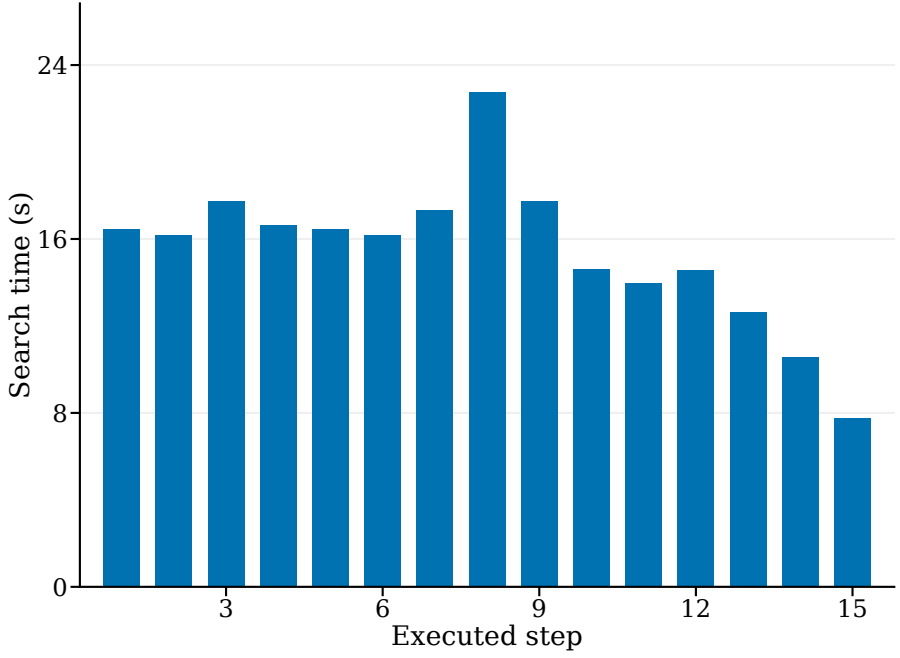
random 13 N2



random 13 N2 | success, 15 steps
N=2, grid=8x6x3, |O|=4, alpha=2

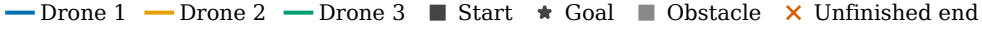


Planning time: mean 15.43 s; max 22.74 s
Experiment: 2,000 simulations, relaxed deadline

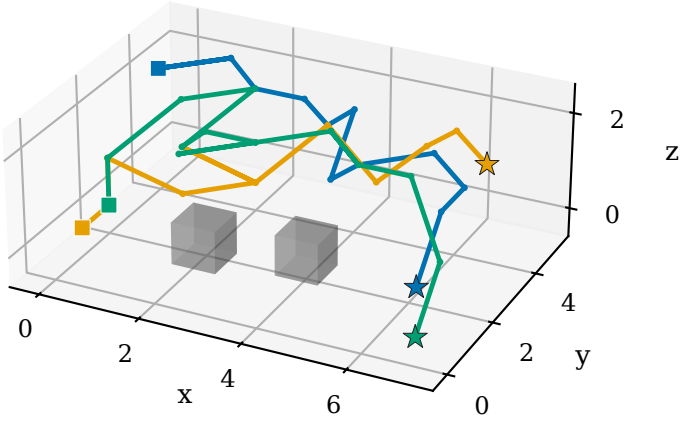


Saved full-solver trace, seed 0: results/trajectories/random_random_13_N2_full_2000.json
Crosses mark returned positions of unfinished drones; collision targets are not logged.

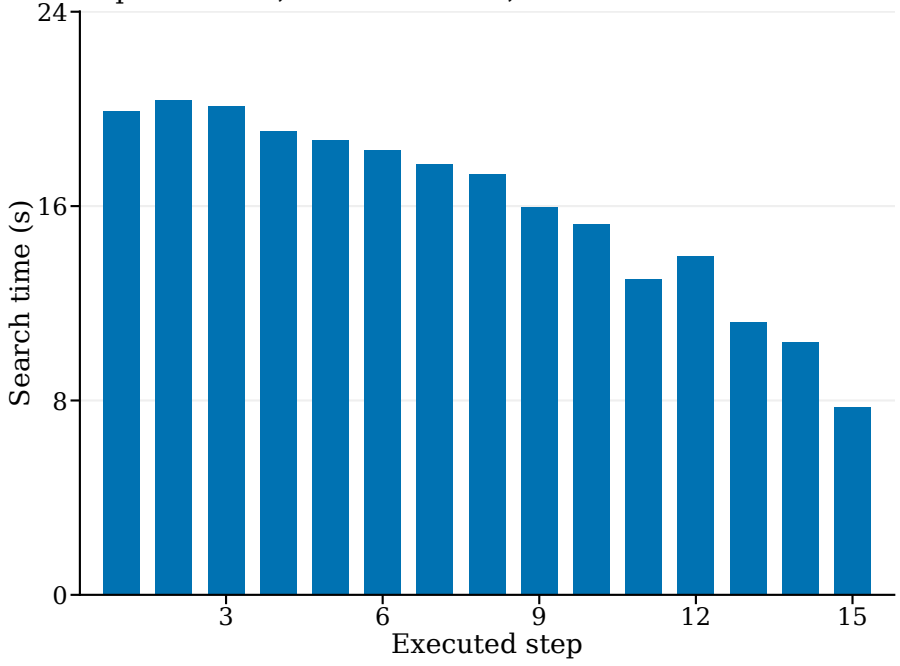
random 14 N3



random 14 N3 | success, 15 steps
N=3, grid=8x6x3, |O|=2, alpha=2



Planning time: mean 15.93 s; max 20.35 s
Experiment: 2,000 simulations, relaxed deadline

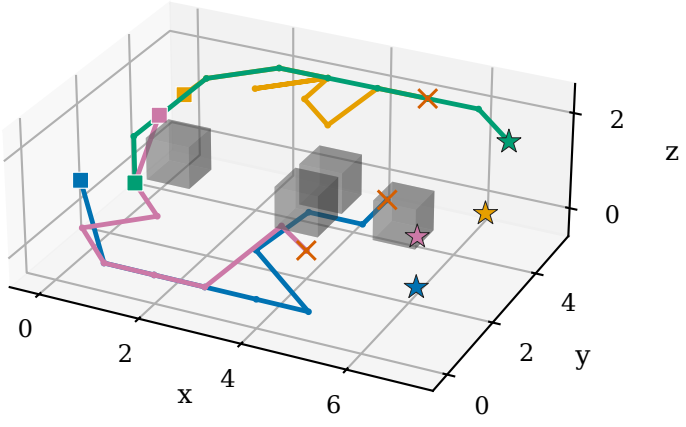


Saved full-solver trace, seed 0: results/trajectories/random_random_14_N3_full_2000.json
Crosses mark returned positions of unfinished drones; collision targets are not logged.

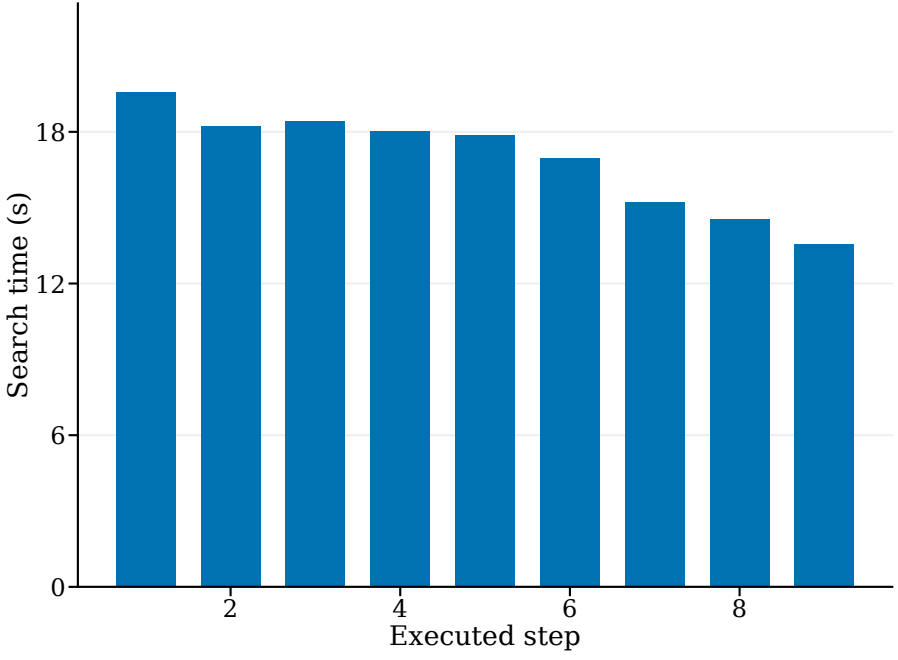
random 15 N4



random 15 N4 | collision, 9 steps
N=4, grid=8x6x3, |O|=4, alpha=2

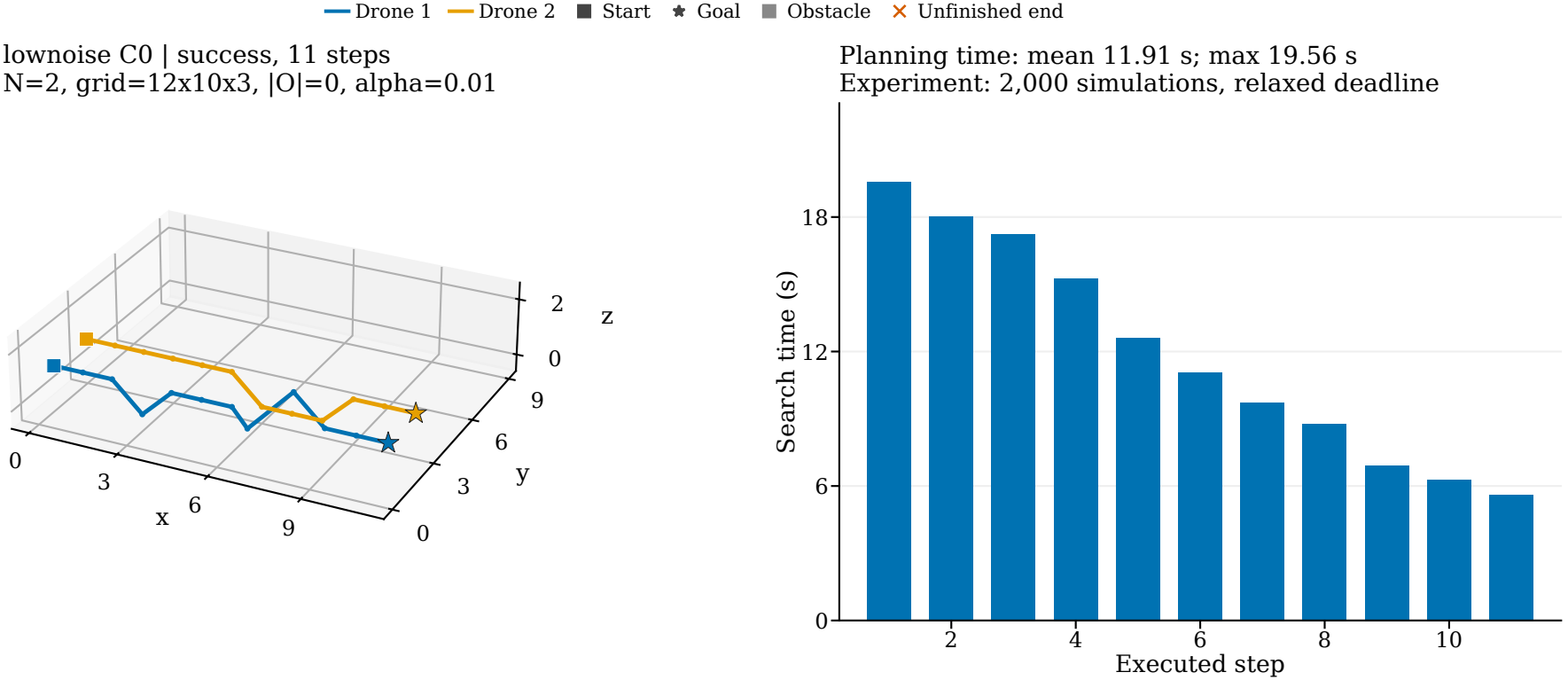


Planning time: mean 16.93 s; max 19.56 s
Experiment: 2,000 simulations, relaxed deadline



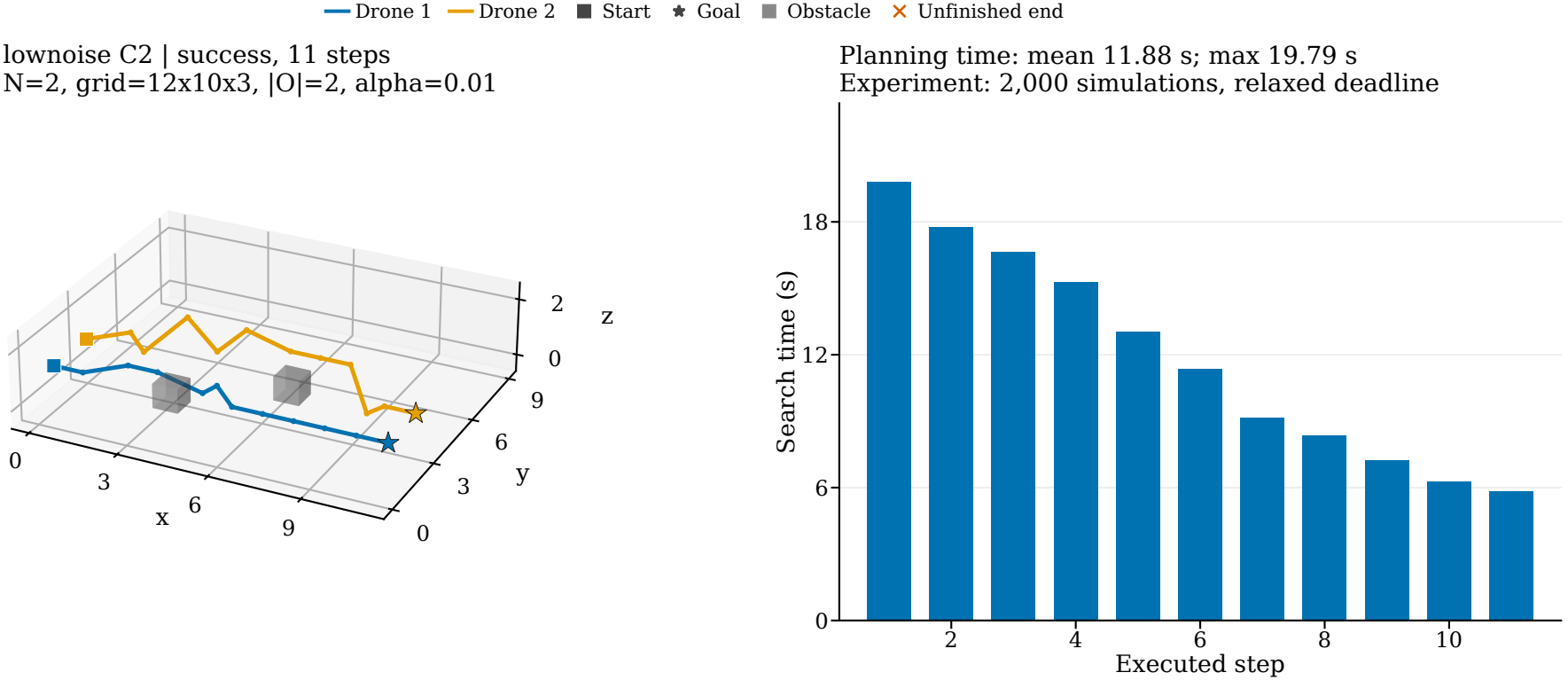
Saved full-solver trace, seed 0: results/trajectories/random_random_15_N4_full_2000.json
Crosses mark returned positions of unfinished drones; collision targets are not logged.

lownoise C0



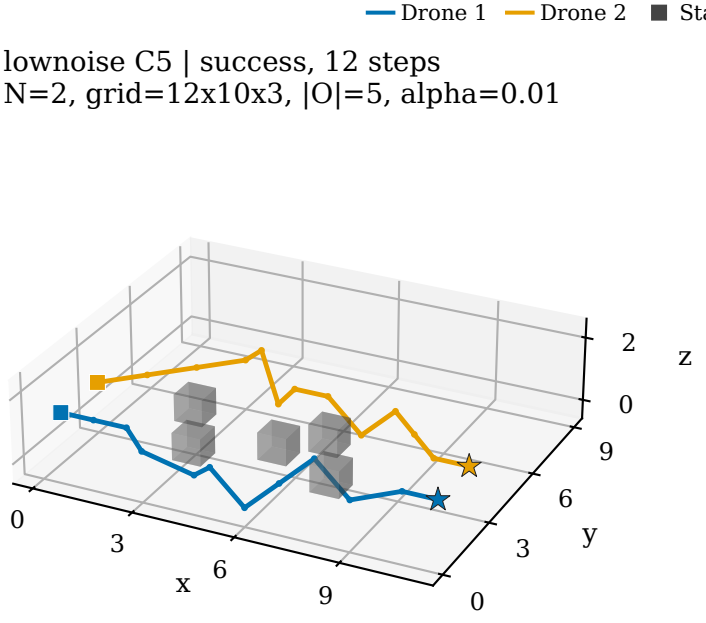
Saved full-solver trace, seed 0: results/trajectories/control_lownoise_C0_full_2000.json
Crosses mark returned positions of unfinished drones; collision targets are not logged.

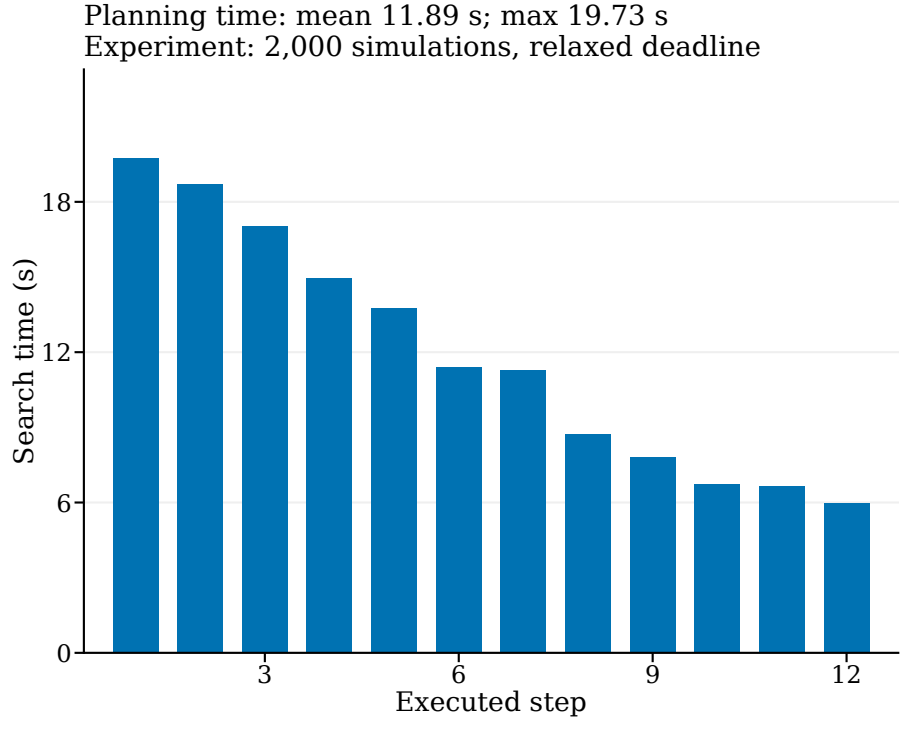
lownoise C2



Saved full-solver trace, seed 0: results/trajectories/control_lownoise_C2_full_2000.json
Crosses mark returned positions of unfinished drones; collision targets are not logged.

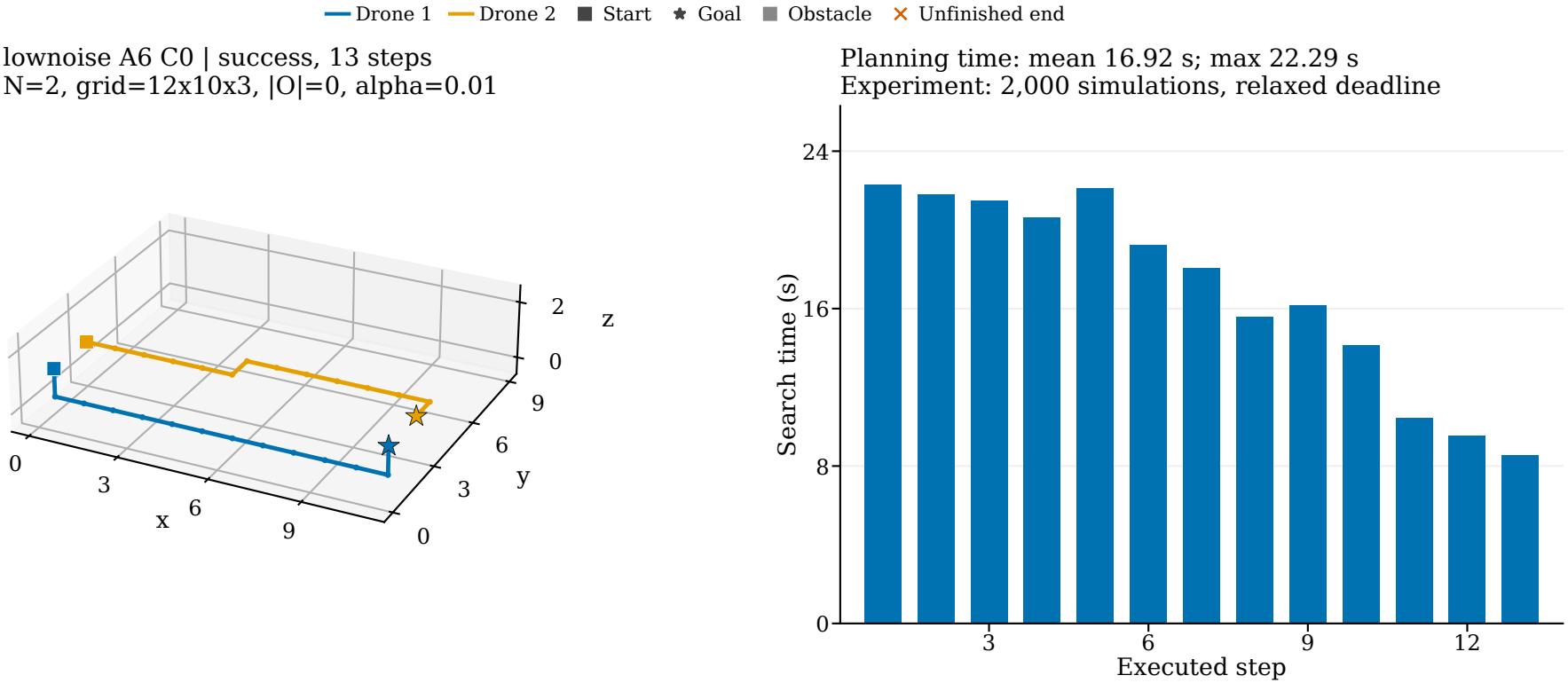
lownoise C5





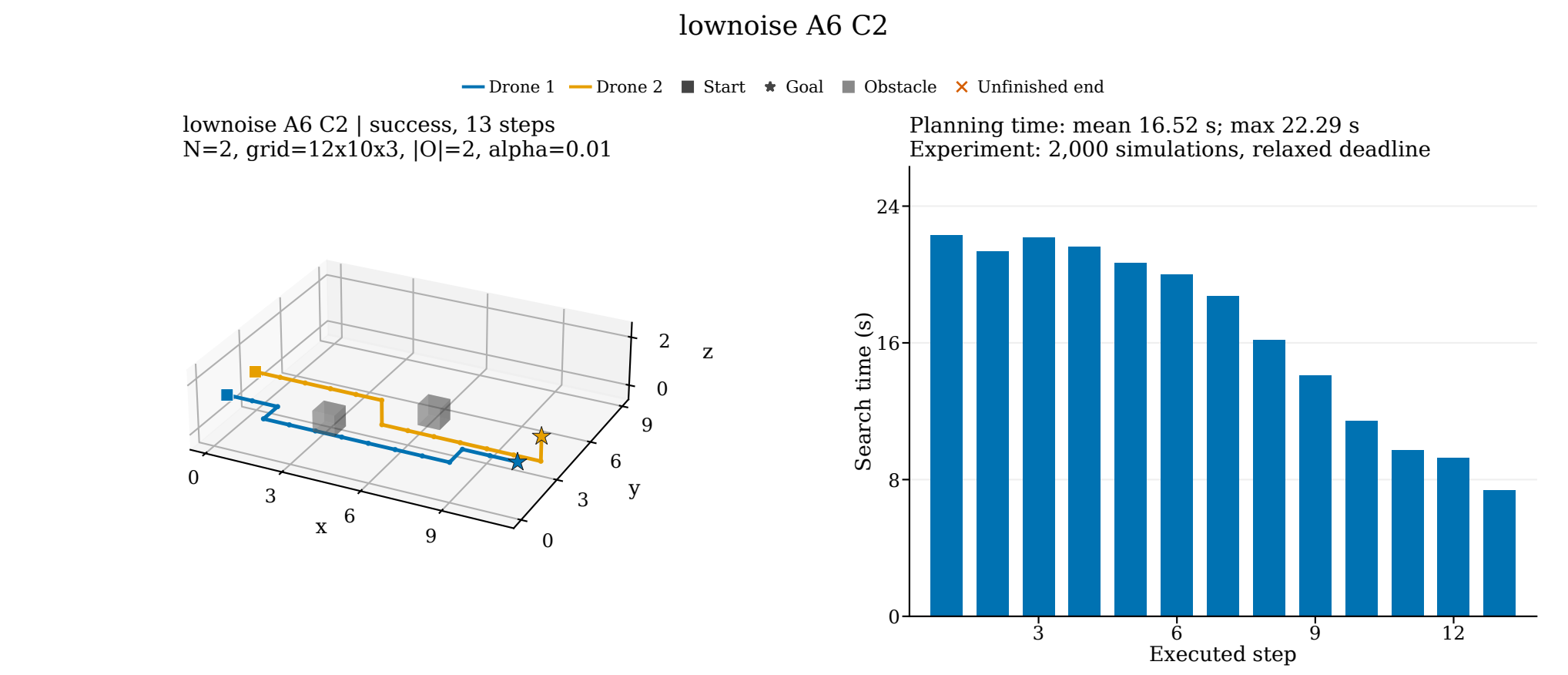
Saved full-solver trace, seed 0: results/trajectories/control_lownoise_C5_full_2000.json
Crosses mark returned positions of unfinished drones; collision targets are not logged.

lownoise A6 C0



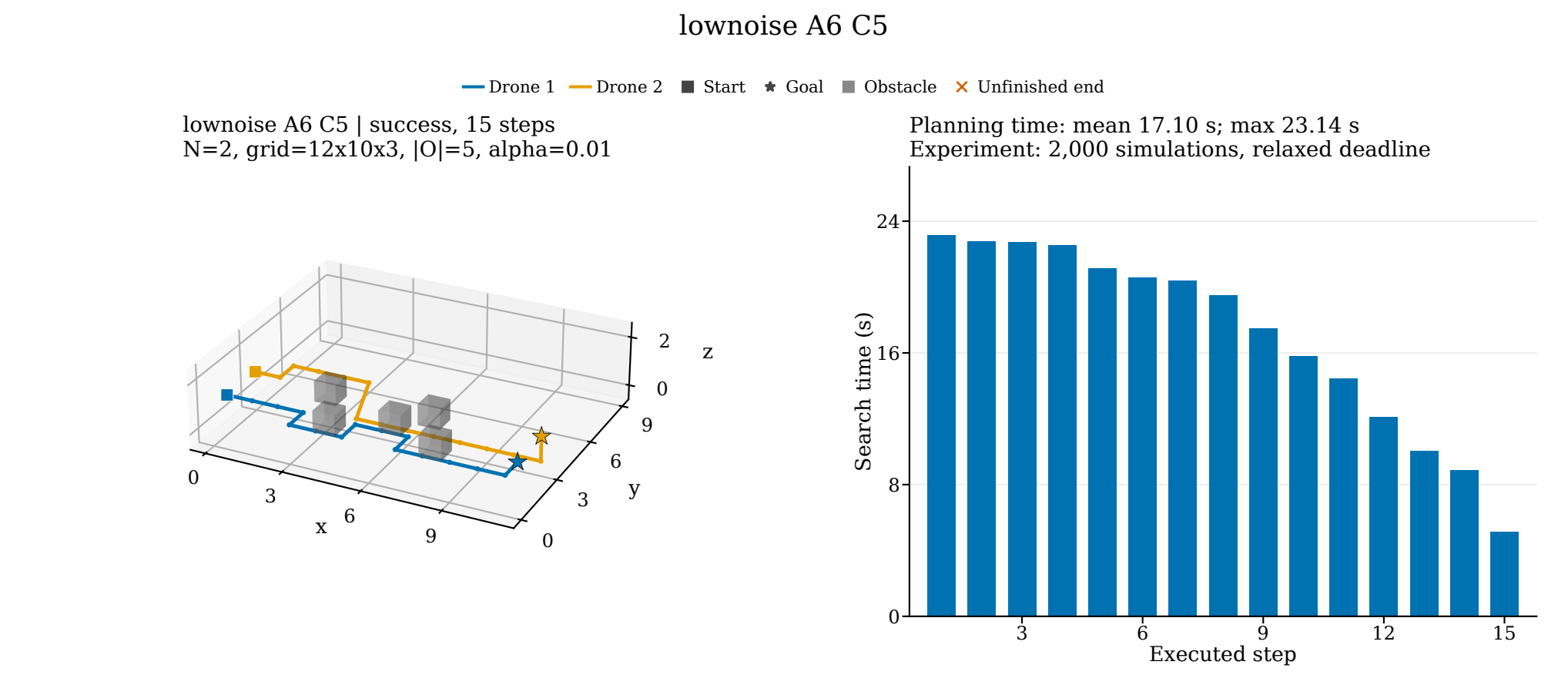
Saved full-solver trace, seed 0: results/trajectories/control_lownoise_A6_C0_full_2000.json
Crosses mark returned positions of unfinished drones; collision targets are not logged.

lownoise A6 C2



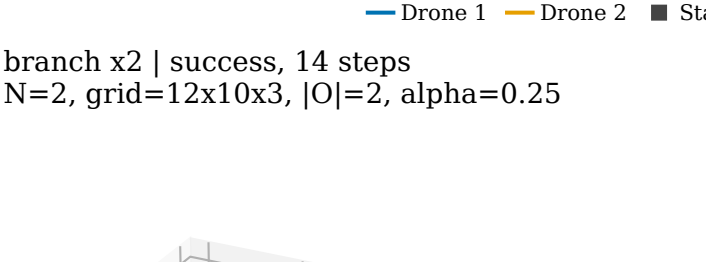
Saved full-solver trace, seed 0: results/trajectories/control_lownoise_A6_C2_full_2000.json
Crosses mark returned positions of unfinished drones; collision targets are not logged.

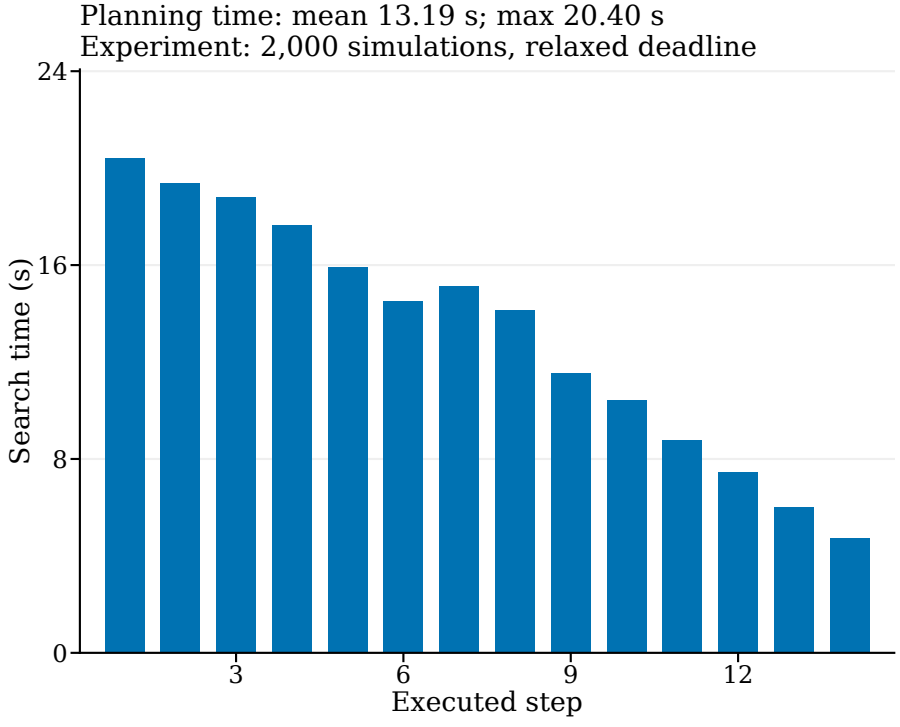
lownoise A6 C5



Saved full-solver trace, seed 0: results/trajectories/control_lownoise_A6_C5_full_2000.json
Crosses mark returned positions of unfinished drones; collision targets are not logged.

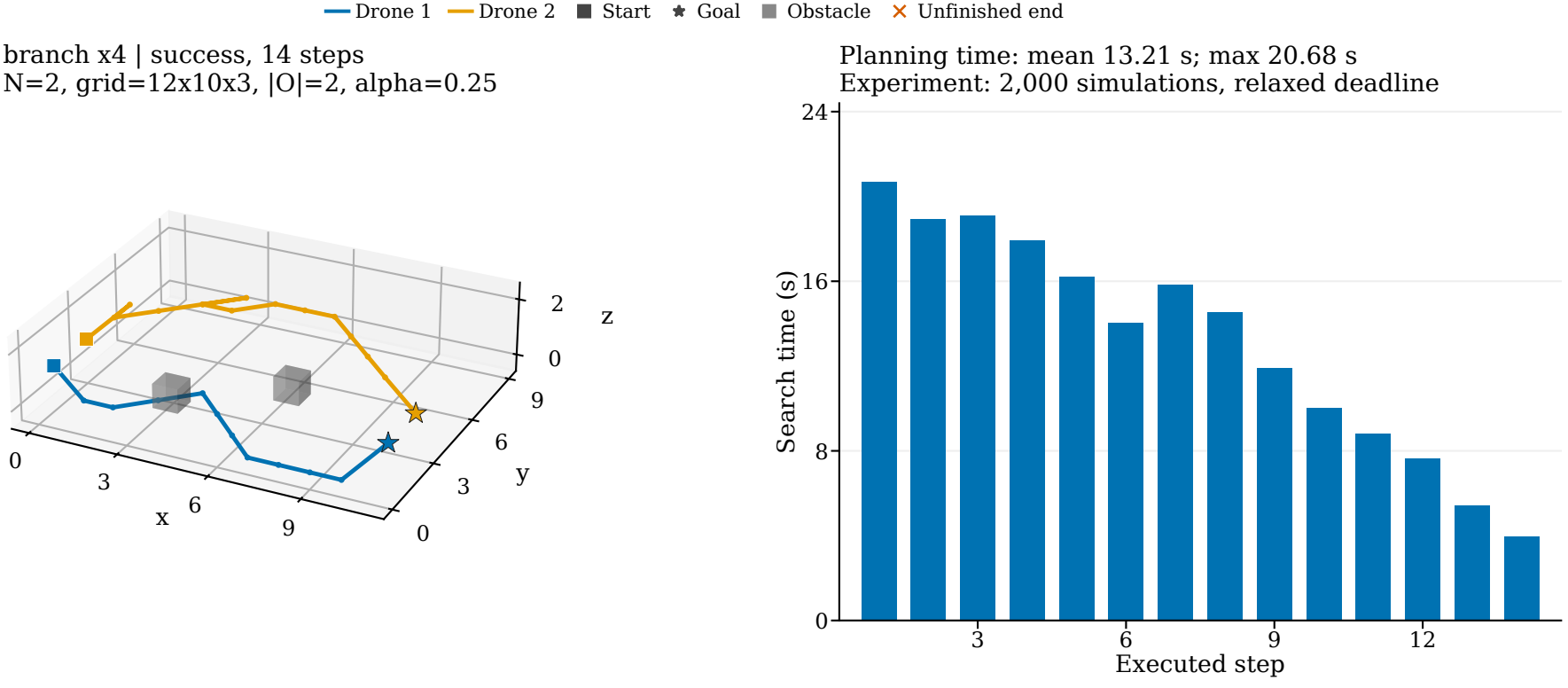
branch x2





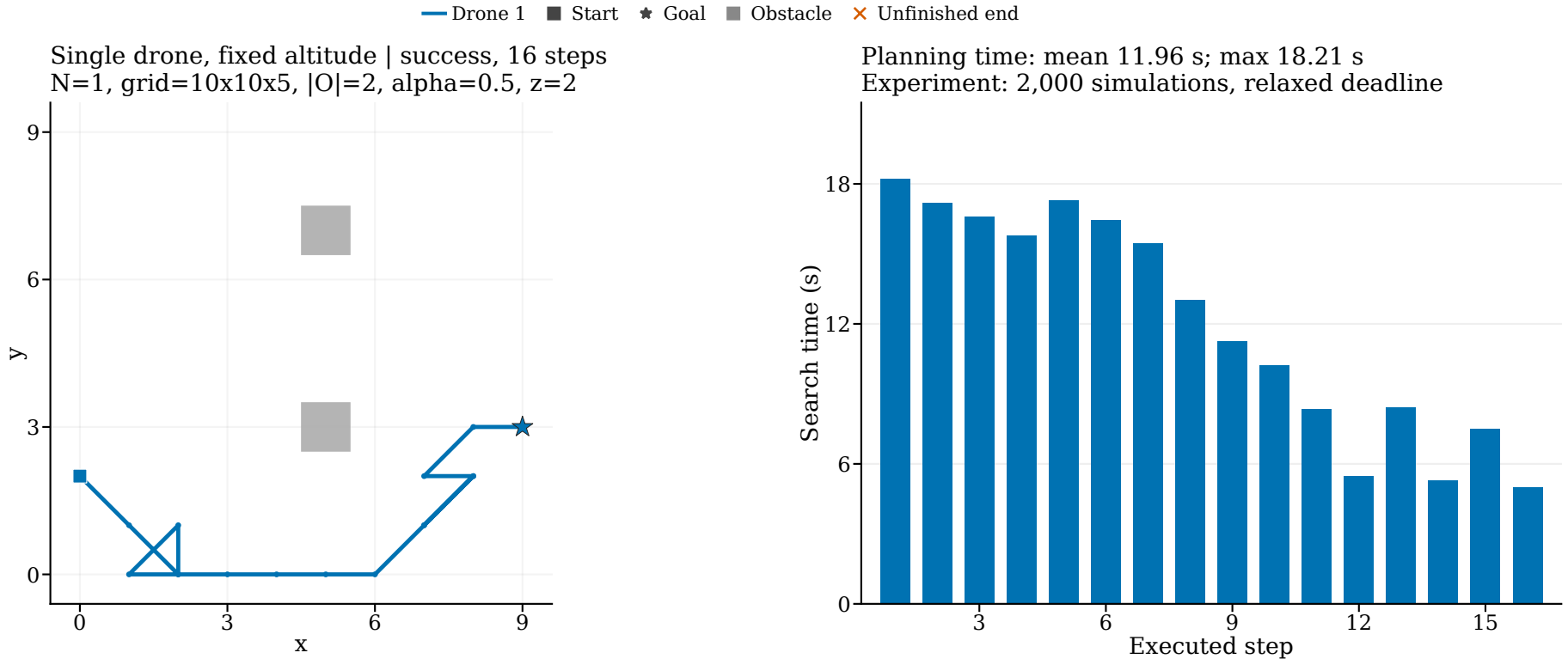
Saved full-solver trace, seed 0: results/trajectories/control_branch_x2_full_2000.json
Crosses mark returned positions of unfinished drones; collision targets are not logged.

branch x4



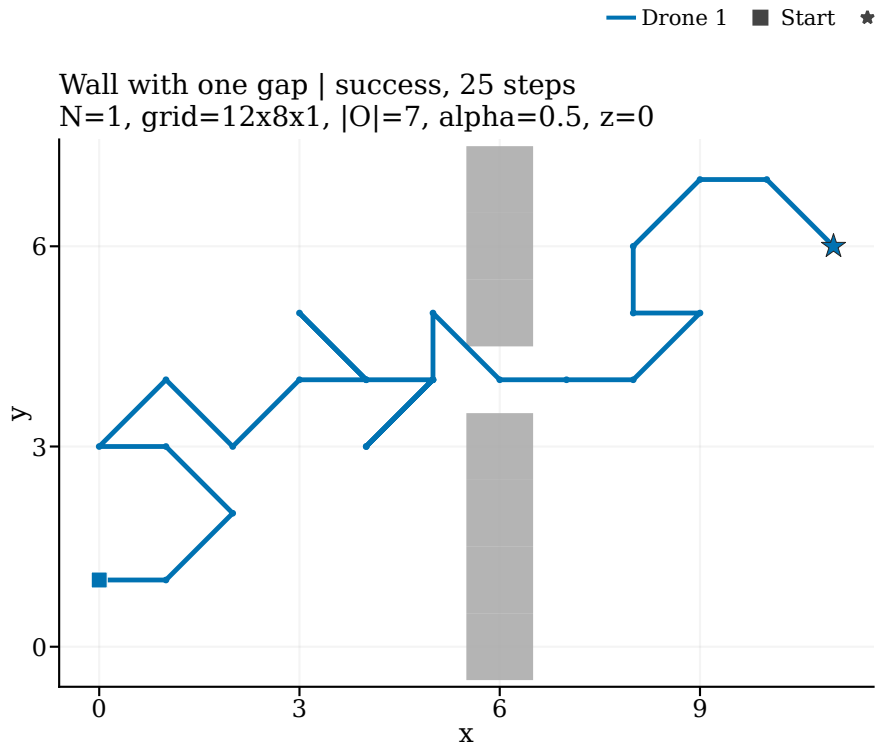
Saved full-solver trace, seed 0: results/trajectories/control_branch_x4_full_2000.json
Crosses mark returned positions of unfinished drones; collision targets are not logged.

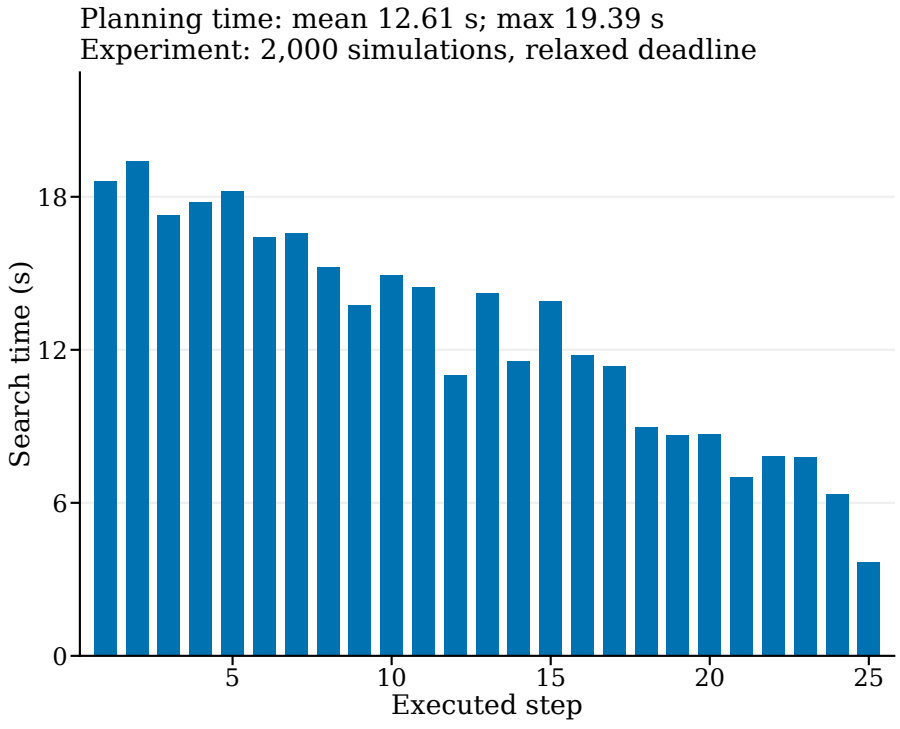
supp single example 2d



Saved full-solver trace, seed 0: results/trajectories/supplement_supp_single_example_2d_full_2000.json
Crosses mark returned positions of unfinished drones; collision targets are not logged.

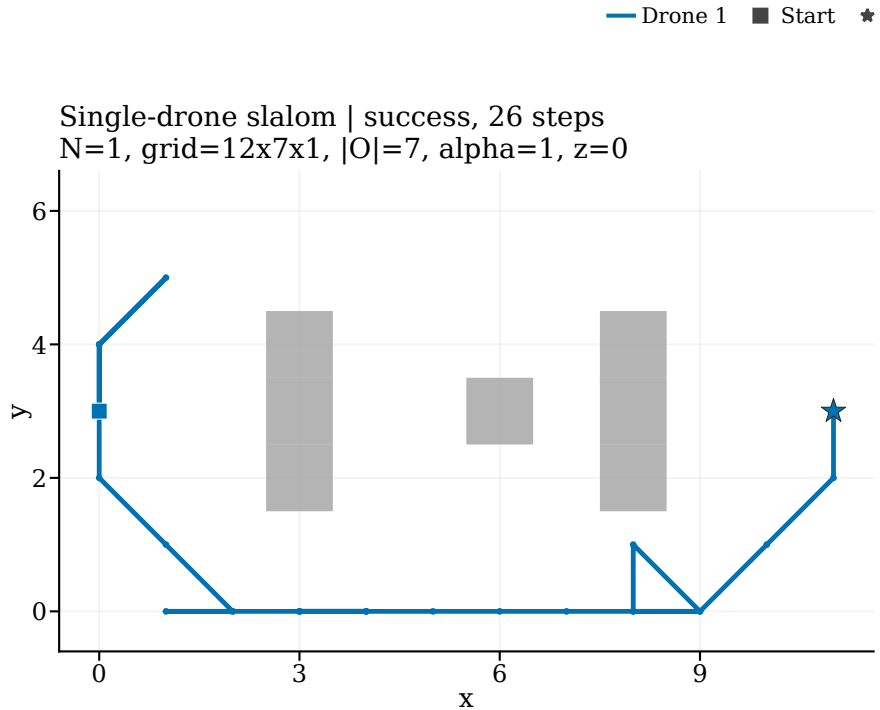
supp single wallgap 2d

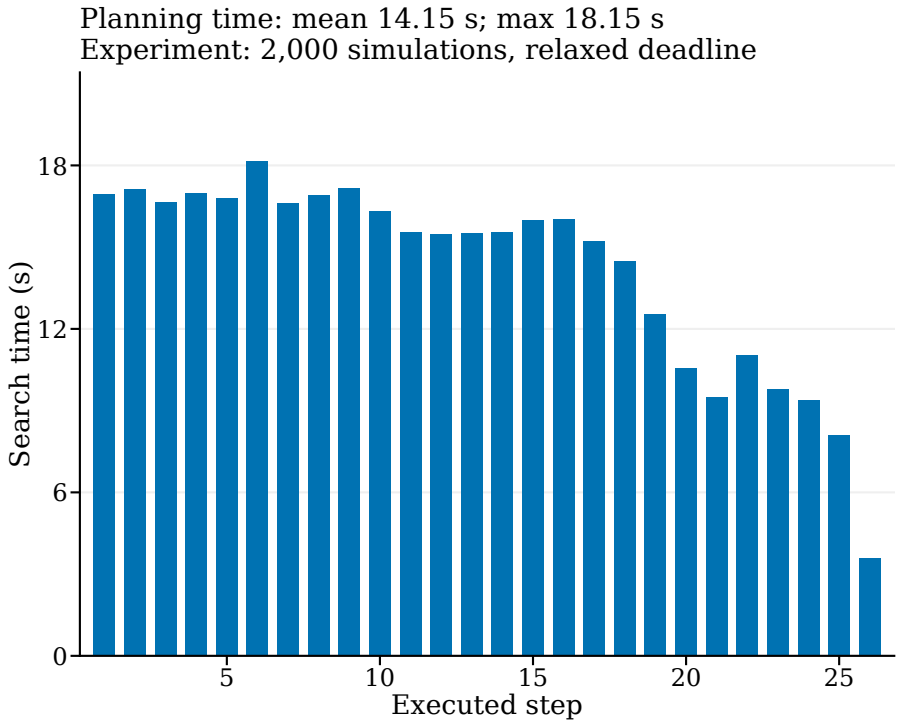




Saved full-solver trace, seed 0: results/trajectories/supplement_supp_single_wallgap_2d_full_2000.json
Crosses mark returned positions of unfinished drones; collision targets are not logged.

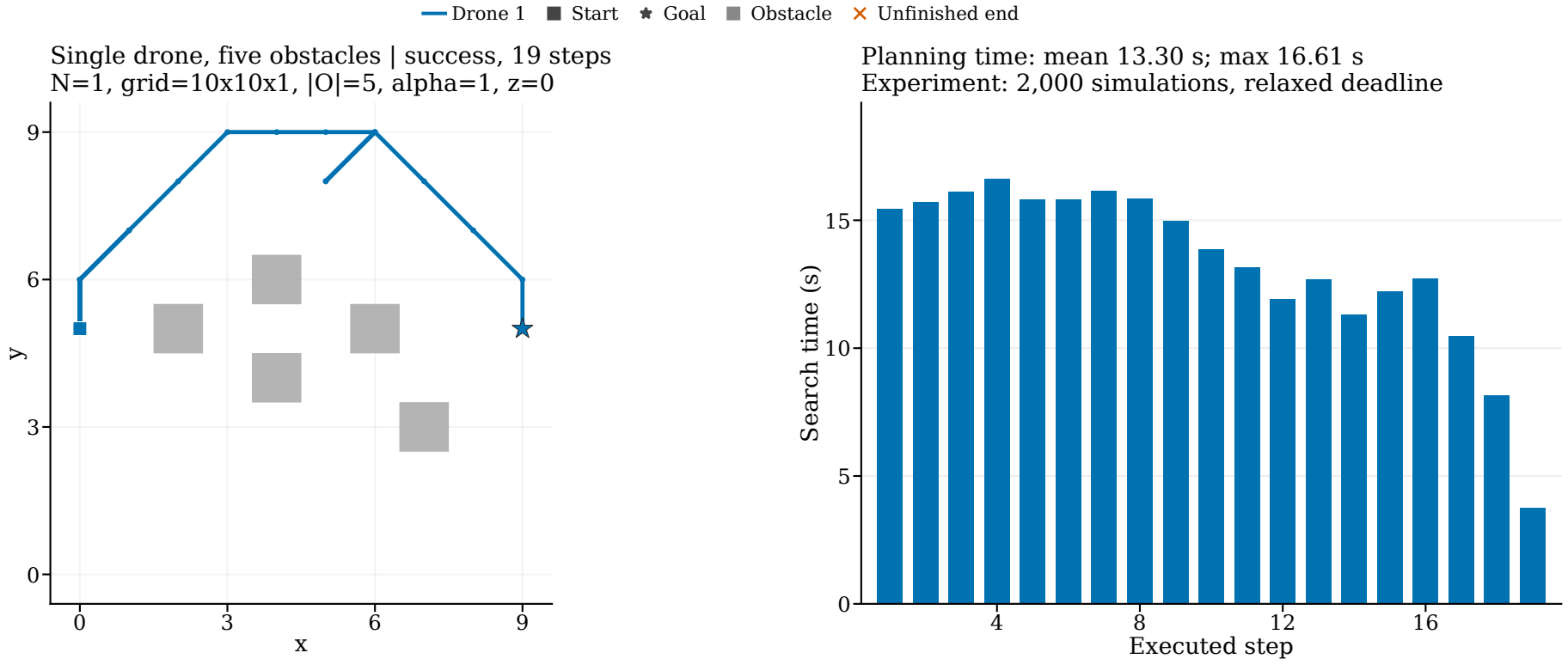
supp single slalom 2d





Saved full-solver trace, seed 0: results/trajectories/supplement_supp_single_slalom_2d_full_2000.json
Crosses mark returned positions of unfinished drones; collision targets are not logged.

supp single clutter 2d

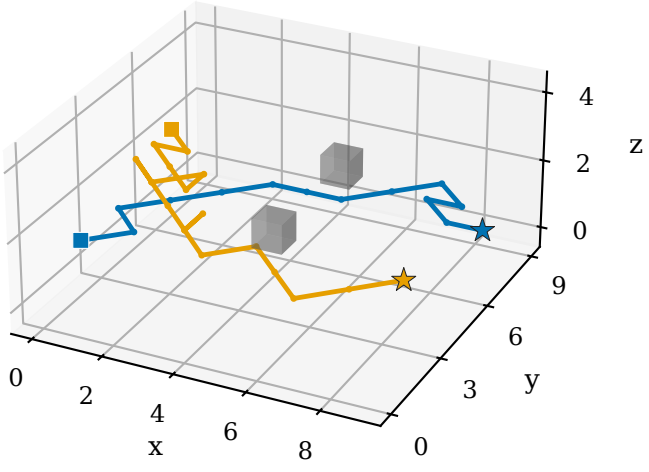


Saved full-solver trace, seed 0: results/trajectories/supplement_supp_single_clutter_2d_full_2000.json
Crosses mark returned positions of unfinished drones; collision targets are not logged.

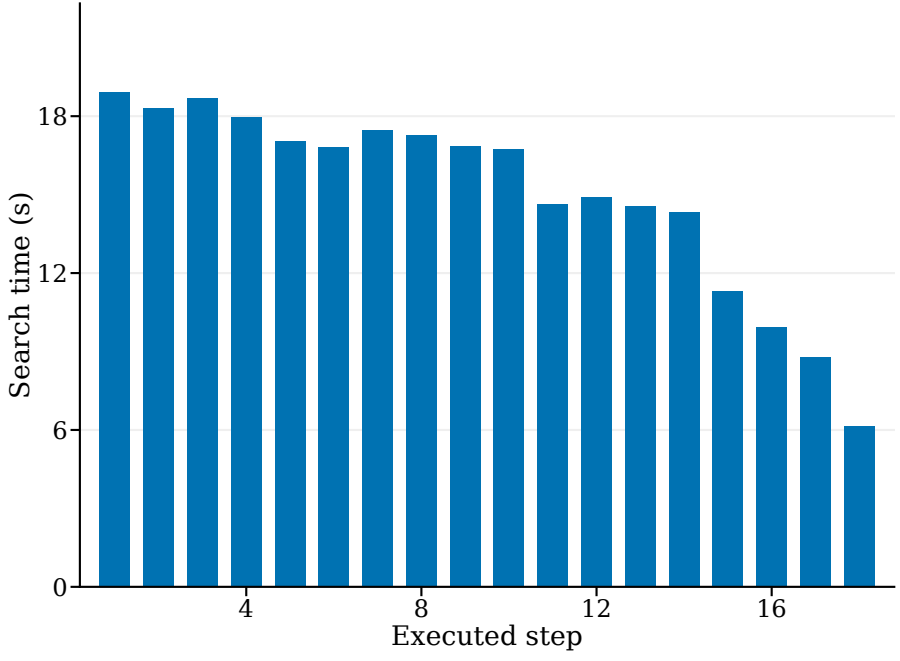
supp layers cross 2d

Drone 1 | Drone 2 | Start | Goal | Obstacle | Unfinished end

Crossing on separate layers | success, 18 steps
N=2, grid=10x10x5, |O|=2, alpha=0.5

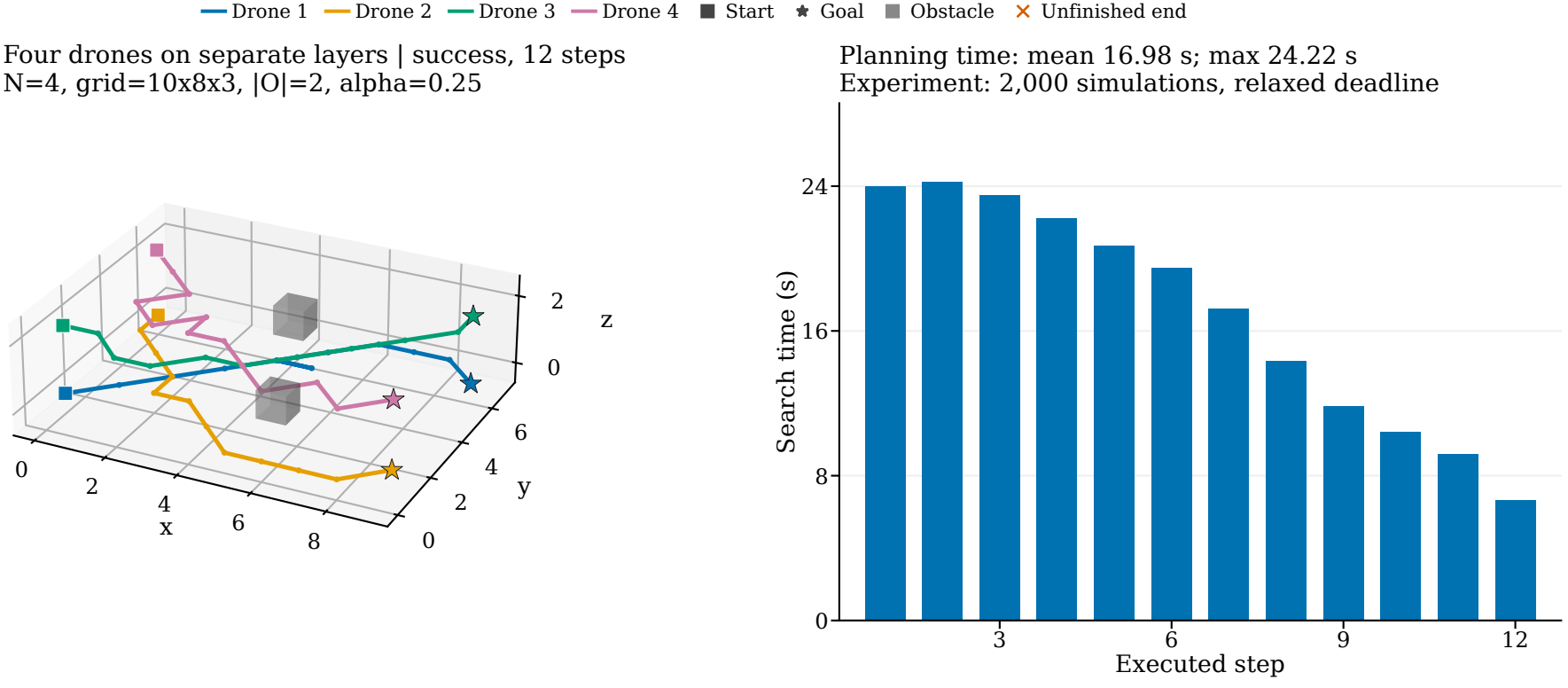


Planning time: mean 15.03 s; max 18.91 s
Experiment: 2,000 simulations, relaxed deadline



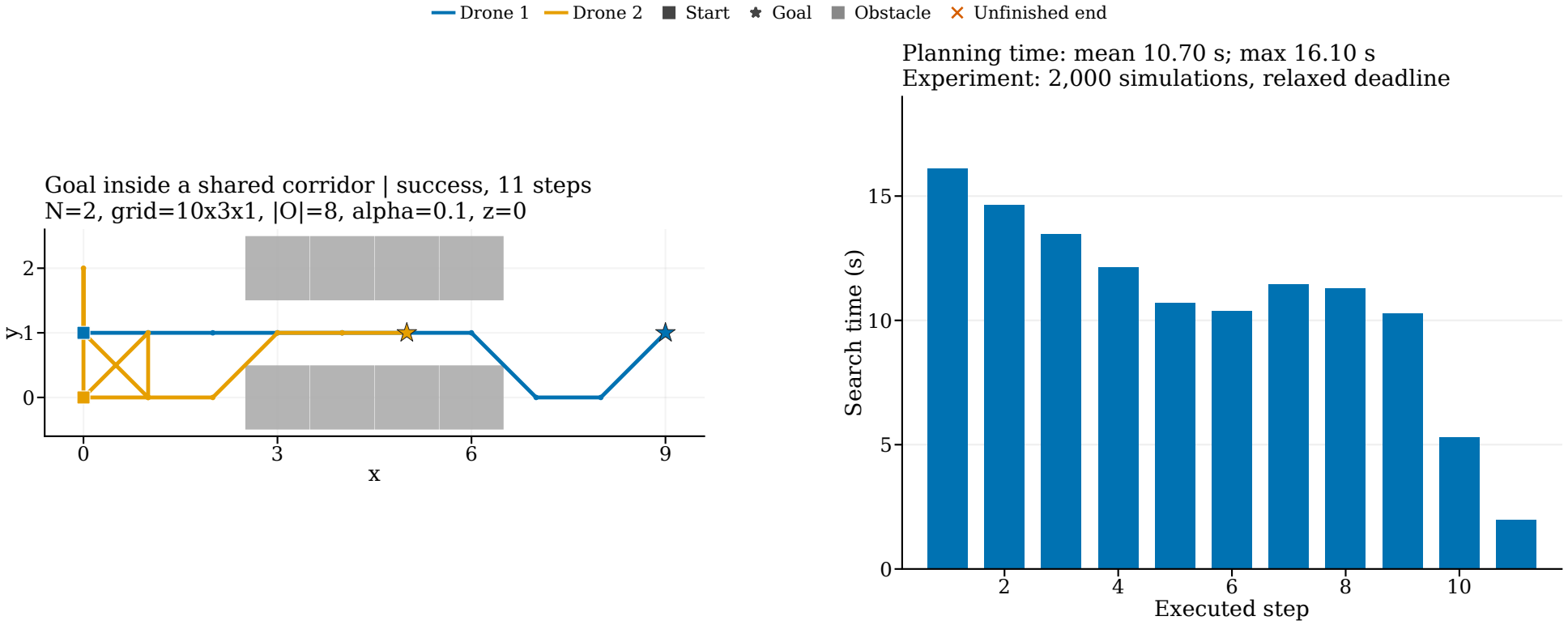
Saved full-solver trace, seed 0: results/trajectories/supplement_supp_layers_cross_2d_full_2000.json
Crosses mark returned positions of unfinished drones; collision targets are not logged.

supp layers four 2d



Saved full-solver trace, seed 0: results/trajectories/supplement_supp_layers_four_2d_full_2000.json
Crosses mark returned positions of unfinished drones; collision targets are not logged.

supp yield corridor



Saved full-solver trace, seed 0: results/trajectories/supplement_supp_yield_corridor_full_2000.json
Crosses mark returned positions of unfinished drones; collision targets are not logged.